



Submittal Review Response

Project Name: Hilo WWTP Rehabilitation and Replacement Project Phase 1
Submittal No.: 11320-001.0
Date: 9/26/2025

Client: County of Hawai'i Carollo Project No.: 203975
Contractor: Nan, Inc.
Submittal Name: Flygt Septage Lift Station
Reviewed By: Francisco Martinez / Christian Wimenta / Tanner Howe / Felicia Fan

SUBMITTAL REVIEW

Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.

Approved	☐	No Exceptions
	☐	Make Corrections Noted - See Comments
	☐	Make Corrections Noted - Confirm
Not Approved	☒	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

Response to Deviations/Exceptions to Specifications (pages 25-26 of 138):

1. Deviation 1: Specs call for cover 6063-T5, but this alloy only applies to the extrusions; standard cover material will be alloy 5052.
 - a. Sump cover shall be aluminum alloy 6061-T6 per Specification Section 05140, 2.01.B. Confirm covers meet loading requirements.
2. Deviation 2: Pump housing is Cast Iron ASTM A-48, Class 35B
 - a. It is acceptable for the pump housing material to be cast iron ASTM A48, Class 35 B per Specification Section 11320, 2.04.
3. Deviation 3: Pump is permanently submersible according to standard IEC 60034 and protection class IP 68
 - a. Confirm that motor construction meets requirements including but not limited to NEMA design type B per Specification Section 11320, 2.14.C.
4. Deviation 4: SUBCAP Pump Cable outer jacket is oil resistant chlorinated polyethylene rubber.
 - a. Confirm that "SUBCAP" is a typo and the proposed pump cable is the Flygt SUBCAB cable. Provide corresponding product data in resubmittal and confirm that the pump cable meets or exceeds the specs of the type SPC with Hypalon/Buna N or the chloroprene rubber jacket that is called out per Specification Section 11320, 2.14.D.2
5. Deviation 7: Float switch in stator chamber is not available
 - a. Moisture detection shall be provided per Specification Section 11320, 2.14.F.1.b.

6. Deviation 15050 2.24: Not applicable. Parts are manufactured as one unit. Separating parts will void warranty.
 - a. Equipment shall be shipped per Specification Section 15050, 2.24. It is acknowledged that in cases where parts are manufactured as one unit, packing parts separately is not possible without voiding warranty. In cases such as this, Contractor shall provide other protection during transport per Specification Section 15050, 2.24.A.1.a.
7. Deviation 15958 3.03.B: Closest HI Level to specification is Grade 2B Tolerance due to small pump's relative performance band
 - a. Provide proposed performance guarantees (tolerances) for each pump selection and how much they deviate from the specified tolerances.

General Notes:

8. Bookmark the submittal per Specification Section 01330, 1.03.C.
9. Sequentially number the submittal pages per Specification Section 01330, 1.03.F.
10. Pages 78-79 of 138 and 97-100 of 138 are oriented incorrectly. Revise as necessary to ensure that all submittal pages are oriented correctly.
11. It is understood that the submitted installation, operation, and maintenance manual (pages 63-138 of 138) are a draft version. The final vendor operation and maintenance manuals are to be provided in a separate submittal per Specification Section 01782.

FRP Sump, Sump Cover, and Sump Hatch:

12. General Notes on pages 33-35 of 138 of the submittal do not include all specified Prefabricated FRP Sump materials. Submit all materials per Specification Section 11320.2.03.A for review and approval.
13. Confirm that 316 SST lifting lugs are to be provided per Specification Section 11320, 2.03.D. Note that each lifting lug shall be capable of handling the entire weight of the station.
14. Confirm that 316 SST anchoring lugs are to be provided per Specification Section 11320, 2.03.E.
15. Calculations shall be submitted for review and approval per Specification Section 11320, 1.04.D.
16. Aluminum surfaces shall be coated (as needed per Section 05140) as specified in Specification Section 09960. Refer to Attachment B - Coating Detail Sheets in Specification Section 09960 for coating requirements. Submit the proposed coating system for review and approval.
17. Confirm that a curved 316 SST transition plate shall be provided and installed on the inside edge of the cable tray to support the cables and transition them from the horizontal to vertical direction without resulting in point stresses, per Specification Section 11320, 2.03.B.3.
18. The orientation of the cable tray for the Headworks Drain Sump lift station, as shown on the submitted shop drawings, does not align with the Contract Drawings. Revise as necessary to ensure the cable tray orientation matches the Contract Drawings.
19. Submit sump cover and hatch detail shop drawings for Septage Sump lift stations 1 and 2 per Specification Sections 11320, 2.03.B and 11320, 2.03.C for review and approval.
20. Submittal page 52 of 138 indicates that Pump Model No. 3085 requires a Flygt Lift System with Chain Sling: 442 18 06 and Grip Eye: 620 09 00. Submittal page 49 of 138 indicates that only the grip eye/chain combinations listed in the provided table are approved for function and safety. Confirm that the proposed grip eye and chain sling combination is approved by the manufacturer for function and safety.
21. Contract Drawings 01-M-01-101 and 01-M-01-301 show foul air duct penetrations at the top of each packaged lift station. Revise shop drawings as necessary to ensure that the FRP duct penetrations, as shown in Typical Detail H142, are compatible with the packaged lift station. All modifications to the packaged lift station required to accommodate the duct penetrations shall be shown on shop drawings in resubmittal.
22. The Headworks Drain Sump shop drawing (25-186) on submittal page 33 of 138 does not include the 4" SPD connection. Include the 4-inch SPD per 01-M-01-301.
23. It is unclear if the submitted cable tray dimensions match those of the respective concrete trench at the edge of the sump. Provide confirmation that the submitted cable tray dimensions match those of the respective concrete trench at the edge of the sump as required per Specification Section 11320, 2.03.B.3.
24. Power and control cable product data is missing from the submittal. Provide power and control cable product data and show compliance with specification requirements as listed in Specification Section 11320 - 2.14.D.

25. Contractor to confirm that all power and control conductors shall terminate at terminal blocks in the local control panel or junction box as required per Specification Section 11320, 2.14.D.7.
26. Contractor to confirm that stainless steel cable and stainless steel wire braid sleeves to support power cable shall be provided as required per Specification Section 11320, 2.14.D.9.

Pump:

27. Submit proposed coating systems as required in Specification Section 11320, 2.19 for review and approval per Specification Section 09960.
28. Product data including but not limited to Pump and Motor Casing, Impellers, Wear Rings, Pump Shafts, Bearings, Shaft Seals, Discharge Base and Elbow, and Motor and Power Cables are not provided. Revise as necessary to include all product data per Specification Section 11320, 1.04.B to allow verification that Specification is satisfied.
29. Spare parts shall be provided per Specification Section 11320, 2.20.
30. Spare Parts Inventory List shall be provided per Specification Section 01600, 2.04.A.1.
31. It is understood that Commissioning submittals are to be submitted for review and approval per Specification Section 11320, 1.04.F in a forthcoming submittal after approval of this submittal.
32. It is understood that Warranty is to be submitted for review and approval per Specification Section 11320, 1.07 in a forthcoming submittal.
33. It is understood that Package Control System is to be submitted for review and approval per Specification Section 11320, 2.15 in a forthcoming submittal.

Concrete Ballast:

34. Provide calculations for resistance to ground water buoyancy uplift effect. Ground water elevation shall be assumed at grade level per Specification Section 01850, 2.04.C, and Section 11320, 2.02.B.3.a.1.c.2&3.
 - a. As a minimum, concrete ballast at the base of the tanks shall be per the Contract Drawings B/01-S-01-101 and Section D/ 01-S-01-301. Additional concrete weight shall be provided as required to resist buoyancy uplift.
 - b. Submit shop drawings for reinforcement in the concrete ballast, minimum reinforcement shall be per the Contract Drawings.
35. Submittal drawing number 25-186 & 25-187 detail "2" indicates $\frac{3}{4}$ " x 4-3/4" wedge anchor. Provide supporting calculations for anchorage design per Specification Section 01850, 2.04.A.1.c and 2.04.E.3.
36. Submit calculations for the design of the submitted septage sumps per Specification Section 11320, 2.02.B.3.a.1.c.
37. Stamp and seal design calculations and drawings per Specification Section 01357, 1.03.D.2.

High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. Project No.: WW-4705R
Project Name: Hilo WWTP Phase 1 Submittal Number:
Submittal Title: For Information Only
TO:
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec:	Paragraph:
Authorized By:	Date Submitted:

Submittal Certification		
Check Either (A) or (B):		
☐ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .	
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.	
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.		
General Contractor's Reviewer's Signature: 		
Printed Name and Title:		
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.		
Firm:	Signature:	Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED _____
SPECIFICATION SECTION # _____
SPECIFICATION _____
PARAGRAPH _____
DRAWING _____
SUBCONTRACTOR _____
SUPPLIER _____
MANUFACTURER _____

CERTIFIED BY CQCM or Designee : 



1082 Makepono St. Honolulu, HI 96819 Ph:(808)841-0033 Fax:(808)841-2534 info@hiengineering.com

AREA USED:

1. 01 - Septage Receiving
2. 02 - Headworks

Replace Entire Section

SECTION 11320

PACKAGED SEWER LIFT STATION

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes: Pump systems including packaged submersible sump pump systems.

1.02 REFERENCES

- A. American Bearing Manufacturers Association (ABMA):
 - 1. 9 - Load Ratings and Fatigue Life for Ball Bearings.
 - 2. 11 - Load Ratings and Fatigue Life for Roller Bearings.
- B. American Society of Mechanical Engineers (ASME):
 - 1. B 16.1 - Cast Iron Pipe Flanges and Flanged Fittings, Class 25, 125, 250, and 800.
- C. ASTM International (ASTM):
 - 1. A48 - Standard Specification for Gray Iron Castings.
 - 2. A278 - Specification for Gray Iron Castings for Pressure-Containing Parts for Temperatures Up to 650 degrees Fahrenheit.
- D. Hydraulic Institute (HI):
 - 1. 9.1-9.5 - Pumps - General Guidelines.
- E. CSA International (CSA).
- F. FM Global (FM).
- G. Insulated Cable Engineer's Association (ICEA).
- H. National Electrical Code (NEC).
- I. National Electrical Manufacturers Association (NEMA):
 - 1. 250 - Enclosures for Electrical Equipment (1000 V Maximum).
- J. Underwriters Laboratories, Inc. (UL):
 - 1. 508 - Standard for Industrial Control Equipment.
 - 2. 508A - Standard for Industrial Control Panels.
 - 3. 698A - Standard for Industrial Control Panels Relating to Hazardous (Classified) Locations.
 - 4. 1283 - Standard for Electromagnetic Interference Filters.
 - 5. 1449 - Surge Protective Devices.

1.03 DEFINITIONS

- A. Pump Head (Total Dynamic Head, TDH), flow capacity, pump efficiency, net positive suction head available (NPSHa), and net positive suction head required (NPSHr): As defined in HI 9.1-9.5, 14.1-14.2, 14.3, 14.4, 14.6, and as modified in this Section.
- B. Tolerances: As stipulated in the listed HI standards, unless specified more restrictively.

1.04 SUBMITTALS

- A. Submit as specified in Section 01330 - Submittal Procedures.
- ✓ B. Product data: As specified in Section 15050 - Common Work Results for Mechanical Equipment.
- ✓ C. Shop drawings: As specified in Section 15050 - Common Work Results for Mechanical Equipment.
- D. Calculations: As specified in Section 15050 - Common Work Results for Mechanical Equipment.
 - 1. Wet well design: Design wet well to include the following criteria.
 - a. Piping penetrations for all pipe sizes shown on the Drawings.
 - b. Cable wall penetrations.
 - c. Anti-floatation ring:
 - 1) Packaged sewer lift station supplier shall design the wet well and anti-floatation ring to resist floatation assuming the groundwater level is at-grade.
 - d. Pipe supports and bracing inside the wet well.^{AD4}
- ~~D-E.~~ Vendor operation and maintenance manuals: As specified in Section 01782 - Operation and Maintenance Manuals.

 ~~E-F.~~ Commissioning submittals:

- 1. Provide Manufacturer's Certificate of Source Testing as specified in Section 01756 - Commissioning.
- 2. Provide Manufacturer's Certificate of Installation and Functionality Compliance as specified in Section 01756 - Commissioning.

1.05 PROJECT CONDITIONS

- A. Environmental requirements: As specified in Section 01850 - Design Criteria.
- B. As specified in Section 15050 - Common Work Results for Mechanical Equipment.
- C. All devices mounted within or around the equipment shall be suitable for the area classification in accordance with Section 16050 - Common Work Results for Electrical.

Will be submitted in a
seperate submittal

1.06 SEQUENCING AND SCHEDULING

- A. Coordinate work with restrictions specified in Section 01140 - Work Restrictions.

1.07 WARRANTY

- A. ✓ Provide warranty as specified in Section 01783 - Warranties and Bonds.

PART 2 PRODUCTS

✓ 2.01 MANUFACTURERS

- A. Packaged system consisting of pumps, motors and components, including base elbows, guide rails and lifting devices (where indicated on the Drawings), electrical devices internal to pump housing, all submersible cabling for power and control conductors, sump level measurement and control components, Vendor Control Panels (VCP), fiberglass sump (tank), [access hatch and fall protection grating](#), [sealed piping penetrations](#),^{AD4} and inserts:
 - 1. Flygt, TOP basin, no equal, to match existing equipment at Owner's facilities for standardization.

✓ 2.02 DESIGN AND PERFORMANCE CRITERIA

- ✓ A. Submersible sump pumps with components: Pumps, motors, base elbows, guide rails and lifting devices, electrical devices internal to pump housing, all submersible cabling for power and control conductors, sump level transducers and control components, Vendor Control Panels, fiberglass pump manhole inserts, [access hatch and fall protection grating](#), [sealed piping penetrations](#),^{AD4} and other items as required for a complete and operational system.
- ✓ B. Design requirements:
 - 1. Pump performance characteristics: Refer to ~~Sump~~^{AD4}-Pump Schedule appended to this Section.
 - 2. Motor characteristics: Refer to ~~Sump~~^{AD4}-Pump Schedule appended to this Section.
 - 3. Sump requirements:
 - a. Design the lift station as indicated on the Drawings and as follows:
 - 1) Provide underground storage tank with following design criteria:
 - a) Interior diameter as specified on the Drawings.
 - b) Tank height: as specified on the Drawings:
 - (1) The Septage Sumps and Headworks Drain Sump shall share a common ballast as indicated in the Drawings.
 - c) Structural Requirements:
 - (1) Loads against walls. Include effects of groundwater and seismic accelerations on earth pressures:
 - (a) Equivalent Lateral Pressure: Triangular distribution: 60 pounds per square foot per foot of depth (lateral triangular distribution).

- (b) Equivalent Lateral Pressure: Rectangular distribution backfill-induced live load surcharge: 250 pounds per square foot (lateral uniform rectangular pressure).
- (2) Groundwater effects: Include transient hydrostatic pressure from groundwater effects on lateral earth pressure loads using design elevation to site finished grade elevation:
 - (a) Use equivalent lateral pressure of 122 pounds per square foot per foot of depth (triangular distribution) for soil below the design groundwater elevation.
 - (b) Groundwater effects do not need to be combined with seismic loads.
- (3) Groundwater and flood loads:
 - (a) As specified in Geotechnical Report for design groundwater and design flood elevations.
 - (b) Lateral pressure effects: Determine based on groundwater and flood elevations specified.
- (4) Seismic acceleration effects:
 - (a) As specified in Section 01850 - Design Criteria.
 - (b) On opposite sides of the structure, uniform equivalent lateral pressure in pressure type distribution, with a pressure of $29H$ in pounds per square foot, where " H " is the depth of structure from the ground surface.
 - (c) Adding lateral force for soil accelerating toward structure: Direct rectangular uniform pressure distribution toward the wall, effectively increasing the static lateral soil pressure.
 - (d) Reducing lateral force for soil accelerating away from structure: Direct inverted pressure distribution away from the wall, effectively reducing the static lateral soil pressure.
- (5) Vertical cylinder made integrally with a reinforced bottom capable of withstanding hydrostatic pressure with water to grade while the station is completely empty.

See
Deviation/Exception
sheet



- C. Product requirements as specified in Sections 01600 - Product Requirements and 15050 - Common Work Results for Mechanical Equipment.

2.03 PREFABRICATED FRP SUMP

A. Materials:

- ✓ 1. Interior finish: Premium isophthalic NPG gelcoat or white pigmented premium grade isophthalic polyester, at minimum.
- ✓ 2. Structural laminate:
 - ✓ a. Resin: Premium grade isophthalic polyester, at minimum.
 - b. Glass fiber reinforcing: Commercial Grade "E" type glass.
 - c. Corrosion liner on the interior surface.
- ✓ 3. Corrosion Liner: The surface of the liner exposed to the corrosive medium shall be resin rich reinforced with a "C" grade surfacing veil. The veil shall be saturated with white pigmented resin or an ISO-NPG white gelcoat. This layer is to be .01 inch minimum thickness. The liner behind the surfacing veil shall have a minimum thickness of .10 inch and shall be reinforced with not less

than 20 percent and not more than 30 percent by weight of non-continuous chopped strand mat. The inside surface is to be a smooth molded surface with a bright white finish. The corrosion liner shall be free of air and voids for optimum corrosion resistance.

4. Surface Finish:

a. Interior: All inside surfaces should be smooth and free of cracks and crazing. The inside surface shall be pigmented, 4545 or gel coated to a bright white finish. All surfaces other than those made in contact with the mold surface shall be coated with air-inhibited resin or gelcoat; this includes any cut edges of laminates.

5. Exterior: All external surfaces are to be resin coated with an air inhibited resin coat, including any drilled holes, ground areas or cut edges.

1

B. Provide sump with cover that is flush with the surrounding slab. Refer to Drawings for elevations:

1. Material: 6063 T5 aluminum checkered plate.
2. Rated to support 300 pounds per square-foot minimum.
3. Provide integral cable tray on the under-side of the sump cover with a removable, flush, and pedestrian rated 6063 T5 aluminum checkered plate cover. Cable tray shall be compatible with sump hatch, permitting the removal of the pump cable without threading or disassembly of the cable. The cable tray shall be routed from the sump hatch to the concrete trench at the edge of the sump as indicated on the Drawings. Cable trench dimensions shall match those of the respective concrete trench at the edge of the sump. A curved 316 SST transition plate shall be provided and installed on the inside edge of the cable tray to support the cables and transition them from the horizontal to vertical direction without resulting in point stresses.

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C. Provide flush to cover, hinged, and lockable hatch as indicated on the Drawings:

1. Type 6063 T5 aluminum construction with Type 316 SST hardware and the following features:
 - a. Door design utilizing structural shapes and reinforced checkered plate.
 - b. Safe-Hatch access cover.
 - c. Minimum metal thickness: 1/4 inch for structural shapes and plates.
 - d. Torsion bar or other device to assist opening.
 - e. Automatic hold-open arm.
 - f. Retractable handle.
 - g. Recessed padlock.
 - h. Strap anchors bolted to frame exterior.
 - i. Lifting cable hook and upper bracket mounting support.
 - j. Rated to support 300 pounds per square-foot minimum.



D. Provide minimum (2) lifting lugs, each capable of handling the entire weight of the station:

1. Material: 316 SST.



E. Provide anchoring lugs to secure the lift station to the concrete ballast:

1. Material: 316 SST.
2. Designed to fit over 7/8-inch diameter "J" type anchors embedded in the concrete ballast.

- ✓ F. Provide tank with fiberglass reinforced plastic self-cleaning basin as indicated in the Drawings:
1. Integral, hopper-shaped lift station bottom with a flat surface area that minimizes the area directly influenced by pump suction.
 2. The bottom surface area shall have a ratio of 1:4 as it relates to the cross-sectional area of the pump station.
 3. Designed for mounting and removal of compatible submersible pump with guide rails.
 4. Compatible with pump discharge base and elbow.

2

2.04 PUMP MATERIALS

- A. General: When materials are referenced in this Section or on the pump schedule, the compositions shall be the UNS Alloys, Types, or Grades in this Article unless specified or scheduled otherwise.
- B. **Cast iron: ASTM A48, Class 35 B minimum.**
- C. Nickel cast iron: ASTM A48, Class 35 minimum with 3 percent nickel added.
- D. Steel: ASTM A108, Grade or UNS Alloy as specified or scheduled.
- E. Stainless steel: ASTM A276 or ASTM A582, Type or UNS Alloy as specified or scheduled.
- F. Bronze: ASTM B505 or ASTM B584, UNS Alloy C83600.
- G. Zincless bronze: ASTM B505 or ASTM B584, Leaded Tin Bronze, UNS Alloy C92700.
- H. Aluminum bronze: ASTM B148, ASTM B505 or ASTM B584, UNS Alloy C95200.
- I. Fasteners: Stainless steel, ASTM F593 or ASTM F594, type or grade as specified.

✓ **2.05 GENERAL PUMP CONSTRUCTION**

- A. The pump shall be a single-stage, end-suction submersible unit, capable of handling raw unscreened sewage and stormwater, with fibrous materials like wet wipes without clogging.

✓ **2.06 PUMP AND MOTOR CASING**

- A. Type: Watertight, air filled.
- B. Material: ASTM A48, minimum Class 35 B, cast iron.
- C. Watertightness: Able to run submerged up to 50 feet; and to run dry.
- D. Motor cooling: Adequate for continuously submerged or unsubmerged operation as scheduled.

- E. Design working pressure: Minimum 1.10 times maximum shutoff total dynamic head with maximum diameter impeller at maximum operating speed plus maximum suction static head or minimum 20 pounds per square inch gauge.
- F. O-ring seals: Capable of sealing mated surfaces (major components) watertight; with the following features:
 - 1. Machined surfaces and grooves.
 - 2. O-ring contact on 4 surfaces and O-ring compression on 2 surfaces.
 - 3. Does not require specific fastener torque or tension to obtain watertight joint.
 - 4. Does not require secondary sealing compounds, gasket, grease, or other devices.
- G. Testing: Perform 5-minute hydrostatic test of pump casing at minimum 1.5 times Design Working Pressure.

✓ 2.07 IMPELLERS

- A. General:
 - 1. Water passages: Smooth enough to prevent clogging by stringy or fibrous materials.
 - 2. Passage sizes: Large enough to pass solids encountered in raw unscreened sewage and stormwater.
 - 3. Casting: 1 piece, free of cracks and porosity.
 - 4. Method for securing impeller to shafts:
 - a. Conical sleeve and impeller screw.
- B. Rotation: As indicated on the Drawings; clockwise looking from top when not indicated.
- C. Type: As specified in the Pump Schedule:
 - 1. N-style impeller:
 - a. The impeller shall be wear resistant and made of high chromium cast iron with at least 24 percent chrome against sand and grit. Impellers that have surface hardening (by thermal, coating, etc.) will not be allowed.
 - b. Dynamically balanced, semi-open, multi-vane, backswept, non-clog design.
 - c. The impeller blades shall be self-cleaning upon each rotation as they pass across a sharp relief groove in the Insert ring and shall keep the impeller blades clear of debris.
 - d. The insert ring shall have a guide pin which moves fibers from the center of the impeller to the leading edges of the impeller.
 - e. Designed to move axially upwards to allow larger debris to pass through and immediately return to normal operating position.
 - f. The clearance between the insert ring and the impeller leading edges shall be adjustable.

✓ 2.08 WEAR RINGS

- A. Provide a wear plate system:
 - 1. Material: High chromium cast iron with at least 24 percent chrome.
 - 2. Attached to the pump volute to prevent rotation.

3. Replaceable and adjustable from impeller to maintain proper clearances between the wear plate and the impeller.

✓ 2.09 PUMP SHAFTS

- A. Material: Type 431 Stainless Steel; turned, ground, and polished.
- B. Features:
 1. Strength: Able to withstand minimum of 1.5 times maximum operating torque and other loads.
- C. Tapered to fit at impeller with conical sleeve and bolt for securing impeller.
- D. Pump and motor shaft shall be a solid continuous shaft.

✓ 2.10 BEARINGS

- A. Pump shaft shall rotate on a minimum of 2 permanently sealed, grease lubricated bearings:
 1. Upper bearing for radial forces shall be self-aligning spherical roller type.
 2. Lower bearing for combined axial and radial forces shall be angular contact ball type.
- B. Bearing Type: Anti-friction meeting ABMA standards.
- C. Bearing lubrication system shall be sized to safely absorb heat energy normally generated in bearing under maximum ambient temperature of 60 degrees Celsius.
- D. Bearing life: One of the following whichever provides longer bearing life in intended service:
 1. Minimum L-10 bearing life shall be 50,000 hours in accordance with ABMA 9 or ABMA 11.

✓ 2.11 SHAFT SEALS

- A. Upper seal unit material: 1 stationary tungsten-carbide ring and 1 positively driven rotating carbon ring.
- B. Lower seal unit material: 1 stationary and 1 positively driven rotating tungsten-carbide ring.
- C. Features:
 1. Tandem arrangement running in an oil chamber.
 2. Design oil chamber to ensure that air is left in the oil chamber to absorb the expansion of the oil due to temperature variations.
 3. Oil in oil chamber shall be FDA approved, paraffin type, colorless, odorless, and non-toxic.
 4. Independent spring system between seal interfaces able to withstand maximum suction submergences.
 5. Does not require pressure differential to effect sealing.
 6. Does not use pumped media for lubrication.

7. Lower mechanical seal effectively lubricated from oil chamber housings.
 8. Not damaged when pump is run dry (unsubmerged) for extended periods.
 9. Springs and other hardware: Stainless steel, 300 or 400 series.
- D. Moisture sensing system: Intrinsically safe type that signals seal leakage.
- E. Provide oil chamber with manufacturer's standard drain and inspection plug, with positive anti-leak seal, easily accessible from the outside.

✓ 2.12 DISCHARGE BASE AND ELBOW

- A. Materials: Cast iron ASTM A-48, Class 35B.
- B. Features:
1. Mounted on a sloped wall so solids can't settle below the discharge pipes.
 2. Structurally capable of firmly supporting guide rails, discharge piping and pumping unit under operating conditions.
 3. One or more integral support legs or pads with bolting to sump floor provisions.
 4. Incorporates 90 degree flanged elbow that receives horizontal flow from pump and discharges flow vertically.
 5. Discharge interface:
 - a. Sealing of the pumping unit to the discharge connection shall be accomplished by a machined metal to metal watertight contact.
 - b. Self-aligning without having to enter the wet well.
 - c. Discharge elbow to mate to pump discharge and transition to discharge piping.
 6. Piping: Provide flanged piping connections as scheduled in Section 15052 - Common Work Results for General Piping and as indicated on the Drawings in accordance with ASME B 16.1, minimum Class 125 or as required to meet specified design pressure, whichever is greater.

✓ 2.13 GUIDE RAILS, BRACKETS, FASTENERS, AND LIFTING CHAIN FOR EACH PUMP

- A. Material: Type 316 stainless steel, with the following features:
1. Dual pipes or dual rails that extend from discharge base to upper bracket unless scheduled otherwise.
 2. Rail wall thickness sufficient to suspend pump unit between brackets plus minimum 50 percent safety factor.
 3. Sized to fit discharge base and sliding bracket of pump.
 4. Integral, self-aligning, guide rail sliding brackets that seal pump to discharge base under operating conditions.
 5. Upper guide rail bracket.
 6. Intermediate guide rail brackets where indicated on the Drawings or at 10 foot maximum intervals.
 7. Lifting chain of sufficient strength and length to permit safe removal of pump unit from sump.

2.14 MOTOR AND POWER CABLES

- A. Horsepower:

1. As scheduled.
2. Listed driver horsepower is the minimum to be supplied:
 - a. Increase driver horsepower if required to prevent driver overload while operating at any point of the supplied pump operating head-flow curve including runout.
 - b. When scheduled driver is a motor, increase motor horsepower if required to prevent operation in the service factor.
 - c. Make structural, mechanical, and electrical changes required to accommodate increased horsepower.

B. Motors:

1. Revolutions per minute: As scheduled.
2. Enclosure: As scheduled.
3. Electrical characteristics: As scheduled.

3

C. Motor construction:

1. Squirrel cage induction motor, shell type design.
2. If explosion proof motor is scheduled, provide motor that is UL or FM listed for NEC Class I, Division 1, Groups C and D service, whether submerged or unsubmerged.
3. NEMA design type: B.
4. Motor insulation, either one:
 - a. Class F, moisture resistant, rated for 155 degrees Celsius.
 - b. Class H, moisture resistant, rated for 185 degrees Celsius.
5. The motor shall be designed for continuous duty handling pumped media of 40 degrees Celsius and capable of a minimum of 30 evenly spaced starts per hour.
6. The motor shall be capable of continuous operation under load with the motor submerged, partially submerged, or exposed, without derating the motor.
7. Motor cooling, one of the following:
 - a. Motor cooled by the surrounding media.
 - b. Cooling jacket:
 - 1) Non-clogging ports and channels that use pumped fluid as the cooling media.
 - 2) Functional with motor submerged, partially submerged, or exposed.
 - 3) Provide capability to relieve entrapped air from the system.
 - c. Spray systems, air moving equipment or other secondary cooling systems are not acceptable.
8. Motor sealing: Design motor case and seals to withstand 65 feet of submergence.
9. As scheduled and as specified in sections listed in the Pump Schedule.

4

D. Power and control cables:

1. Submersible to same water depth as motor casing.
2. Type SPC with Hypalon/Buna N or chloroprene rubber jacket.
3. Insulation rated for 90 degrees Celsius.
4. Non-wicking fillers.
5. Provide Meltric plug for connection to Meltric receptacle as indicated on the Drawings.
6. Length: Sufficient to connect to Meltric receptacle as indicated on the Drawings or 30 feet, whichever is greater.

5

7. All power and control conductors shall terminate at terminal blocks in the local control panel or junction box.

6

8. Sized in accordance with NEC, ICEA, and CSA specifications.
9. Provide stainless steel cable and stainless steel wire braid sleeve to support power cable from underside of wet well roof slab or access frame

E. Cable entry/junction chamber:

1. Cable entry seal design shall not require specific torque requirements to insure a watertight and submersible seal.
2. Cable entry seal shall consist of a single cylindrical elastomer grommet, flanked by stainless steel washers.
3. The entry body shall perform compression and strain relief that is separate from the sealing function.
4. The cable entry junction chamber shall be separate from the motor chamber to prevent foreign material from gaining access to the motor interior through the top of the pump.

F. Control/protection module:

1. Each pump shall be supplied with its own self-contained control/protection module to provide for the direct connection to internal pump monitoring devices, including:
 - a. Thermal protection: Provide automatic reset motor stator temperature detectors, 1 switch in each phase winding. If any detector is activated, the sensor shall activate an alarm and shut down the motor. The thermal detectors shall activate when the stator temperature exceeds 125 degrees Celsius.
 - b. Moisture detection: one of the following:
 - 1) Provide a capacitive type leakage sensor to detect water in the oil chamber or a moisture sensor in the lower motor chamber to detect any fluid in the motor. The sensor shall have an explosion-proof rating.
 - 7 2) Provide a small float to detect the presence of water in the stator chamber.
2. The module shall signal an alarm condition if any of the internal monitoring devices is activated.
3. Install module in manufacturer-supplied VCPs (if scheduled in the Pump Schedule) and provide necessary modules to the Contractor for installation in control panels not provided by the manufacturer.

2.15 PACKAGE CONTROL SYSTEM

This will be submitted
in a separate
submittal

A. General:

1. This Section summarizes key control system components for vendor control panels (VCP):
 - a. Referenced sections can be obtained from the Contractor for complete control system requirements.
2. Provide instruments and other components performing similar functions of the same type, model, or class, and from one manufacturer:
 - a. Coordinate with Contractor to align make and model with Contractor-furnished equipment.

3. Provide a separate control panel for each vendor packaged system such that control panels will have fully independent control of the functions and operations for their respective system.
 4. VCP shall be completely preprogrammed, prewired, and factory tested for proper operation prior to shipment:
 - a. Additional testing requirements as specified in Section 17950 - Commissioning for Instrumentation and Controls.
 5. Provide necessary control hardware, software, and components as required for a fully functional and operational installation.
 6. Design and size panels:
 - a. Design panels to fit in the space as indicated on the Drawings.
- B. Panels, Enclosures, and Panel Components:
1. Manufacturers: One of the following or equal:
 - a. nVent/Hoffman.
 - b. Saginaw Control & Engineering.
 2. Provide components and equipment with UL 508 listing.
 3. Provide control panels with UL 508A labeled.
 4. Provide fuses for equipment that is not UL listed or recognized.
 5. Provide intrinsically safe circuits and equipment in accordance with UL 698A.
 6. Provide control panel power circuits in accordance with the panel power wiring diagram included with the contract documents:
 - a. Design devices and instruments to fail in a safe condition.
 7. Accept single 480 VAC, 3-phase power source:
 - a. Equipped with a UL 1449 rated SPD.
 - b. Terminate incoming power feed on main circuit breaker as specified in Section 16412 - Low Voltage Molded Case Circuit Breakers:
 - 1) Rated for the available fault current as calculated as specified in Section 16305 - Electrical System Studies.
 - c. Control power transformer:
 - 1) Provide a single power source.
 - 2) Control power transformers as required:
 - a) Primary voltage: 460 VAC, 3 phase, 60 hertz.
 - b) Secondary voltages:
 - (1) 120 VAC for controls.
 - (2) Derive additional voltages as required by the application.
 - (3) Sized for all panel components plus 10 percent spare capacity.
 - (4) Primary and secondary fuses.
 - d. FVNR motors starters as specified in Section 16422 - Motor Starters.
 - e. Protective barriers and touch-safe components for 480 VAC equipment.
 - f. Derive additional voltages as required by the application.
 - g. Conform panel enclosure rating to the area classifications specified in Section 16050 - Common Work Results for Electrical.
 8. Conform panel enclosure rating to the area classifications as specified in Section 16050 - Common Work Results for Electrical.

9. Control panel shall be rated as follows:
 - a. Indoor panels:
 - 1) Electrical/control rooms: NEMA 12, powder-coated mild steel.
 - 2) Process areas: NEMA 4X:
 - a) Non-Corrosive Areas: Type 316 stainless steel.
 - b) Corrosive Areas: Polycarbonate or fiberglass reinforced polyester (FRP).
 - b. Outdoor panels: NEMA 4X, Type 316 stainless steel.
10. Perform and submit thermal management calculations for each control panel.
 - a. Provide heating, cooling, and dehumidifying devices to maintain instrumentation and control devices within their rated operational temperature range.
11. Pad lockable flange mounted disconnect switch to disconnect power from the panel:
 - a. Interlock with the panel door.
12. Provide nonmetallic ducts for routing and organization of conductors and cables.
13. Provide NEC Class 2 power to devices requiring NEC Class 2 or SELV supply power.
14. Provide UL 1449 rated SPD and UL 1283 rated EMI filter on 120 VAC panel power source.
15. Provide GFCI duplex 120 VAC convenience receptacle.
16. Factory-assemble and wire the control panel so field wiring only requires connections to terminals.
17. Additional requirements as specified in Section 17710 - Control Systems - Panels, Enclosures, and Panel Components.

2.16 CONTROL REQUIREMENTS

- A. General:
 1. This section describes components, features and control functionality of the vendor control panel (VCP).
- B. Pilot Devices:
 1. Controls:
 - a. HAND-OFF-AUTO selector switch.
 - b. RESET pushbutton.
 - c. OVERRIDE START pushbutton – Active in AUTO mode only (Allows bypassing the preset Start Level setpoint.).
 2. Lights:
 - a. Status:
 - 1) Running Indication.
 - 2) Stopped Indication.
 - 3) Fail light.
 - 4) High motor temperature.
 - 5) Moisture detection.
 - b. Alarms:
 - 1) Fault.
 - 2) High level.
 - 3) Low-Low level.
 - 4) General alarm beacon.

3. Horns:
 - a. None.
- C. Monitoring and control:
 1. Dry contact rating: 120 VAC at 10 A.
 2. VCP Outputs:
 - a. Status:
 - 1) Running Indication.
 - 2) In Auto.
 - 3) Analog level signal from level sensors: Indicated in the Pump Schedule.
 - b. Alarms:
 - 1) Fault.
 - 2) High level.
 - 3) LowLow level.
- D. Hardwired control functions:
 1. Include a Pump Protection Module as specified in this Section.
 2. Operation:
 - a. With the HAND/OFF/AUTO selector switch in the HAND position, the pump will run.
 - b. With the HAND/OFF/AUTO selector switch in the OFF position, the pump is prevented from running.
 - c. With the HAND/OFF/AUTO selector switch in the AUTO position, the pump will run based on sump level as specified below:
 - 1) Simplex AUTO operation:
 - a) At low level (LSL), the pump is off.
 - b) At pump start level (LSA), the pump starts:
 - (1) The pump continues to run until the low level (LSL) is reached.

When the OVERRIDE START pushbutton is activated the preset Start Level setpoint (LSA) will be bypassed and the pump will start if the water level in the sump is above the Low Level setpoint (LSL).
 - 2) Duplex AUTO operation:
 - a) At low level (LSL), both pumps are off.
 - b) At lead pump start level (LSA), the lead pump starts.
 - c) At lag pump start level (LSB), the lag pump starts.
 - d) Both pumps continue to run until the low level (LSL) is reached.
 3. Interlocks:
 - a. A hardwired low-low level interlock will prevent the pump(s) from operating at or below this level:
 - 1) In both the HAND and AUTO modes.
 - 2) Activated by the low-low level setpoint (LALL).
 - b. A motor high-temperature interlock will prevent the pump from running in all modes:
 - 1) Alarm is latched until the RESET pushbutton is pressed.
 - c. For Area 23 North Drainage Pump Station only: A hardwired Headworks Drain Sump high-high level interlock will prevent the pump from running in all modes.^{AD4}
 4. High level alarm is generated by the high-level setpoint (LAH):

- a. Alarm is latched until the RESET pushbutton is pressed.
 - b. High level alarm turns on the alarm beacon:
 - 1) The beacon shall remain lit until the high-level alarm is reset.
- 5. Low-low alarm is generated by the low-low level setpoint (LALL).
 - a. Indicates sump level lower than safe level for pump operation.

✓ 2.17 LEVEL SENSORS

- A. Submersible pressure transducer as specified in Section 17407 - Pressure Measurement: Submersible.
- B. Level sensors used in hazardous areas, either Class I Division 1 Group D or Class I Division 2 Group D, shall have Intrinsic Safety Barriers meeting NEC and specification requirements.
- C. Furnish Type 316 stainless steel cable support.

✓ 2.18 ACCESSORIES

- A. Lifting device:
 - 1. Type: Compatible lifting system complete with stainless steel guide rope and chain with grip eye.
 - 2. Lifting device shall be sized for combined weight of pump and motor.
 - 3. All other accessories as required to lift the pump/motor to grade by a crane without disconnecting the attached power and control cables from motor.

2.19 FINISHES

- A. Pump manufacturer to provide factory-applied coating system as follows:
 - 1. Provide Tankguard 412 2-component solvent free polyamine cured epoxy coating system on pump and motor housing.
 - 2. Coating shall be applied on 2 layers for a total dry film thickness of 14 to 16 mils.
 - 3. Pump manufacture or coating system applicator must be an approved applicator by the coating system manufacturer and shall follow all of the surface preparation and application requirements.
 - 4. Pump manufacturer shall provide an adequate amount of additional Tankguard 412 coating for field touchups and repairing any damages to factory applied coating.
- B. Pump manufacturer to provide factory prime for discharge elbow and field coating as specified in Section 09960 - High-Performance Coatings.

2.20 SPARE PARTS

- A. Spare parts: Deliver the following as specified in Section 01600 - Product Requirements:
 - 1. Thrust bearing set.
 - 2. Radial bearing set.
 - 3. Mechanical seal: 1 each size and type.
 - 4. O-ring set.

- 5. Power cable entry seal set.
- 6. Wear plate (stationary).
- ✓ 7. One extra pump identical to the septage pumps.

PART 3 EXECUTION

3.01 COMMISSIONING

- 8 A. As specified in Section 01756 - Commissioning and this Section.
- B. Manufacturer services:
 - 1. Provide certificates:
 - a. Manufacturer's Certificate of Source Testing.
 - b. Manufacturer's Certificate of Installation and Functionality Compliance.
 - 2. Manufacturer's Representative on-site requirements: Require manufacturer's representative to perform the following services as described below and as specified in Section 01756 - Commissioning. The specified durations are the minimum required time on the job site. Additional services and/or longer durations shall be provided as needed at no cost to the Owner to meet the required quality of work. Work to be done in a minimum of 4 trips:
 - a. Installation assistance: As required:
 - 1) Advise/observe the Contractor on the installation of the equipment.
 - 2) Provide additional assistance as required.
 - b. Installation Inspection: 1 trip, 3 days minimum.
 - c. Functional Testing: 1 trip, 3 days minimum.
 - d. Post start-up field visit: 1 trip; 1 workday.
 - 3. Training:
 - a. Operations: 2 hours per session, 2 sessions.
 - b. Mechanical Maintenance: 2 hours per session, 1 session.
 - c. Combined Electrical and I&C Maintenance: 3 hours per session, 1 session.
 - 4. Final acceptance checkout: 1 workday (trip may be combined with training).
- C. Source testing: As specified in Pump Schedule and in Section 15958 - Mechanical Equipment Testing.
- D. Functional testing: As specified in Pump Schedule and in Section 15958 - Mechanical Equipment Testing.
- E. Refer to Section 16950 - Field Electrical Acceptance Tests and Section 17950 - Commissioning for Instrumentation and Controls for additional requirements.

3.02 PUMP SCHEDULE

Tag Numbers	01-PMP-1010/01-VCP-1010 01-PMP-1020/01-VCP-1020	02-PMP-1712 02-PMP-1722 02-VCP-1710	23-PMP-2812 23-PMP-2813
General Characteristics			
Service	Septage	Headworks Drainage	Sanitary Sewage / Grit Dumpster
Quantity	2 installed + 1 uninstalled spare	2	2
Level Device	01-LT-1010 01-LT-1020	02-LT-1710	23-LE-2810
Maximum Noise, dBA at 3 Feet	Not Required	Not Required	Not Required
Minimum Pumped Fluid, Degrees Fahrenheit	60	60	60
Normal Pumped Fluid, Degrees Fahrenheit	70	70	70
Maximum Pumped Fluid, Degrees Fahrenheit	85	85	85
Installation Configuration	Wet Pit	Wet Pit	Wet Pit
Torsional Analysis	Not Required		Not Required
Pump Characteristics			
Impeller Type	N-Non Clog	N-Non Clog	N-Non Clog
Speed Control	Constant	Constant	Constant
Maximum Pump Speed, rpm	1750	1750	1750
Minimum Pump Speed, rpm	N/A	N/A	N/A
Rated Design Point: (at Maximum Revolutions per Minute):			
Flow, gpm	200	200	200
Head, feet	25.5	25.5	30.0
Minimum Hydraulic Efficiency, Percent	52	52	56
Required Condition 2 (at Maximum Revolutions per Minute):			
Flow, gpm	130	130	100
Head Range, feet	27-29	27-29	38-46

Tag Numbers	01-PMP-1010/01-VCP-1010 01-PMP-1020/01-VCP-1020	02-PMP-1712 02-PMP-1722 02-VCP-1710	23-PMP-2812 23-PMP-2813
Minimum Hydraulic Efficiency, percent	40	40	44
Required Condition 3 (at Maximum Revolutions per Minute):			
Flow, gpm	300	300	240
Head Range, feet	21-23	21-23	23-27
Minimum Hydraulic Efficiency, percent	61	61	55
Required Condition 4 (at Maximum Revolutions per Minute):			
Flow, gpm	400	400	N/A
Head Range, feet	15-19	15-19	N/A
Minimum Hydraulic Efficiency, percent	61	61	N/A
Other Conditions			
Maximum Shutoff Head, feet	36	36	56
Maximum NPSHr at Every Specified Flow, feet	15	15	14
Minimum Discharge Size, inches	3	3	3
Motor Characteristics			
Minimum Submergence	Exposed	Exposed	Exposed
Maximum Driver Speed, rpm	1800	1800	1800
Motor Horsepower, Minimum	3	3	7.5
Voltage/Phases/Hertz	460/3/60	460/3/60	460/3/60
Service Factor	1.15	1.15	1.15
Starting Current, Amperes	22	22	58
Rated Current, Amperes	4.3	4.3	9.6
Motor Efficiency (At 100 Percent Load), Minimum	78.1	78.1	83.8
Enclosure Type	Submersible, Explosion-Proof	Submersible, Explosion-Proof	Submersible, Explosion-Proof

Tag Numbers	01-PMP-1010/01-VCP-1010 01-PMP-1020/01-VCP-1020	02-PMP-1712 02-PMP-1722 02-VCP-1710	<u>23-PMP-2812</u> <u>23-PMP-2813</u>
NEMA Design Type	B	B	<u>B</u>
Vendor Control Panel			
Required	Yes, QTY 2	Yes, QTY 1	<u>Yes, QTY 1</u>
Enclosure Type	NEMA 4X Stainless Steel	NEMA 4X Stainless Steel	<u>NEMA 4X</u> <u>Stainless Steel</u>
Level Control Type	submersible pressure sensor	submersible pressure sensor	<u>Floats</u>
Operation	Simplex	Duplex	<u>Simplex</u>
Accessories			
Basin Cover	Required	Required	<u>Required</u> <u>(Pedestrian</u> <u>Rated, 4"</u> <u>discharge pipe</u> <u>penetrations as</u> <u>shown on the</u> <u>Drawings).</u>
Lifting Device	Required	Required	<u>Required</u>
Source Testing			
Test Witnessing	Not Required	Not Required	<u>Not Required</u>
Performance Test Level	1	1	<u>1</u>
Vibration Test Level	Not Required	Not Required	<u>Not Required</u>
Noise Test Level	Not Required	Not Required	<u>Not Required</u>
Functional Testing			
Performance Test Level	1	1	<u>1</u>
Vibration Test Level	Not Required	Not Required	<u>Not Required</u>
Noise Test Level	Not Required	Not Required	<u>Not Required</u> ^{AD4}

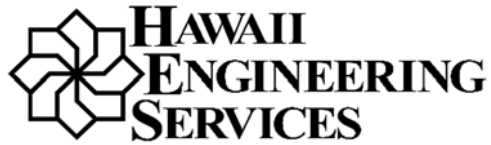
3.03 SETPOINTS TABLE

No.	Setpoint Description	Range	Unit	Initial Setting	Operator Adjustable or Hard Coded	Comments
Septage Pumps						
LALL	Sump Low Level	0-14	feet	1	HC	
LAL	Sump Low Level	0-14	feet	2	OA	
LSA	Sump Start Level	0-14	feet	5	OA	
LAH	Sump High Level	0-14	feet	11	OA	

No.	Setpoint Description	Range	Unit	Initial Setting	Operator Adjustable or Hard Coded	Comments
Headworks Drain Pumps						
LALL	Sump Low Level	0-14	feet	1	HC	
LAL	Sump Low Level	0-14	feet	2	OA	
LSA	Sump Start Level - Lead	0-14	feet	4	OA	
LSB	Sump Start Level - Lag	0-14	feet	5	OA	
LAH	Sump High Level	0-14	feet	5.5	OA	
North Drainage Pumps						
<u>LALL</u>	<u>Sump Low Level</u>	<u>0-14</u>	<u>feet</u>	<u>2</u>	<u>HC</u>	
<u>LAL</u>	<u>Sump Low Level</u>	<u>0-14</u>	<u>feet</u>	<u>4</u>	<u>OA</u>	
<u>LSA</u>	<u>Sump Start Level - Lead</u>	<u>0-14</u>	<u>feet</u>	<u>8</u>	<u>OA</u>	
<u>LAH</u>	<u>Sump High Level</u>	<u>0-14</u>	<u>feet</u>	<u>10</u>	<u>OA</u> ^{AD4}	
Note: All set point levels are measured from bottom of sump.						

END OF SECTION

^{AD4} Addendum No. 4 - June 2024



SUBMITTAL

PROJECT INFORMATION	
Submittal No.: Septage and Headworks TOP Station Job Name: Hilo WWTP Rehabilitation and Replacement Project PH1 Location: Hilo WWTP, 150 Kekuanaoa Place Quote No: RS10331 Purchase Order No: 24077-00026	
CONTRACTOR	
Contractor: Nan Inc. Project Manager: Phone:	Address: 636 Laumaka St Honolulu, HI 96819
MANUFACTURER'S REP	
Supplier: Hawaii Engineering Services Project Manager: Rick Pitts Phone: (907) 232-2505	Address: 1082 Makepono Street Honolulu, HI 96819
Transmittal	
☒ FOR REVIEW AND APPROVAL ☐ APPROVED ☐ REJECTED ☐ REVISE AND RESUBMIT ☐ ACKNOWLEDGE RECEIPT ☐ SEE REVIEW COMMENTS/RESPONSE BY: David Tam DATE: August 26, 2025 ***NOTE*** On approval orders only, written submittal approval must be received before production can be authorized.	



1082 Makepono St. Honolulu, HI 96819 Ph:(808)841-0033 Fax:(808)841-2534 info@hiengineering.com

Hilo WWTP Rehabilitation and Replacement Project – Phase 1

Deviations/Exceptions to Specifications

Item	Section	Comments
1	11320 2.03.B & 2.03.C	Specs call for cover 6063-T5, but this alloy only applies to the extrusions; standard cover material will be alloy 5052.
2	11320 2.04	Pump housing is Cast Iron ASTM A-48, Class 35B
3	11320 2.14.C.3	Pump is permanently submersible according to standard IEC 60034 and protection class IP 68
4	11320 2.14.D.2	SUBCAP Pump Cable outer jacket is oil resistant chlorinated polyethylene rubber.
5	11320 2.14.D.7	By Others
6	11320 2.14.D.9	Supplied by Others
7	11320 2.14.F.1.b.2	Float switch in stator chamber is not available
8	11320 3.01	By Contractor
	15050 1.06.C.2	Not applicable
	15050 1.06.D.2-7	Not applicable
	15050 1.06.E.3	Not applicable, project is for wastewater.
	15050 1.06.F.2	Not applicable, bearing are proprietary to pump manufacture
	15050 1.06.H	By Contractor
	15050 2.01.A.4	Not applicable. There are submersible pumps.
	15050 2.01.A.5	Not applicable
	15050 2.01.A.6	Not applicable
	15050 2.01.B	Not applicable/required
	15050 2.01.C.2-5	Not applicable
	15050 2.02	Not applicable
	15050 2.03.B	Not applicable
	15050 2.03.C	Not applicable
	15050 2.04	Not applicable
	15050 2.07	Not applicable
	15050 2.08	Not applicable
	15050 2.10	Not applicable
	15050 2.11	Not applicable
	15050 2.12	By Contractor
	15050 2.13	See Pump O&M Manual
	15050 2.14	Not applicable
	15050 2.15	Not applicable
	15050 2.16	By Contractor
	15050 2.17	Not applicable



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	15050 2.20	By Others
	15050 2.21	By Others
	15050 2.24	Not applicable. Parts are manufactured as one unit. Separating parts will void warranty.
	15050 Part 3	By Contractor
	15958 1.03.A	By Contractor
	15958 1.03.B	By Contractor
	15958 1.03.C	By Contractor
	15958 1.03.E	Not applicable
	15958 1.03.F	By Contractor
	15958 3.01	By Contractor
	15958 3.02	Not applicable
	15958 3.03.B	Closest HI Level to specification is Grade 2B Tolerance due to small pump's relative performance band
	15958 3.04	Not applicable; motor is integrated to pump
	15958 3.05	Not applicable; submersible pump
	15958 3.06	Factory pump hydrostatic test shall be supplied
	15958 3.07	Not applicable; motor is integrated to pump
	15958 3.08	Not applicable
	15958 3.09	By Contractor
	15958 3.10	By Contractor
	15958 3.11	By Contractor
	01600 2.02	Not applicable
9	01600 2.04.B	Special tools not available for supply. Factory forbids use of special tools by non-factory trained operator/technician. Warranty will be void.
9	01600 2.04.D	By Contractor
9	01600 3.02.E	Special tools not available for supply. Factory forbids use of special tools by non-factory trained operator/technician. Warranty will be void.
		Storage by Contractor
	01600 3.03	Storage by Contractor
	01600 3.04	Installation by Contractor
	01600 3.05	By Contractor
	01783 1.05 D	Spare pump is included; pump repair facility located in Honolulu.
	01783 1.07	By Contractor
	01783 1.08	By Contractor
	01783 1.09	By Contractor
	17950	As applies to VCP based on 11320 2.15.B



TOP Pre-engineered Fiberglass Pump Station

THE OPTIMUM PUMP STATION

FLYGT
a xylem brand

Flygt pump station controls



Xylem offers a fully engineered control panel solution. Our integrated, purposely designed control panels provide an intuitive user interface with the reliability you have come to expect from the leader in submersible pumping.

Standard Control Features

- UL 508 listed
- NEMA 4X 304 Stainless Steel enclosure with aluminum dead front inner door
- Lockable enclosure
- Hand/Off/Auto Selector switches
- Full voltage across-the-line starting
- Main incoming power circuit breaker
- Individual pump circuit breakers
- NEMA rated motor starters w/overloads
- Mini-CAS II pump seal & motor thermal protection
- APP 521 duplex pump controller
- Current transformers
- 24VDC power supply
- ENM-10 float regulators

Available Options

- Generator receptacle and plug assembly with manual transfer switch
- Solid state reduced voltage starting
- LS-100 submersible pressure transducer
- MIO module and multi-sensor level probe
- Horn or bell audible alarm
- Anti-condensation heater and thermostat
- Back up floats (2 x ENM-10, when transducer or probe option is selected)
- Elapsed time meters for pumps
- TD-33 Telephone modem
- 12" x 10" space in panel reserved for future telemetry

TOP Station

Premium pre-engineered pump station

The Flygt TOP fiberglass pump station from Xylem is a premium, pre-engineered and factory built packaged pump station that utilizes advanced features to provide customers with superior pump station performance.

The innovative, self-cleaning, TOP Station sump bottom directs the solids and debris normally found in waste-water to the inlet of the Flygt N-Pumps where they can be effectively pumped away.

The interior of the pump station has a smooth finish which helps inhibit the build-up of grease and sludge.

The outside diameter of the station is equipped with an integral anti-flotation ring utilized to secure the station in place.

The aluminum pump station lid utilizes an integral Safe-Hatch access cover that provides personnel fall-through protection when the aluminum access door is opened. The raised frame provides a kick plate surround eliminating the possibility of tools or debris rolling into the pump station.

During normal inspection, individual pumps can be raised and placed upon one of the closed Safe-Hatch grates and washed-down. The debris will fall back down into the sump resulting in a clean pump to check.



Fully sealed station wall penetrations can be factory installed for the influent pipe, discharge pipes, and electrical connection points. Depending on pipe diameter, properly selected fiberglass hubs with link seals are utilized. Influent pipe wall penetration can be shop installed or field located.

Features & benefits

- Pre-engineered, factory built pump station
 - Available in 4-ft, 5-ft or 6-ft diameters
- Heavy-wall filament-wound fiberglass tank
- Exclusive self-cleaning TOP sump bottom
- Flygt heavy-duty submersible N-Pumps
 - Clog-free, innovative technology
 - 3-hp through 35-hp motors
 - Self-cleaning N-Impeller
 - Sustains high hydraulic efficiency
- Flygt mix-flush valve
 - Provides sump mixing
 - Re-suspends solids
- 2", 3", 4" or 6" diameter discharge pipe
 - PVC discharge pipe
 - Stainless steel discharge pipe
- Stainless steel guide bars
- Stainless steel upper guide bar bracket
- Stainless steel cable holder hooks
- Integral Safe-Hatch aluminum access cover
- Flygt Grip-eye easy lift pump retrieval system
- 4-in diameter PVC station vent pipe
- Pump station level control choices
 - Flygt ENM-10 ball float-type
 - Flygt LS-100 pressure transducer-type
 - Flygt probe-type
- Duplex Flygt pump station controls
 - NEMA-4 enclosure
 - Several enclosure material choices
 - UL listed control available
 - NEMA or IEC rated starters available
 - Standard and custom controls
- Single lift, easy station installation
- Single-source responsibility



Flygt TOP pre-engineered pump station

Grip-eye lifting device

Safe-Hatch access cover

Flygt N-Pump

Flygt Mix-flush valve

4" station vent pipe

Stainless steel guide bars

Heavy-wall construction

Level control system

TOP self-cleaning basin



Xylem |'zīləm|

- 1) The tissue in plants that brings water upward from the roots;
- 2) a leading global water technology company.

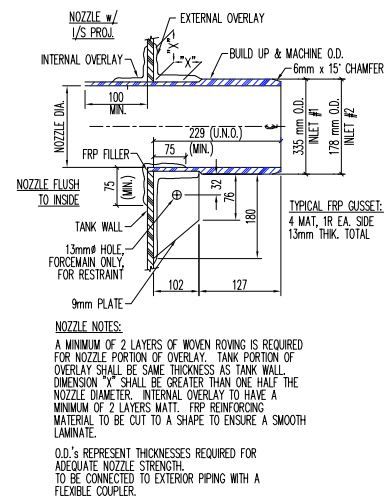
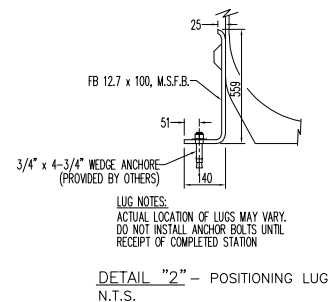
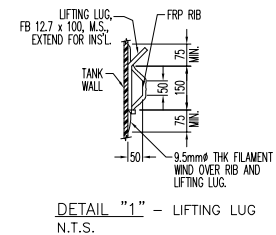
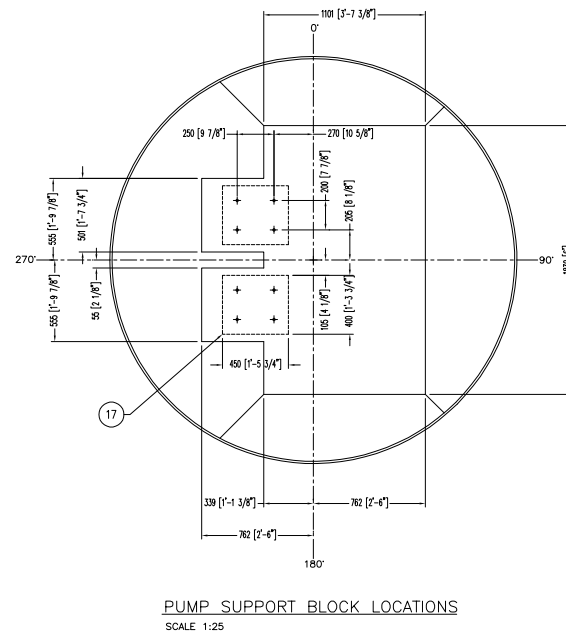
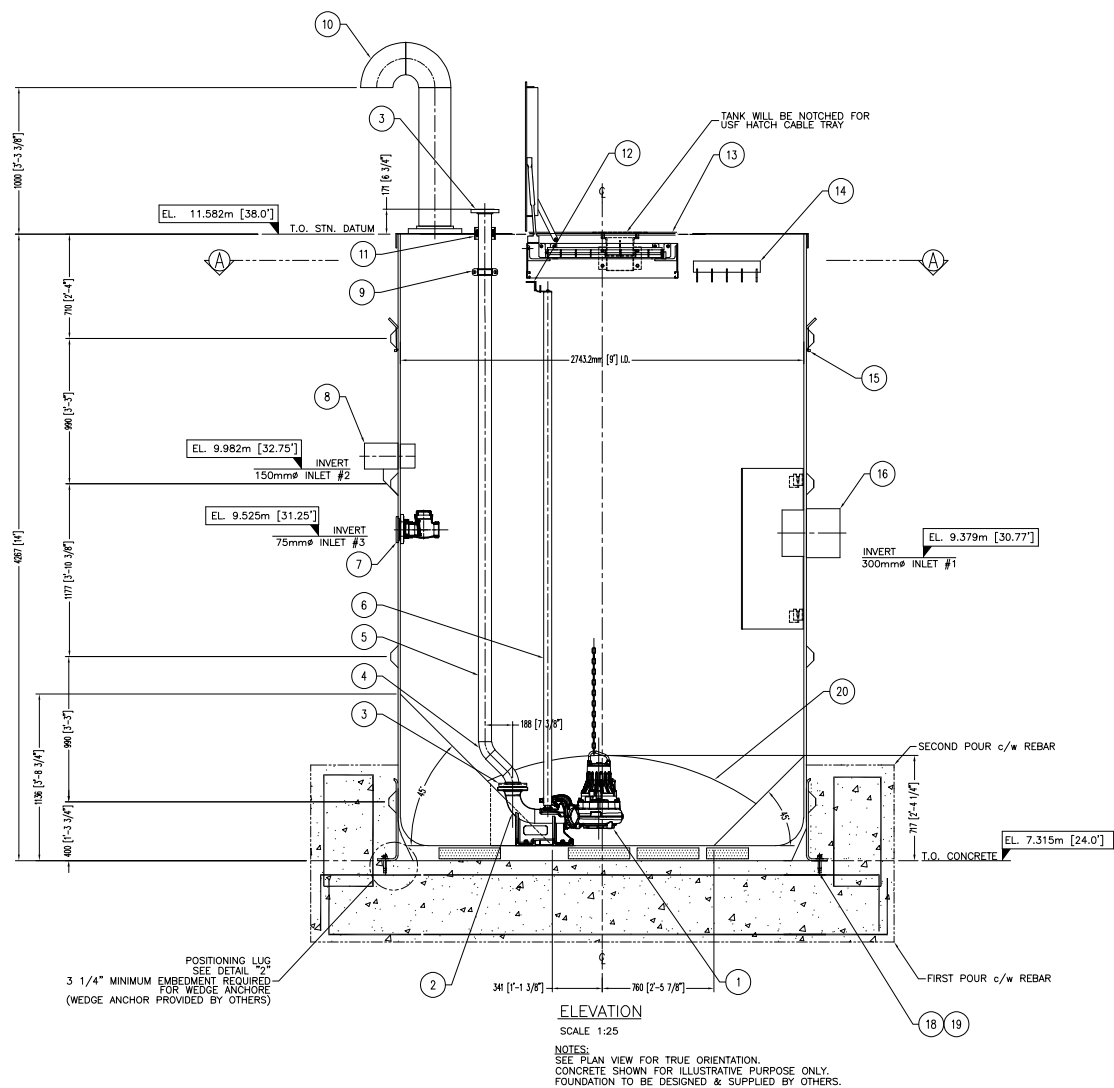
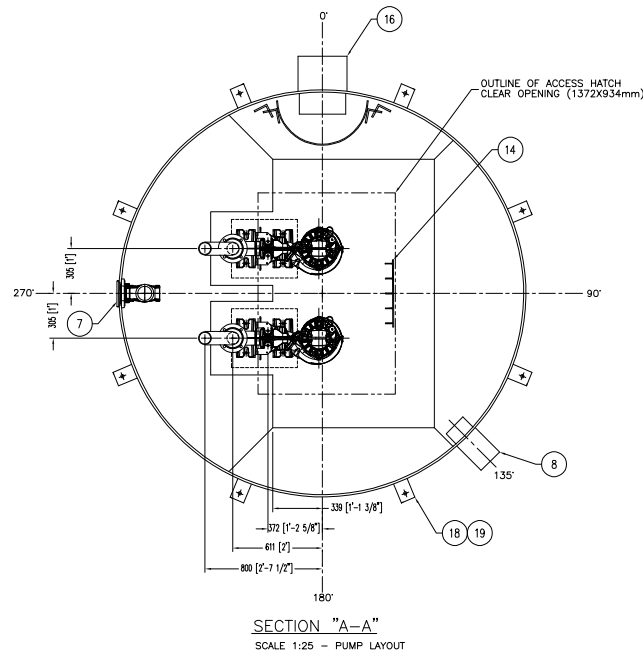
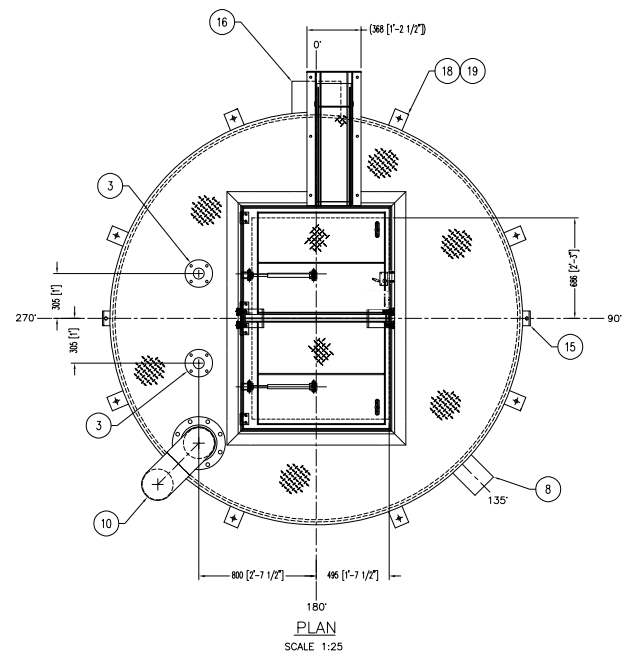
We're 12,000 people unified in a common purpose: creating innovative solutions to meet our world's water needs. Developing new technologies that will improve the way water is used, conserved, and re-used in the future is central to our work. We move, treat, analyze, and return water to the environment, and we help people use water efficiently, in their homes, buildings, factories and farms. In more than 150 countries, we have strong, long-standing relationships with customers who know us for our powerful combination of leading product brands and applications expertise, backed by a legacy of innovation.

For more information on how Xylem can help you, go to www.xyleminc.com



Xylem, Inc.
14125 South Bridge Circle
Charlotte, NC 28273
Tel 704.409.9700
Fax 704.295.9080
www.xyleminc.com

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Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 8/28/2025
SPECIFICATION SECTION # 11320
SPECIFICATION Packaged Lift Sewer Station
PARAGRAPH 1.04 - C
DRAWING Shop Drawings
SUBCONTRACTOR N/A
SUPPLIER N/A
MANUFACTURER FLYGT

CERTIFIED BY: M.Chun, QC Engineer

BILL OF MATERIALS			
QTY	DESCRIPTION		
1	FLYGT PUMPS, NP-3085 MT, DN80 [3"]	(BY XYLEM)	2
2	STANDARD DN80 [3"] DISCHARGE ELBOW, P/N 444 68 05		2
3	75mm-150# [3"] RF FLANGE, 316 SS		4
4	75mm# [3"] 45° SR ELBOW, SCH.10, 316 SS		4
5	75mm# [3"] RISER PIPE, SCH.10, 316 SS		2
6	50mm# [2"] GUIDE RAILS, SCH.40, 316 SS		
7	75mm# [3"] INLET #3, NPT BULKHEAD FITTING, FRP c/w 75mm# [3"] SWING CHECK VALVE, SS		1
8	150mm# [6"] INLET #2, MACHINED, FRP (SEE DETAIL "3")		1
9	75mm# [3"] VICTAULIC COUPLING 489SS		2
10	200mm# [8"] GOSSENKNECK VENT, SCH.10 316 SS c/w 180° RETURN BEND & SST BROSSCREEN		1
11	125# I.D. [5"] I.D.] FRP SLEEVE c/w WRAP-IT LINK WL-300, 8 LINK		2
12	50mm# [2"] UPPER GUIDE BAR BRACKET		2
13	U.S.F. HATCH ALUMINUM, 41-1/2" X 61-1/2" OPENING (DWG #1000288089)		1
14	316SS CABLE HANGER		1
15	LIFTING LUGS, EPOXY COATED, U.S. (SEE DETAIL "1")		2
16	300mm# [12"] INLET #1, MACHINED, FRP (SEE DETAIL "3") c/w REMOVABLE INLET BAFFLE		1
17	450x400mm FRP POLYMER BLOCK WITH STUDS (PUMP SUPPORTS)		2
18	POSITIONING LUGS, U.S. EPOXY COATED (SEE DETAIL "2")		8
19	3/4"x4-3/4" WEDGE ANCHOR BOLTS, GALV. CS	(BY OTHERS)	8
20	CUSTOM FRP HYDRAULIC BENCHING		-

FABRICATION DESIGN STANDARDS

1. FLYGT SPECIFICATION GE-1008-04, REVISION MAY 2002
2. AMEC 4S-10.01 MANUFACTURE AND INSTALLATION FOR FRP STRUCTURES
3. AMEC 4S-10.02 FRP PRESSURE PIPE, FITTINGS AND FLANGES

GENERAL NOTES

1. WINDING ANGLE - 75°
2. TANK WALL - VARIES WITH ELEVATION.
3. LINER - C-GLASS VEIL AND (2)-1 1/2 oz. MATT
4. RESIN - ISOPHTHALIC
5. INTERIOR FINISH - WHITE ISOPHTHALIC NPG GELCOAT
6. EXTERIOR (ABOVE GRADE) TO HAVE DARK GREEN URETHANE PAINTED FINISH
7. DIMENSIONS ARE IN MILLIMETERS U.N.O.
8. APPROX. SHIPPING WEIGHT: 3,175 kg (7,000 lbs)

INSTALLATION RECOMMENDATIONS

THE FOLLOWING RECOMMENDATIONS ARE BASED ON FLYGT EXPERIENCE.

1. USE THE LIFTING LUGS PROVIDED FOR VERTICAL HANDLING.
2. USE SLINGS AROUND THE MAIN TANK FOR HORIZONTAL HANDLING.
3. ENSURE UNIT IS STANDING VERTICAL ON CONCRETE PAD.
4. BOLT UNIT FIRMLY AND SQUARELY IN PLACE, SHIM WHERE NECESSARY.
5. ENCASE BOTTOM RIB IN CONCRETE TO A MIN. HEIGHT OF 150mm ABOVE RIB TO PROVIDE ANCHORAGE. REBAR TO CONNECT SECOND POUR TO THE CONCRETE BASE PAD.
6. MAINTAIN A DRY SITE UNTIL BACKFILLING OPERATIONS COMMENCE.
7. PLEASE REFER TO BARSKI BACKFILL PROCEDURE FOR FRP VERTICAL TANKS DOCUMENT
8. PLACE THE BACKFILL IN EQUAL INCREMENTS NOT EXCEEDING 300mm THICK AROUND THE STATION TO PREVENT UNBALANCED LOADS BEING IMPOSED DURING BACKFILLING OPERATIONS. PROGRESSIVELY TAMP BACKFILL AROUND STATION TO FULL HEIGHT TO REDUCE SETTLEMENT TO AN ABSOLUTE MINIMUM.
9. PLEASE CHECK WITH THE MANUFACTURER REGARDING THE ACTUAL DIMENSIONS OF THE LIFT STATION BEFORE DRAFTING SHOP DRAWINGS OF THE CONCRETE FOUNDATION.

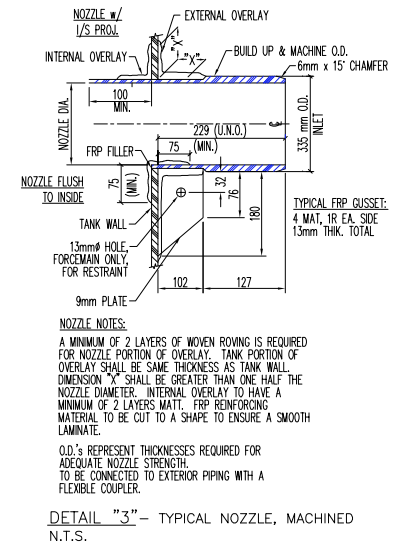
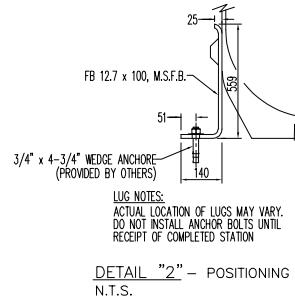
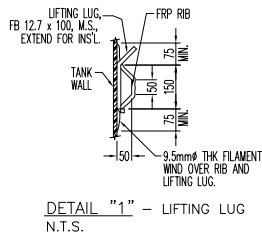
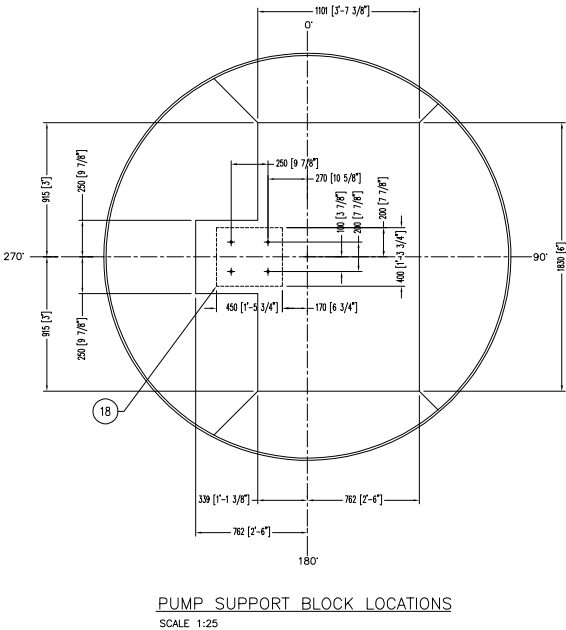
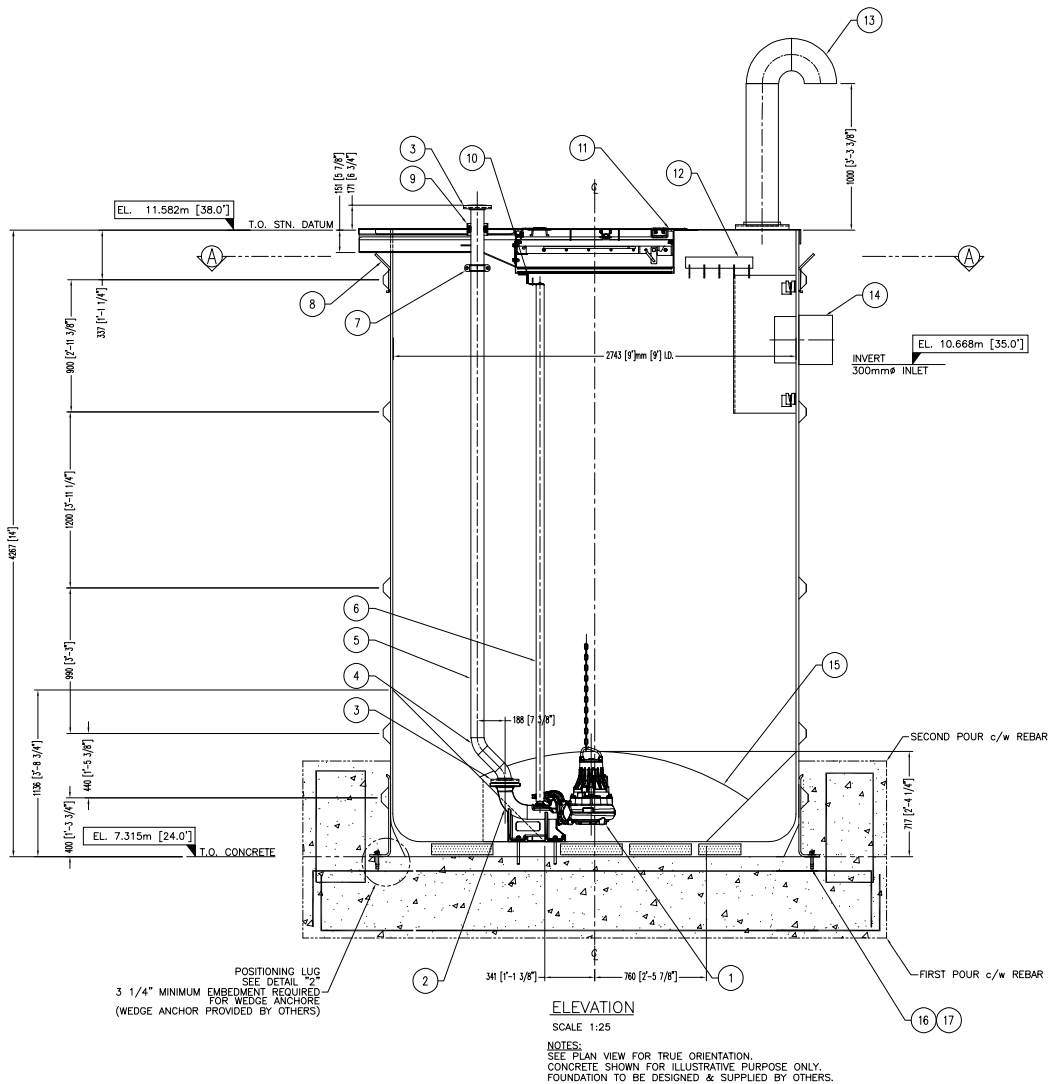
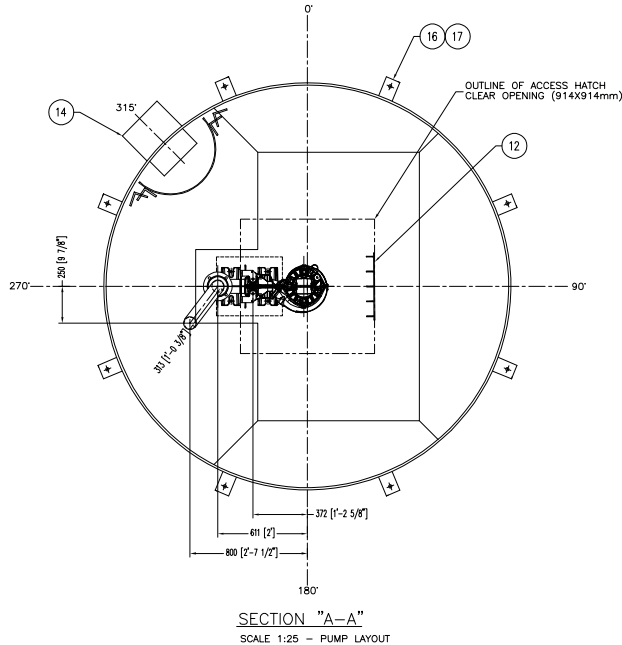
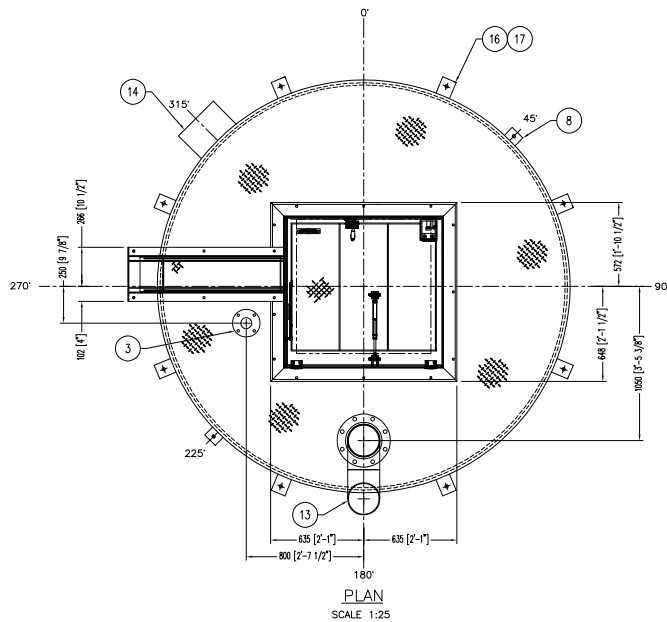
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REV	DATE	DESCRIPTION	BY

	TOLERANCES (U.N.O.) <table> <tr> <td><u>LINEAR</u></td><td><u>ANGULAR</u></td></tr> <tr> <td>$X \pm 1.5$</td><td>$X.X \pm 1/2'$</td></tr> </table>	<u>LINEAR</u>	<u>ANGULAR</u>	$X \pm 1.5$	$X.X \pm 1/2'$
<u>LINEAR</u>	<u>ANGULAR</u>				
$X \pm 1.5$	$X.X \pm 1/2'$				



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CLIENT XYLEM CHARLOTTE, NC		ENGINEER	
PROJECT HILO HEADWORKS DRAIN SEWAGE LIFT STATION			
TITLE 2743.2mm [9.0'] I.D. HILO HEADWORKS DRAIN LIFT STATION			
ENG BY:	DATE:	CAD FILE: HILO	PROJECT:
CAD BY: M. WOODCOCK	DATE: 2025/08/25	SCALE: AS SHOWN	
APP BY:	DATE:	DRAWING NUMBER	
ISSUED BY:	DATE:	25-186	REV 0



BILL OF MATERIALS			
QTY	DESCRIPTION		
1	FLYGT PUMPS, NP-3085 MT, DN80 [3"]	(BY XYLEM)	1
2	STANDARD DN80 [3"] DISCHARGE ELBOW, P/N 444 88 05		1
3	75mm-150q [3"] RF FLANGE, 316 SS		2
4	75mm [3"] 45° SR ELBOW, SCH.10, 316 SS		2
5	75mm [3"] RISER PIPE, SCH.10, 316 SS		1
6	50mm [2"] GUIDE RAILS, SCH.40, 316 SS		2
7	75mm [3"] VICTAULIC COUPLING 489SS		1
8	LIFTING LUGS, EPOXY COATED, M.S. (SEE DETAIL "1")		2
9	125# I.D. [5" I.D.] FRP SLEEVE c/w WRAP-IT LINK ML-300, 8 LINK		1
10	50mm [2"] UPPER GUIDE BAR BRACKET		1
11	U.S.F. HATCH ALUMINUM, 40" X 42" OPENING (DWG #1000285935)		1
12	316SS CABLE HANGER		1
13	200mm [8"] GOOSENECK WENT, SCH.10 316 SS c/w 180° RETURN BEND & SST BROSSCREEN		1
14	300mm [12"] INLET, MACHINED, FRP (SEE DETAIL "3") c/w REMOVABLE INLET BAFFLE		1
15	CUSTOM FRP HYDRAULIC BENCHING		-
16	POSITIONING LUGS, M.S. EPOXY COATED (SEE DETAIL "2")		8
17	3/4"x4-3/4" WEDGE ANCHOR BOLTS, GALV. CS	(BY OTHERS)	8
18	450x400mm FRP POLYMER BLOCK WITH STUDS (PUMP SUPPORTS)		1

FABRICATION DESIGN STANDARDS

- FLYGT SPECIFICATION CE-1008-04, REVISION MAY 2002
- AMEC 45-10.01 MANUFACTURE AND INSTALLATION FOR FRP STRUCTURES
- AMEC 45-10.02 FRP PRESSURE PIPE, FITTINGS AND FLANGES

GENERAL NOTES

- WINDING ANGLE - 75°
- TANK WALL - VARIES WITH ELEVATION
- LINER - C-GLASS VEIL AND (2)-1 1/2 oz. MATT
- RESIN - ISOPHTHALIC
- INTERIOR FINISH - WHITE ISOPHTHALIC NPG GELCOAT
- EXTERIOR (ABOVE GRADE) TO HAVE DARK GREEN URETHANE PAINTED FINISH
- DIMENSIONS ARE IN MILLIMETERS U.N.O.
- APPROX. SHIPPING WEIGHT: 3,175 kg (7,000 lbs)

INSTALLATION RECOMMENDATIONS

THE FOLLOWING RECOMMENDATIONS ARE BASED ON FLYGT EXPERIENCE.

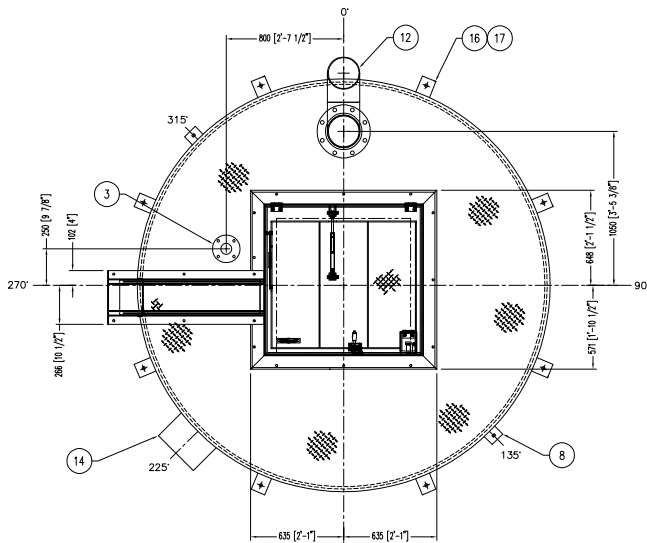
- USE THE LIFTING LUGS PROVIDED FOR VERTICAL HANDLING.
- USE SLINGS AROUND THE MAIN TANK FOR HORIZONTAL HANDLING.
- ENSURE UNIT IS STANDING VERTICAL ON CONCRETE PAD.
- BOLT UNIT FIRMLY AND SQUARELY IN PLACE, SHIM WHERE NECESSARY.
- ENCASE BOTTOM RIB IN CONCRETE TO A MIN. HEIGHT OF 150mm ABOVE RIB TO PROVIDE ANCHORAGE. REBAR TO CONNECT SECOND POUR TO THE CONCRETE BASE PAD.
- MAINTAIN A DRY SITE UNTIL BACKFILLING OPERATIONS COMMENCE.
- PLEASE REFER TO BARSKI BACKFILL PROCEDURE FOR FRP VERTICAL TANKS DOCUMENT
- PLACE THE BACKFILL IN EQUAL INCREMENTS NOT EXCEEDING 300mm THICK AROUND THE STATION TO PREVENT UNBALANCED LOADS BEING IMPOSED DURING BACKFILLING OPERATIONS. PROGRESSIVELY TAMP BACKFILL AROUND STATION TO FULL HEIGHT TO REDUCE SETTLEMENT TO AN ABSOLUTE MINIMUM.
- PLEASE CHECK WITH THE MANUFACTURER REGARDING THE ACTUAL DIMENSIONS OF THE LIFT STATION BEFORE DRAFTING SHOP DRAWINGS OF THE CONCRETE FOUNDATION.

0	2025/08/25	ISSUED FOR APPROVAL	MW
REV	DATE	DESCRIPTION	BY
TOLERANCES (U.N.O.)			
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X ± 1.5		X.X ± 1/2°	

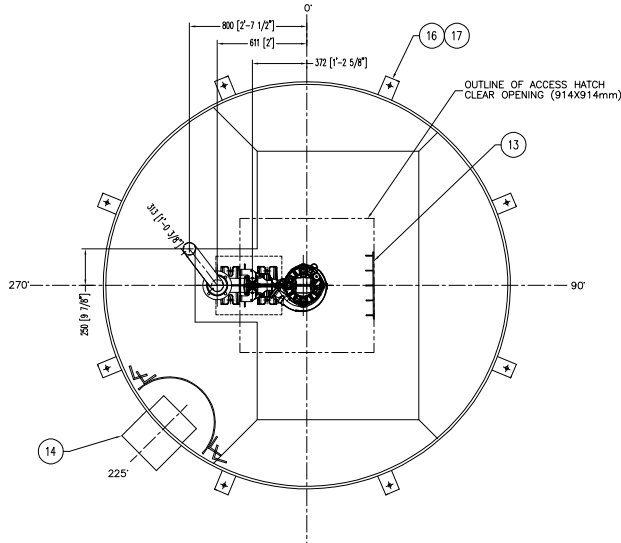


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CLIENT	XYLEM CHARLOTTE, NC		ENGINEER	
PROJECT	HILO WWTP SEPTAGE SUMP 1 SEWAGE LIFT STATION			
TITLE	2743.2mm [9.0"] I.D. HILO SEPTAGE SUMP 1 LIFT STATION			
ENG BY:	DATE:	CAD FILE: HILO	PROJECT:	
CAD BY: M.WOODCOCK	DATE: 2025/08/25	SCALE: AS SHOWN		
APP BY:	DATE:	DRAWING NUMBER		
ISSUED BY:	DATE:	25-187		0

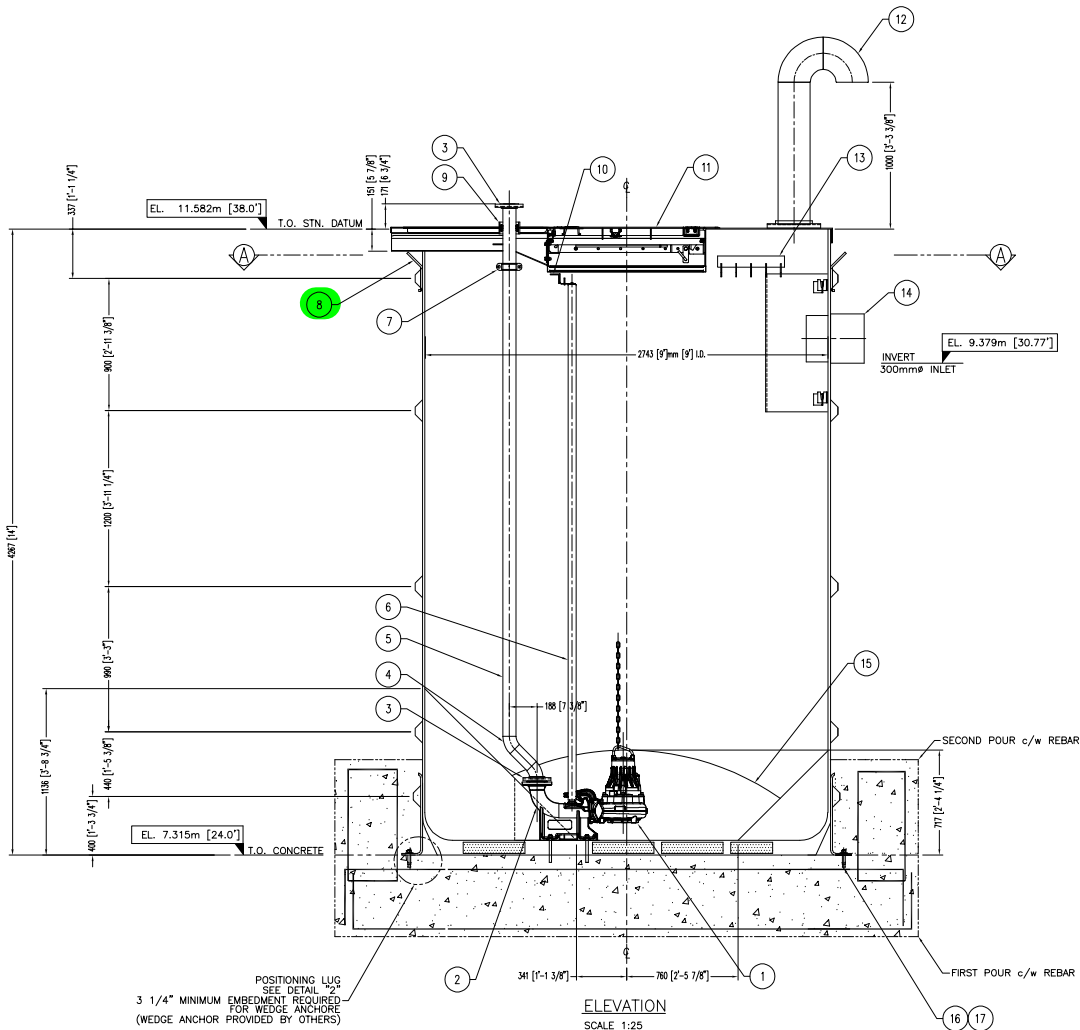


PLAN
SCALE 1:25



SECTION "A-A"

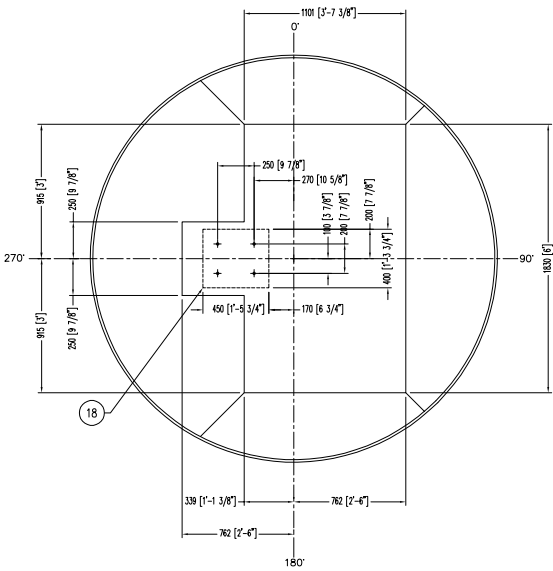
SCALE 1:25 - PUMP LAYOUT



ELEVATION

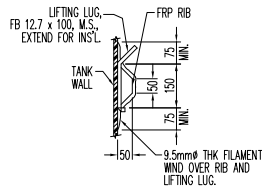
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NOTES:
SEE PLAN VIEW FOR TRUE ORIENTATION.
CONCRETE SHOWN FOR ILLUSTRATIVE PURPOSE ONLY.
FOUNDATION TO BE DESIGNED & SUPPLIED BY OTHERS.



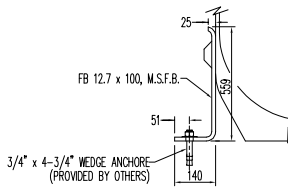
PUMP SUPPORT BLOCK LOCATIONS

SCALE 1:25



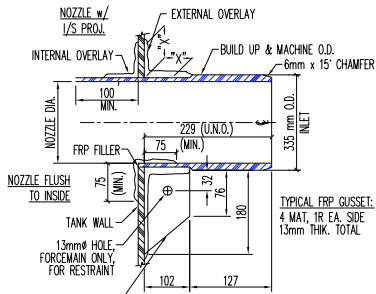
DETAIL "1"

N.T.S.



DETAIL "2"

N.T.S.



DETAIL "3"

N.T.S.

NOZZLE NOTES:
A MINIMUM OF 2 LAYERS OF WOVEN ROVING IS REQUIRED FOR NOZZLE PORTION OF OVERLAY. TANK PORTION OF OVERLAY SHALL BE SAME THICKNESS AS TANK WALL. DIMENSION "X" SHALL BE GREATER THAN ONE HALF THE NOZZLE DIAMETER. INTERNAL OVERLAY TO HAVE A MINIMUM OF 2 LAYERS MAT. FRP REINFORCING MATERIAL TO BE CUT TO A SHAPE TO ENSURE A SMOOTH LAMINATE.
O.D.'s REPRESENT THICKNESSES REQUIRED FOR ADEQUATE NOZZLE STRENGTH. TO BE CONNECTED TO EXTERIOR PIPING WITH A FLEXIBLE COUPLER.

DETAIL "3" - TYPICAL NOZZLE, MACHINED
N.T.S.

BILL OF MATERIALS

QTY	DESCRIPTION	
1	FLYGT PUMPS, NP-3085 MT, DN80 [3"]	(BY XYLEM) 1
2	STANDARD DN80 [3"] DISCHARGE ELBOW, P/N 444 68 05	1
3	75mm-150# [3"] RF FLANGE, 316 SS	2
4	75mm# [3"] 45° SR ELBOW, SCH.10, 316 SS	2
5	75mm# [3"] RISER PIPE, SCH.10, 316 SS	1
6	50mm# [2"] GUIDE RAILS, SCH.40, 316 SS	2
7	75mm# [3"] VCTAULIC COUPLING 489SS	1
8	LIFTING LUGS, EPOXY COATED, M.S. (SEE DETAIL "1")	2
9	125# I.D. [5" I.D.] FRP SLEEVE c/w WRAP-IT LINK WL-300, 8 LINK	1
10	50mm# [2"] UPPER GUIDE BAR BRACKET	1
11	U.S.F. HATCH ALUMINUM, 40" X 42" OPENING (DWG #1000285937)	1
12	200mm# [8"] GOOSENECK VENT, SCH.10 316 SS c/w 180° RETURN BEND & SST BIRDSCREEN	1
13	316SS CABLE HANGER	1
14	300mm# [12"] INLET, MACHINED, FRP (SEE DETAIL "3") c/w REMOVABLE INLET BAFFLE	1
15	CUSTOM FRP HYDRAULIC BENCHING	-
16	POSITIONING LUGS, M.S. EPOXY COATED (SEE DETAIL "2")	8
17	3/4"x4-3/4" WEDGE ANCHOR BOLTS, GALV. CS	(BY OTHERS) 8
18	450x400mm FRP POLYMER BLOCK WITH STUDS (PUMP SUPPORTS)	1

FABRICATION DESIGN STANDARDS

- FLYGT SPECIFICATION GE-1008-04, REVISION MAY 2002.
- AMEC 45-10.01 MANUFACTURE AND INSTALLATION FOR FRP STRUCTURES
- AMEC 45-10.02 FRP PRESSURE PIPE, FITTINGS AND FLANGES

GENERAL NOTES

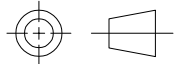
- WINDING ANGLE - 75°
- TANK WALL - VARIES WITH ELEVATION
- LINER - C-GLASS VEIL AND (2)-1 1/2 oz. MATT
- RESIN - ISOPHTHALIC
- INTERIOR FINISH - WHITE ISOPHTHALIC NPG GELCOAT
- EXTERIOR (ABOVE GRADE) TO HAVE DARK GREEN URETHANE PAINTED FINISH
- DIMENSIONS ARE IN MILLIMETERS U.N.O.
- APPROX. SHIPPING WEIGHT: 3,175 kg (7,000 lbs)

INSTALLATION RECOMMENDATIONS

THE FOLLOWING RECOMMENDATIONS ARE BASED ON FLYGT EXPERIENCE.

- USE THE LIFTING LUGS PROVIDED FOR VERTICAL HANDLING.
- USE SLINGS AROUND THE MAIN TANK FOR HORIZONTAL HANDLING.
- ENSURE UNIT IS STANDING VERTICAL ON CONCRETE PAD.
- BOLT UNIT FIRMLY AND SQUARELY IN PLACE, SHIM WHERE NECESSARY.
- ENCASE BOTTOM RIB IN CONCRETE TO A MIN. HEIGHT OF 150mm ABOVE RIB TO PROVIDE ANCHORAGE. REBAR TO CONNECT SECOND POUR TO THE CONCRETE BASE PAD.
- MAINTAIN A DRY SITE UNTIL BACKFILLING OPERATIONS COMMENCE.
- PLEASE REFER TO BARSKI BACKFILL PROCEDURE FOR FRP VERTICAL TANKS DOCUMENT
- PLACE THE BACKFILL IN EQUAL INCREMENTS NOT EXCEEDING 300mm THICK AROUND THE STATION TO PREVENT UNBALANCED LOADS BEING IMPOSED DURING BACKFILLING OPERATIONS. PROGRESSIVELY TAMP BACKFILL AROUND STATION TO FULL HEIGHT TO REDUCE SETTLEMENT TO AN ABSOLUTE MINIMUM.
- PLEASE CHECK WITH THE MANUFACTURER REGARDING THE ACTUAL DIMENSIONS OF THE LIFT STATION BEFORE DRAFTING SHOP DRAWINGS OF THE CONCRETE FOUNDATION.

0	2025/08/25	ISSUED FOR APPROVAL	MW
REV	DATE	DESCRIPTION	BY



TOLERANCES (U.N.O.)

LINEAR
X ± 1.5

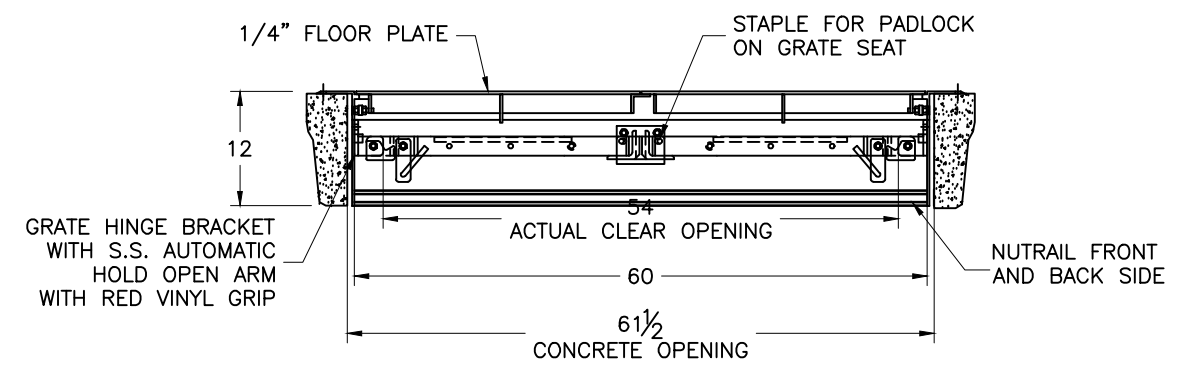
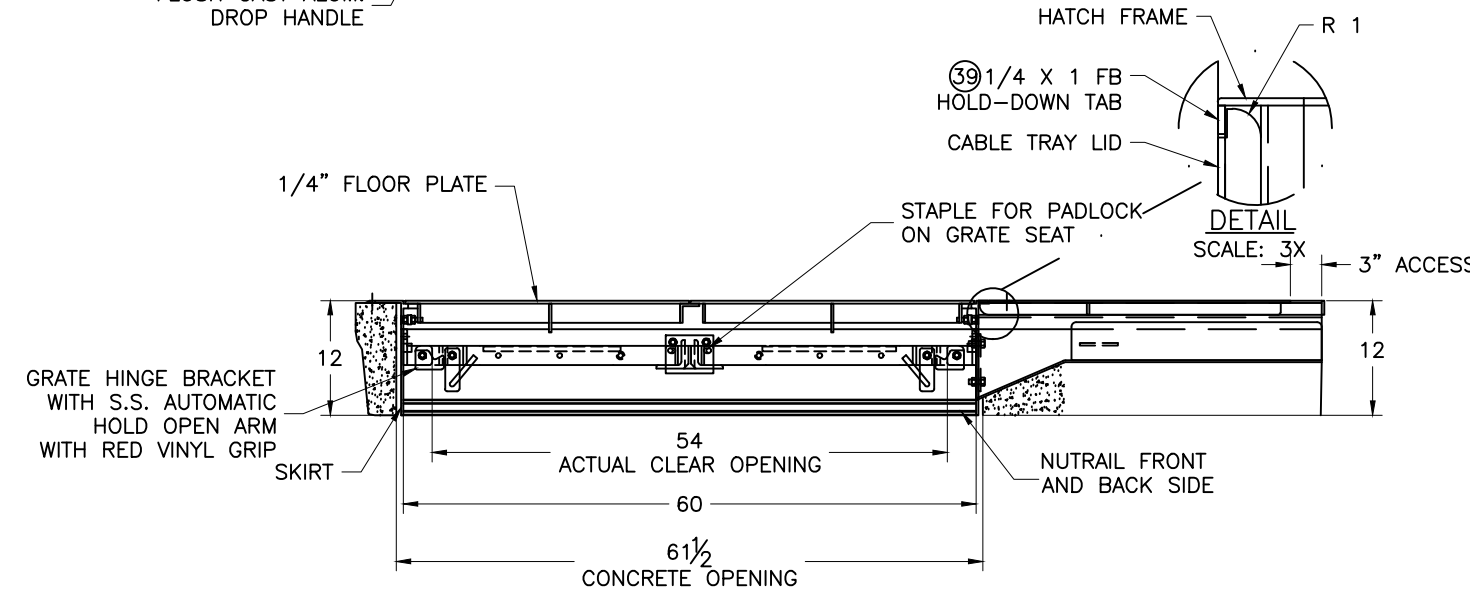
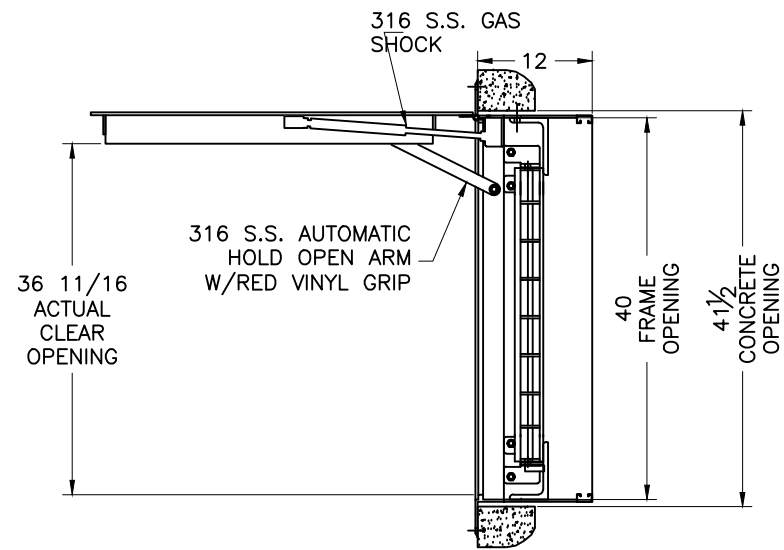
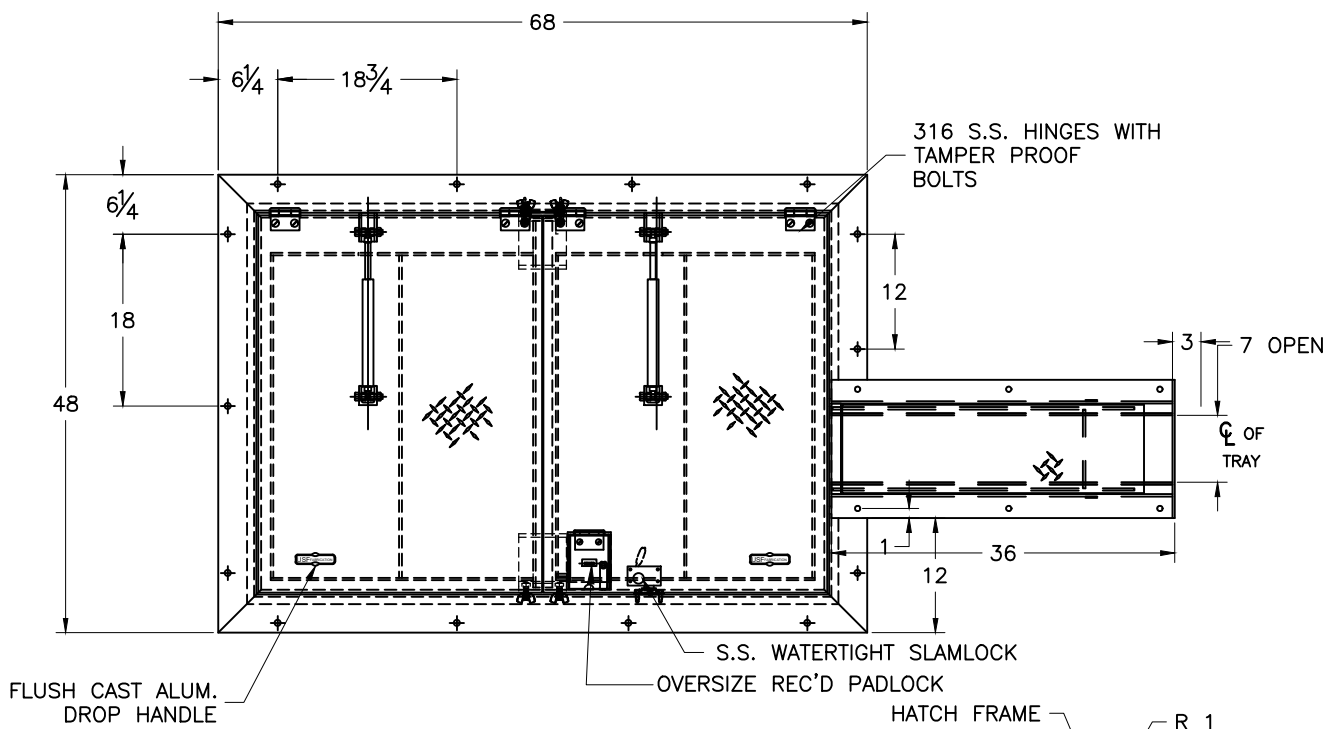
ANGULAR
X.X ± 1/2°



2378 WESTLAKE RD.
KELOWNA, B.C.
V1Z 2V2

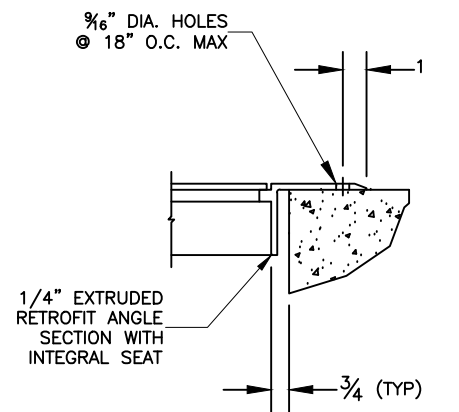
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CLIENT	XYLEM CHARLOTTE, NC	ENGINEER	
PROJECT	HILO WWTP SEPTAGE SUMP 2 SEWAGE LIFT STATION		
TITLE	2743.2mm [9.0'] I.D. HILO SEPTAGE SUMP 2 LIFT STATION		
ENG BY:	DATE:	CAD FILE: HILO	PROJECT:
CAD BY: M.WOODCOCK	DATE: 2025/08/25	SCALE: AS SHOWN	
APP BY:	DATE:	DRAWING NUMBER	REV
ISSUED BY:	DATE:	25-188	0



- SELECTED FEATURES**
- HINGED SAME SIDE
 - S.S GAS SHOCKS
 - SAFETY GRATING
 - BITUMINOUS COATING
 - CABLE TRAY CENTERED ON RIGHT SIDE

- NOTES**
- MATERIAL: ALUMINUM
 - FINISH: MILL
 - LOADING: 300 PSF
 - 316 SS NUTS & BOLTS
 - AREA OF FRAME IN CONTACT WITH CONCRETE TO BE PAINTED WITH BITUMINOUS COATING
 - SAFETY GRATING TO BE PAINTED WITH SAFETY ORANGE POWDER COAT
 - APPROX HATCH WT: 214.63 LBS



FRAME DETAIL

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INTERPRET DIMENSIONS AND TOLERANCES PER ASME Y14.5M		COPYRIGHT © 2020 ALL RIGHTS RESERVED	 U.S.F. FABRICATION INC. HIALEAH, FLORIDA	
BREAK ALL SHARP CORNERS & EDGES TO 0.01				
TOLERANCES UNLESS OTHERWISE SPECIFIED FRACTIONAL <10' = ± 1/8 >10' = ± 1/4			RETROFIT HATCH APD ALUM FOR 41 1/2 X 61 1/2 EXISTING OPENING	
DWN. BY: LOLA		SCALE: 1:20	SHEET: 1 OF 1	DATE: 08/20/25
CHK. BY: LOLA		DWG.# 1000288089	SHEET SIZE: B	REV: 0

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NP 3085 MT 3~ Adaptive 462

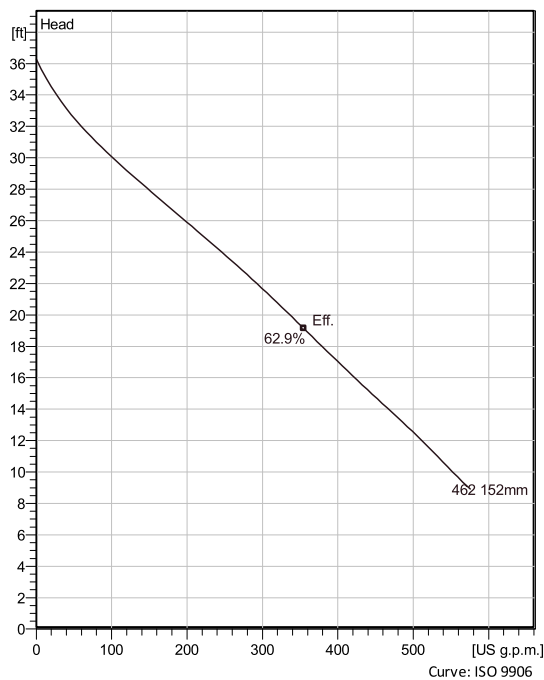
Patented self cleaning semi-open channel impeller, ideal for pumping in waste water applications. Modular based design with high adaptation grade.



Technical specification



Curves according to: Water, pure Water, pure [100%], 39.2 °F, 62.42 lb/ft³, 1.6891E-5 ft²/s



Nominal (mean) data shown. Under- and over-performance from this data should be expected due to standard manufacturing tolerances.
Please consult your local Flygt representative for performance guarantees.

Configuration

Motor number N3085.070 15-10-4AL-W 3hp	Installation type P - Semi permanent, Wet
Impeller diameter 152 mm	Discharge diameter 3 inch

Pump information

Impeller diameter 152 mm
Discharge diameter 3 inch
Inlet diameter 100 mm
Maximum operating speed 1705 rpm
Number of blades 2
Max. fluid temperature 40 °C

Material

Impeller Hard-Iron TM
Stator housing material Grey cast iron

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NP 3085 MT 3~ Adaptive 462

Technical specification



Motor - General

Motor number N3085.070 15-10-4AL-W 3hp	Phases 3~	Rated speed 1705 rpm	Rated power 3 hp
ATEX approved FM	Number of poles 4	Rated current 4.5 A	Stator variant 61
Frequency 60 Hz	Rated voltage 460 V	Insulation class H	Type of Duty S1
Version code 070			

Motor - Technical

Power factor - 1/1 Load 0.81	Motor efficiency - 1/1 Load 77.8 %	Total moment of inertia 0.475 lb ft²	Starts per hour max. 30
Power factor - 3/4 Load 0.74	Motor efficiency - 3/4 Load 78.4 %	Starting current, direct starting 25 A	
Power factor - 1/2 Load 0.62	Motor efficiency - 1/2 Load 76.2 %	Starting current, star-delta 8.34 A	

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NP 3085 MT 3~ Adaptive 462

Performance curve

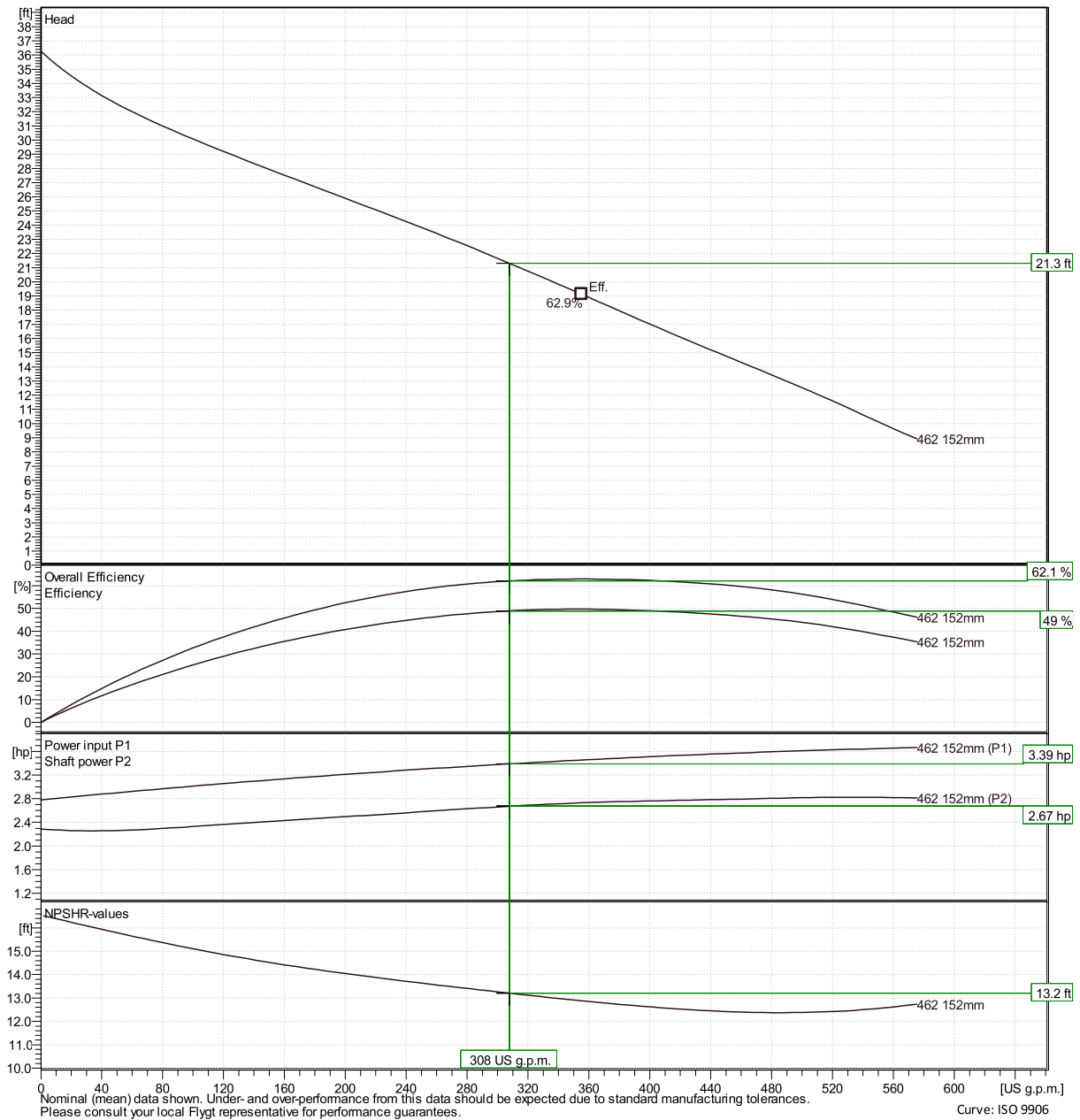


Duty point

Flow
308 US g.p.m.

Head
21.3 ft

Curves according to: Water, pure [100%], 39.2 °F, 62.42 lb/ft³, 1.6891E-5 ft²/s



Xylect-20280277

Ron Saoit HES

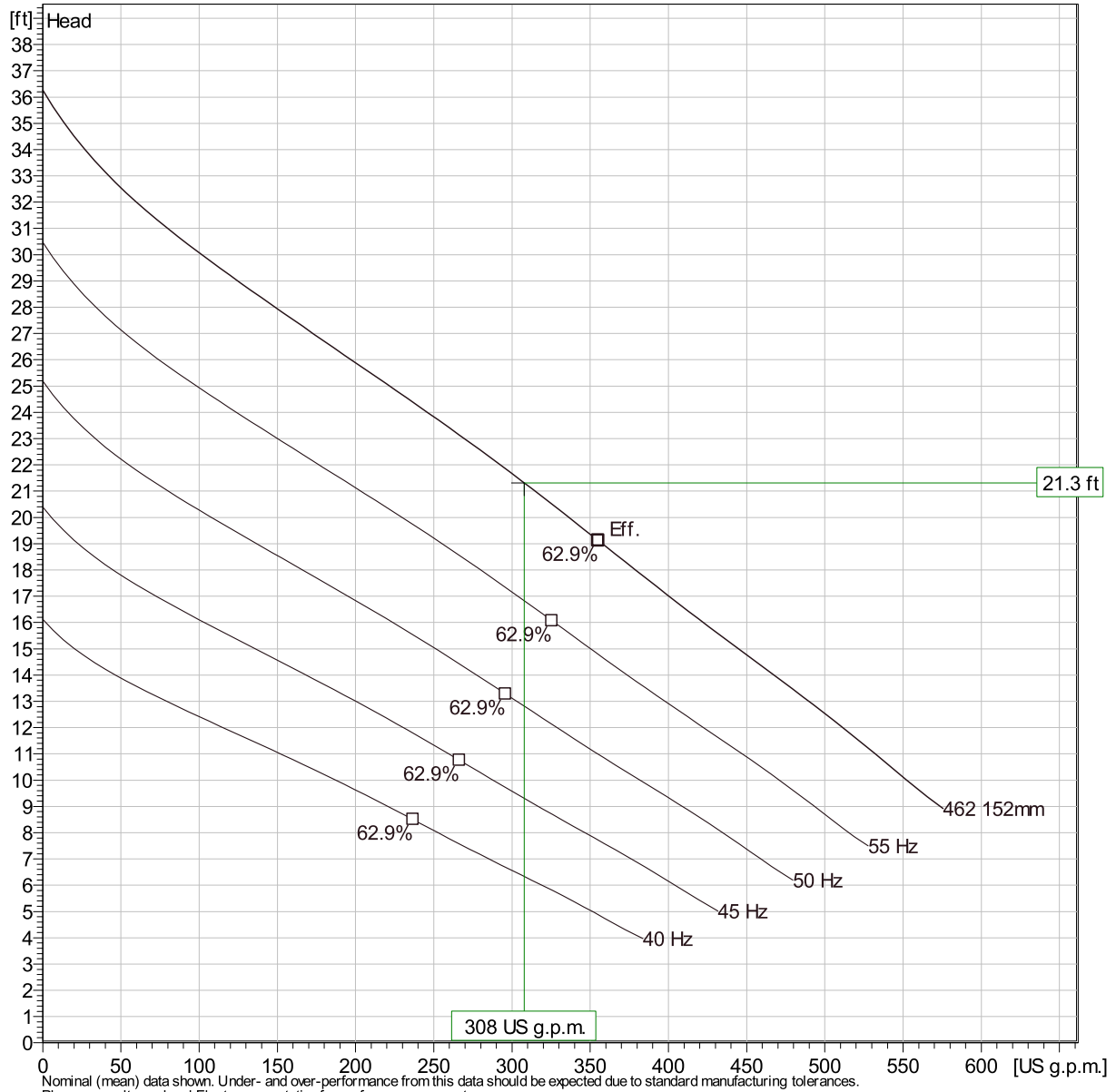
Created on 7/6/2024 Last update 7/6/2024

NP 3085 MT 3~ Adaptive 462

Duty Analysis



Curves according to: Water, pure [100%] ; 39.2°F; 62.42lb/ft³; 1.6891E-5ft²/s



Nominal (mean) data shown. Under- and over-performance from this data should be expected due to standard manufacturing tolerances.
Please consult your local Flygt representative for performance guarantees.

Operating characteristics

Pumps / Systems	Flow	Head	Shaft power	Flow	Head	Shaft power	Hydr.eff.	Spec. Energy	NPSHre
	US g.p.m.	ft	hp	US g.p.m.	ft	hp		kWh/US MG	ft
Required Condition	308	21.3	2.67	308	21.3	2.67	62.1 %	137	13.2
Required Condition	138	28.5	2.39	138	28.5	2.39	41.4 %	279	14.7
Rated Design Point	203	25.8	2.5	203	25.8	2.5	52.9 %	197	14

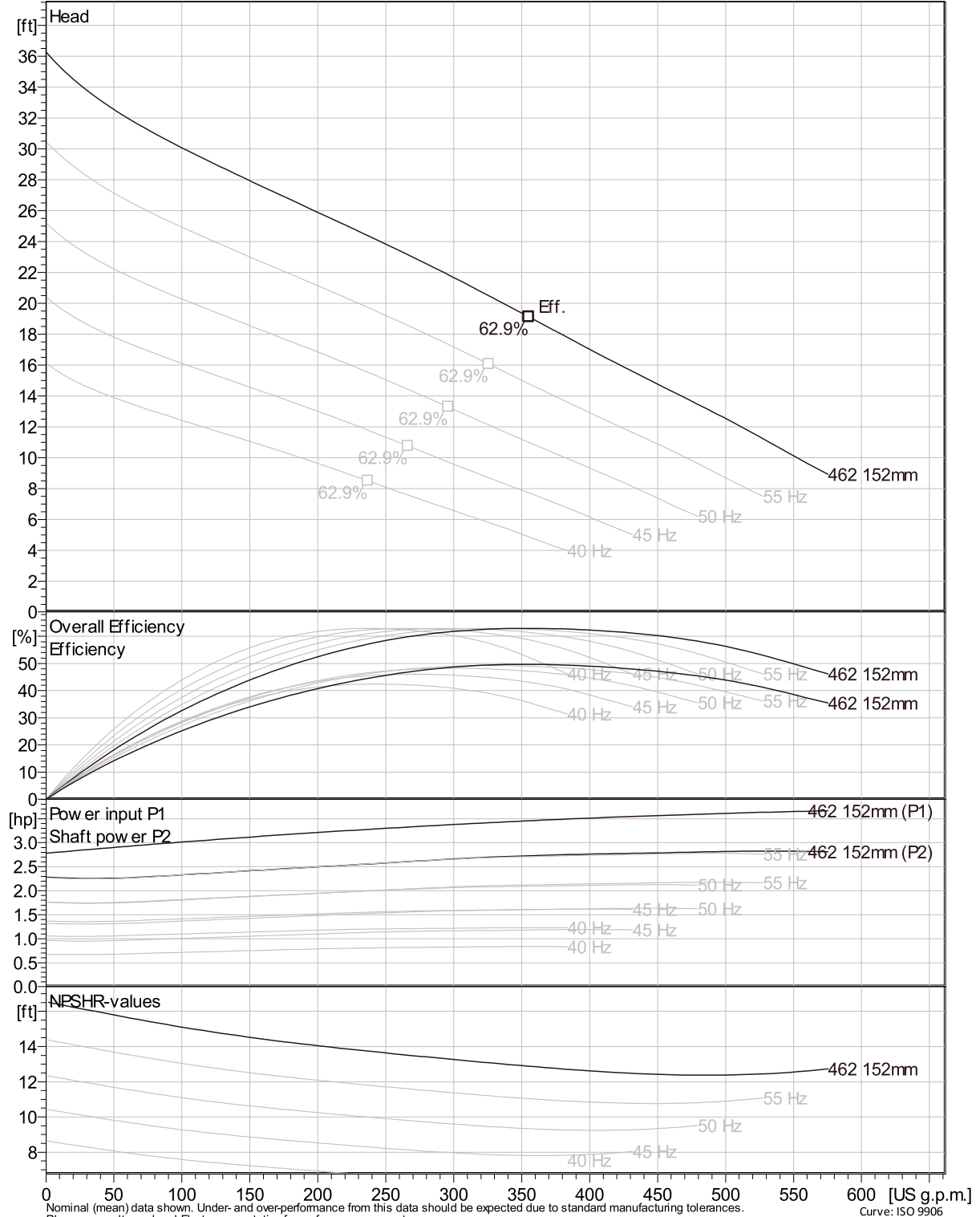
Project		Created by	Ron Saoit HES		
Block	Xylect-20280277	Created on	7/6/2024	Last update	7/6/2024

NP 3085 MT 3~ Adaptive 462

VFD Curve



Curves according to: Water, pure, 39.2 °F, 62.42 lb/ft³, 1.6891E-5 ft²/s

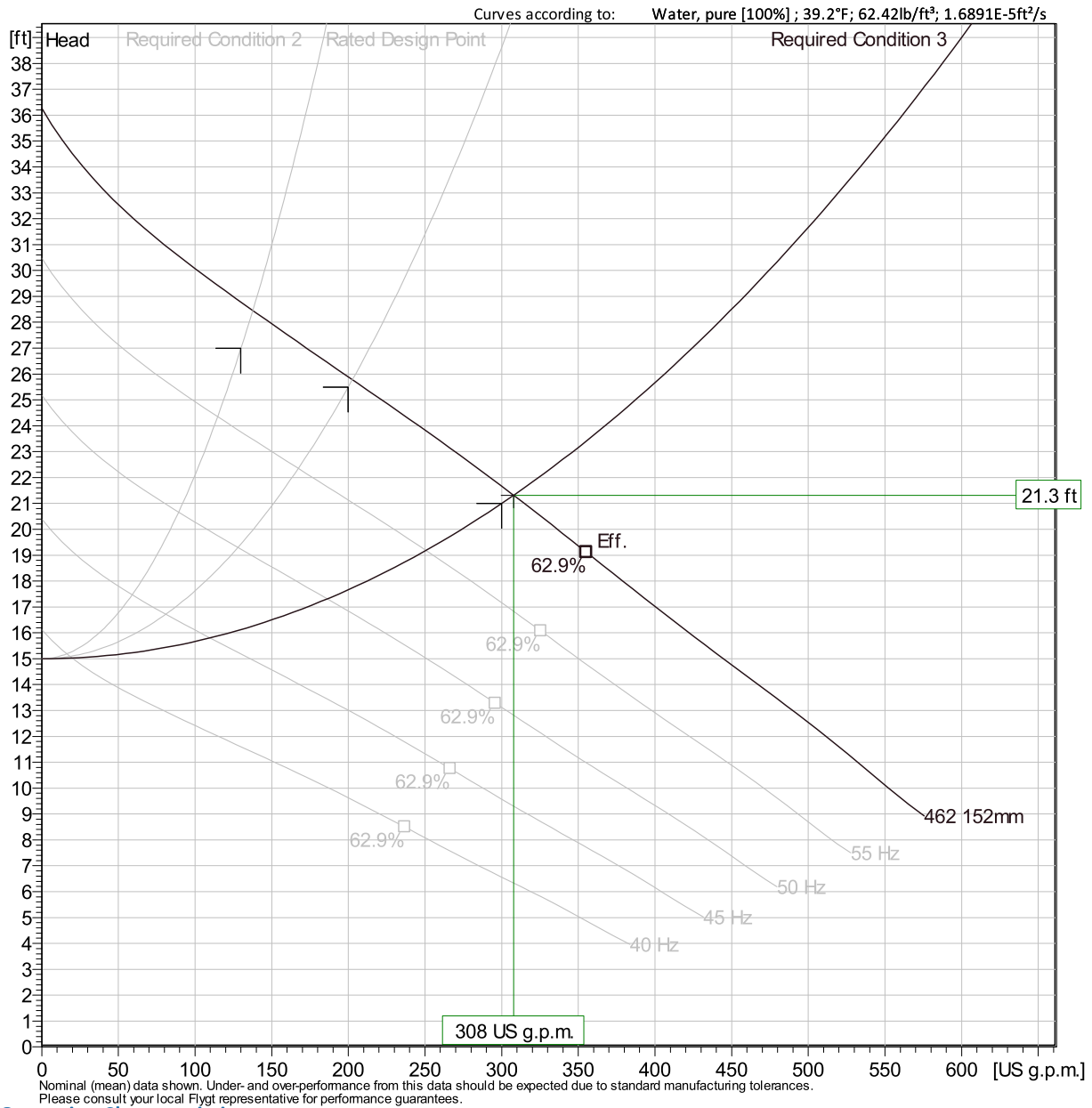


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Block

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NP 3085 MT 3~ Adaptive 462

VFD Analysis



Operating Characteristics

Pumps / Systems	Frequency	Flow	Head	Shaft power	Flow	Head	Shaft power	Hydr. eff.	Specific energy	NPSHr
		US g.p.m.	ft	hp	US g.p.m.	ft	hp		kWh/US MG	ft
Required Condit	60 Hz	308	21.3	2.67	308	21.3	2.67	62.1 %	137	13.2
Required Condit	55 Hz	251	19.2	2.02	251	19.2	2.02	60.3 %	128	11.7
Required Condit	50 Hz	186	17.3	1.47	186	17.3	1.47	55.6 %	129	10.3
Required Condit	45 Hz	110	15.8	1.02	110	15.8	1.02	43.2 %	161	9.19

Project Xylect-20280277

Block

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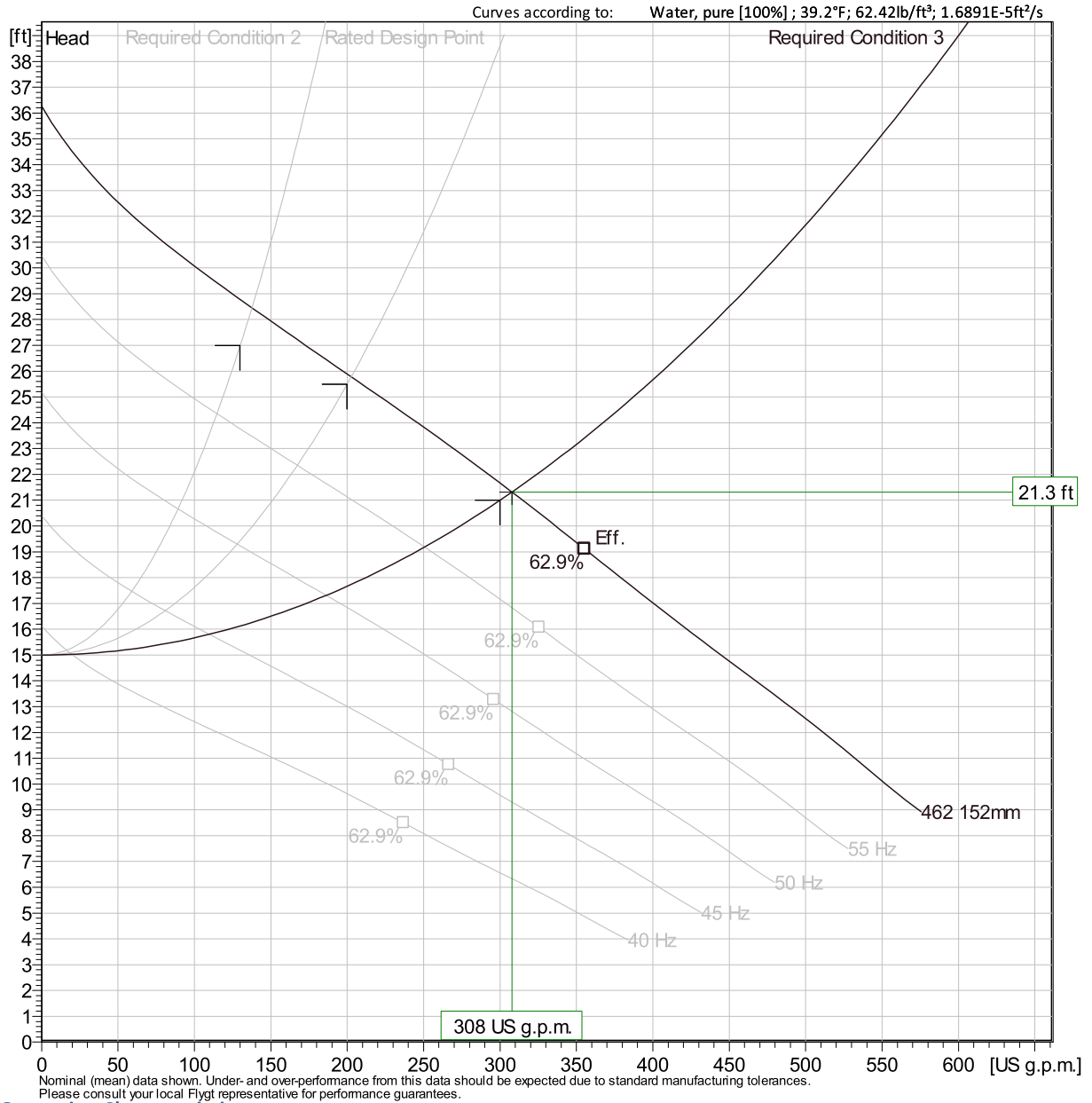
Created on 7/6/2024

Last update

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VFD Analysis



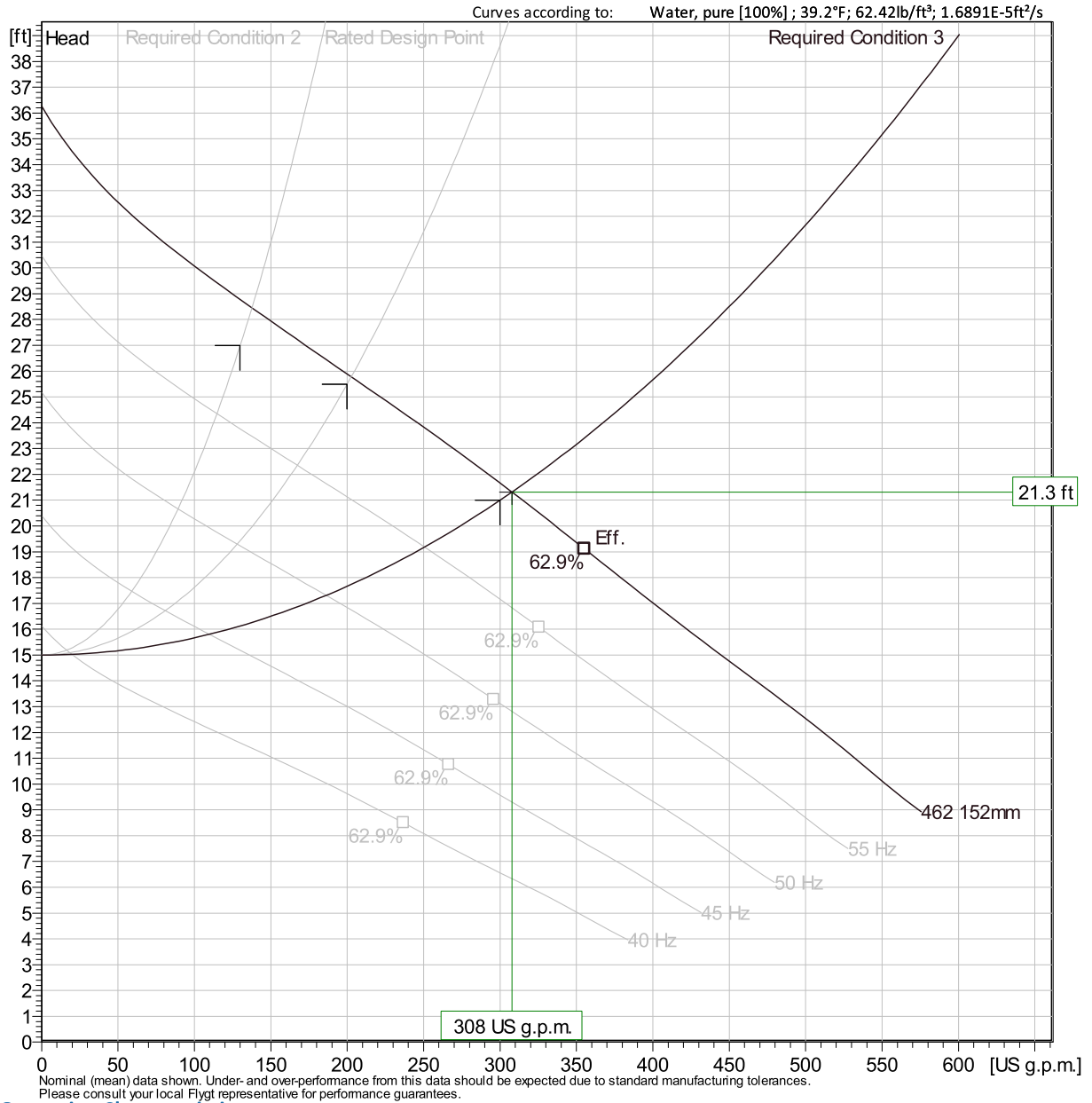
Pumps / Systems	Frequency	Flow	Head	Shaft power	Flow	Head	Shaft power	Hydr. eff.	Specific energy	NPSHre
		US g.p.m.	ft	hp	US g.p.m.	ft	hp		kWh/US MG	ft
Required Condit	40 Hz	19.9	15	0.668	19.9	15	0.668	11.3 %	656	8.41
Required Condit	60 Hz	138	28.5	2.39	138	28.5	2.39	41.4 %	279	14.7
Required Condit	55 Hz	115	24.4	1.83	115	24.4	1.83	38.7 %	255	12.9
Required Condit	50 Hz	89.3	20.7	1.35	89.3	20.7	1.35	34.5 %	251	11.2

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NP 3085 MT 3~ Adaptive 462

VFD Analysis



Operating Characteristics

Pumps / Systems	Frequency	Flow	Head	Shaft power	Flow	Head	Shaft power	Hydr. eff.	Specific energy	NPSHre
		US g.p.m.	ft	hp	US g.p.m.	ft	hp		kWh/US MG	ft
Required Condit	45 Hz	59	17.5	0.968	59	17.5	0.968	26.9 %	289	9.71
Required Condit	40 Hz	16.4	15.2	0.669	16.4	15.2	0.669	9.44 %	796	8.45
Rated Design P	60 Hz	203	25.8	2.5	203	25.8	2.5	52.9 %	197	14
Rated Design P	55 Hz	167	22.4	1.9	167	22.4	1.9	49.8 %	181	12.4

Project Xylect-20280277

Block

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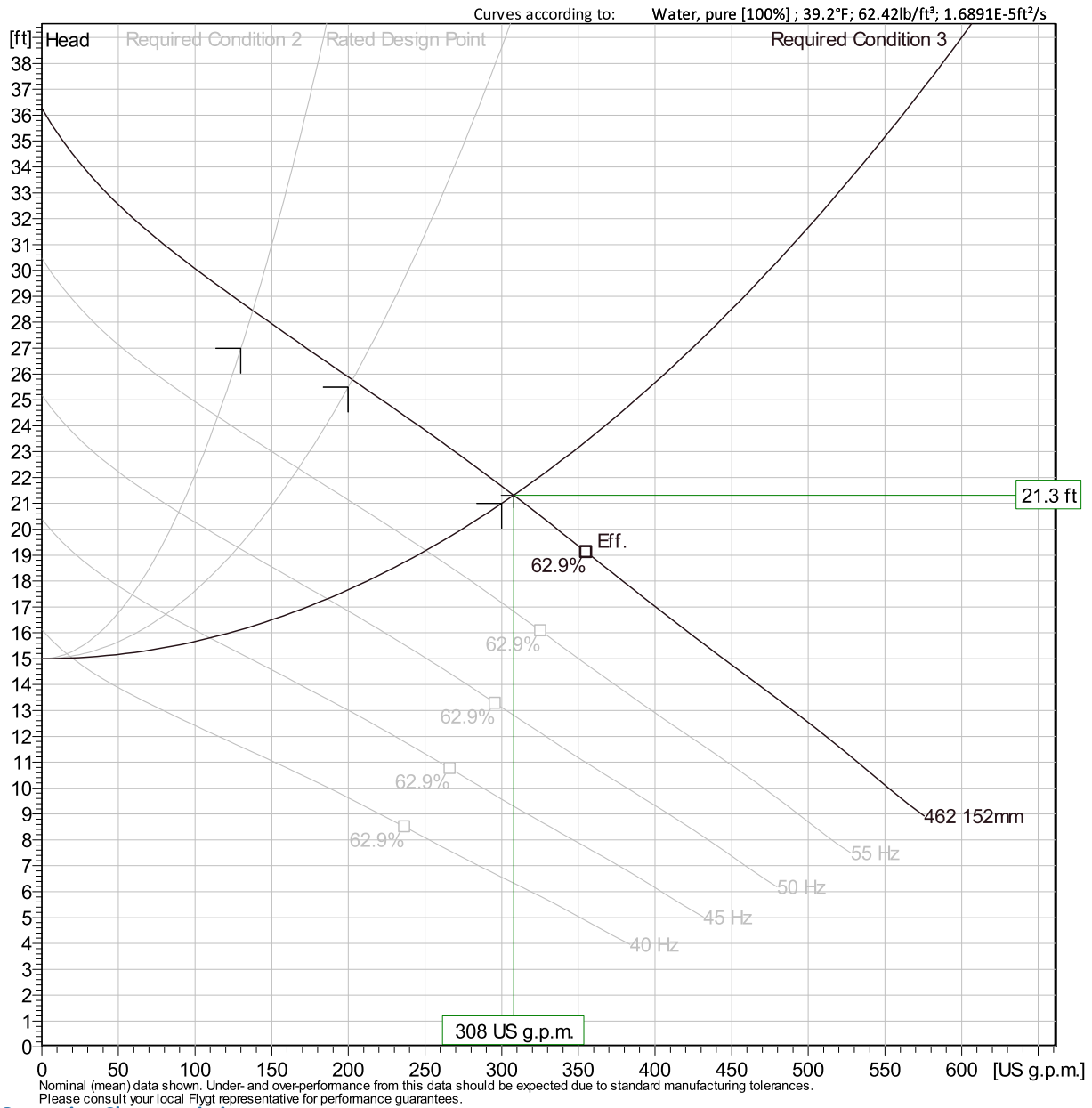
Created on 7/6/2024

Last update

7/6/2024

NP 3085 MT 3~ Adaptive 462

VFD Analysis



Operating Characteristics

Pumps / Systems	Frequency	Flow	Head	Shaft power	Flow	Head	Shaft power	Hydr. eff.	Specific energy	NPSH _r
		US g.p.m.	ft	hp	US g.p.m.	ft	hp		kWh/US MG	
Rated Design P	50 Hz	128	19.3	1.4	128	19.3	1.4	44.6 %	180	10.8
Rated Design P	45 Hz	80.9	16.7	0.988	80.9	16.7	0.988	34.6 %	214	9.47
Rated Design P	40 Hz	18.5	15.1	0.669	18.5	15.1	0.669	10.6 %	705	8.43

Project Xylect-20280277

Block

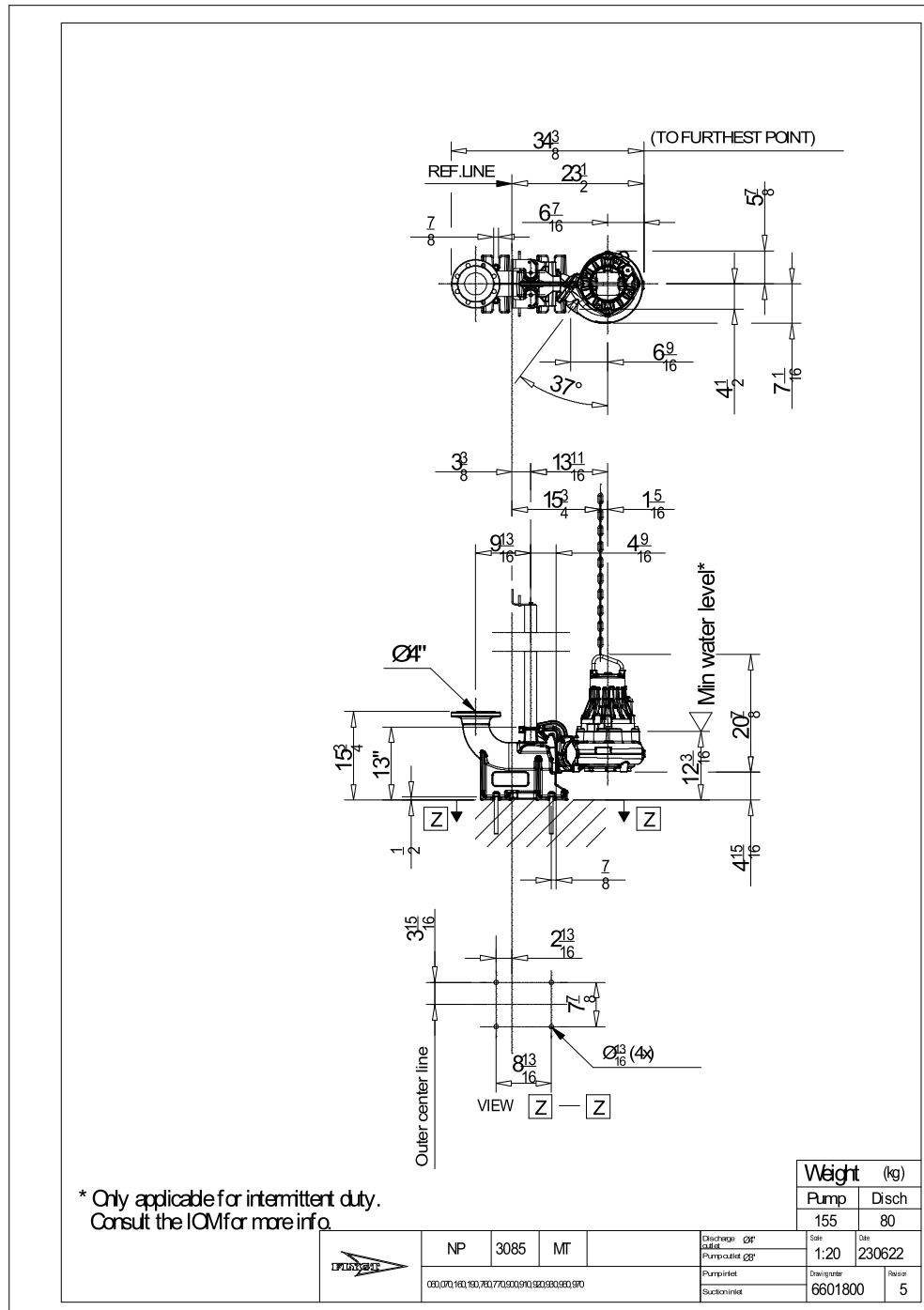
Created by Ron Saoit HES

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Last update

7/6/2024

Dimensional drawing



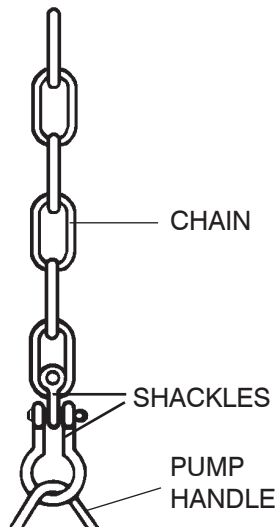
Project	Xylect-20280277	Created by	Ron Saoit HES		
Block		Created on	7/6/2024	Last update	7/6/2024

Lifting Chain (Stainless Steel)

(for: CP/DP/NP, BS/CS/DS/NS installations only)

2000, 3000, 6000 & 8000 Series Pumps

Pump Models:	Stainless Steel Chain Size	Chain Part Number	Chain Fittings Kit Part Number
2730, 2740, 2750, 2830, 2840, 2860, 3060, 3068, 3080, 3085, 3102, 3127, 6020, 8050, 8053, 8056, 8058.	3/16" 316 (stainless steel)	14-48 71 13	14-58 72 08
2870, 3153, 3171.	5/16" 316 (stainless steel)	14-48 71 17	14-58 72 09
3202, 3301.	1/2" 316 (stainless steel)	14-48 71 18	14-58 72 10



Flygt Pump Lift™

The Flygt Pump Lift™ simplifies hoisting submersible pumps. It gives quick pump installation in one continuous hoisting action. The pump is easy to locate also when submerged.

One hoist can be used for several pumps.

Function

- The Pump Lift™ installation consists of a Grip Eye and one or more chain slings:
 - The Grip Eye is fitted to the hook of the hoist
 - Each chain sling is fitted to the handle of the pump. The rope/wire is suspended from the access opening
- The rope/wire guides the Grip Eye to the Pump Lift™ chain. It engages and releases the Grip eye to the chain
- One hoist can handle all the pumps
- The hoist can lift the pump in one go. The clean wire or chain of the hoist works smoothly in the hoist. No problems with dirty lifting chains
- No more “fishing” for the pump. Connect first time regardless of station depth and submergence

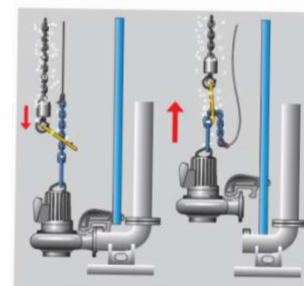
Each individual Flygt lifting chain is tested and CE approved and supplied with a marker plate showing max. load in accordance with the EU Machine directive 98/37/EG.

Note! Only the Grip Eye / chain combinations listed below are approved for function and safety.

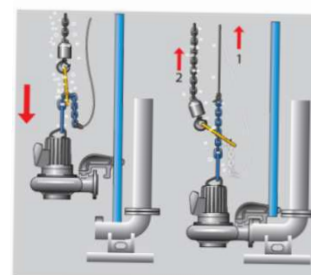
Flygt Pump Lift™ components

- Grip Eye: hangs from the hook of any hoist
- Chain Sling unit: complete with nylon rope or stainless steel 316 wire, a short chain and a master link / shackle for connection to the pump handle
 - The master link fits Flygt 3000 pumps
 - The shackle is required for pumps with drive unit size 600, 700 and 800 (with small handle), depending on pump weight.

Part numbers, materials and dimensions: see table on page 2.



Lifting the pump



Lowering the pump



Guiding the grip eye to the chain

Hoisting

Flygt Pump Lift™

Chain sling and grip eye combinations

Note! Only the Grip Eye/ chain combinations listed below are approved for function and safety.

Load kg	Chain sling					Grip eye	
	P/No.	Chain Material	m	Link / shackle	Rope/wire	P/No.	Material
1 200	442 18 00	Carbon steel	0.3	Link	10m nylon	620 09 00	Carbon steel
4 200	442 18 02	Carbon steel	0.3	Shackle	15m nylon	620 09 01	Carbon steel
540	442 18 06	Stainless steel	0.4	Link	10m nylon	620 09 00	Carbon steel
2 000	442 18 05	Stainless steel	0.7	Link	10m nylon	620 09 01	Carbon steel
540	442 18 16	Stainless steel	0.4	Link	10m SS 316	620 09 00	Carbon steel
540	442 18 17	Stainless steel	0.4	Link	7.6m SS 316	620 09 00	Carbon steel
540	442 18 18	Stainless steel	0.4	Link	6m SS 316	620 09 00	Carbon steel
540	442 18 19	Stainless steel	0.4	Link	4.6m SS 316	620 09 00	Carbon steel



P/No. 620 09 00



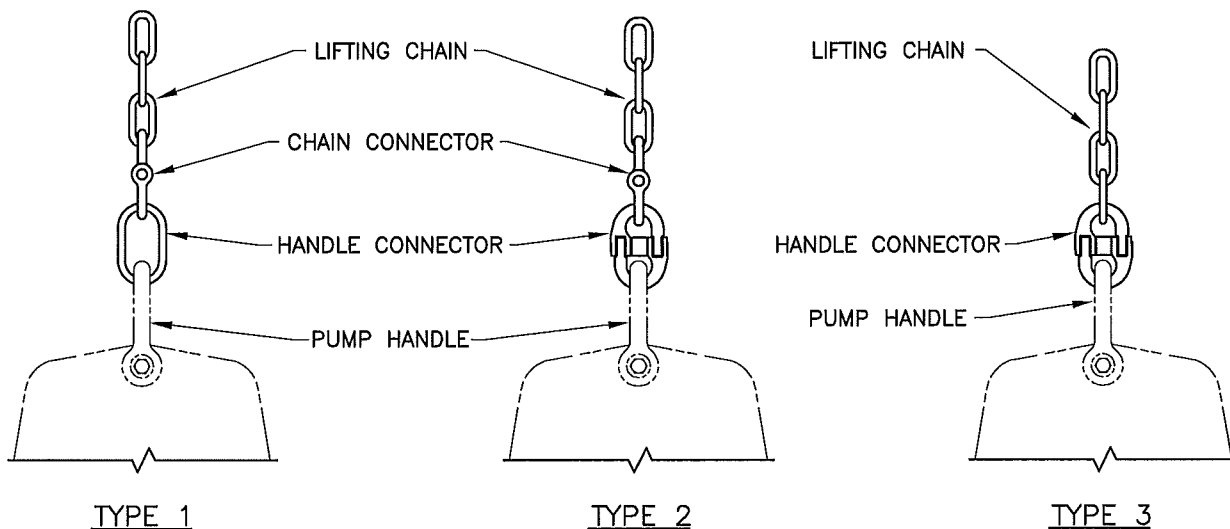
P/No. 442 18 00 / 05-19



P/No. 620 09 01



P/No. 442 18 02



ALL ACCESSORIES ARE SUPPLIED LOOSE.
INSTALLATION ON PUMP BY CUSTOMER.

LIFTING ARRANGEMENT		CHAIN		CHAIN CONNECTOR		HANDLE CONNECTOR		SYSTEM CAPACITY kg.(lbs.)
TYPE	CATEGORY	SIZE	P/N	SIZE	P/N	SIZE	P/N	
1	A	3/16"	13-45 00 57	3/16"	13-44 00 57	3/8"	13-44 00 76	499 (1100)
	B *	9/32"	13-45 00 58	9/32"	13-44 00 58	1/2"	13-44 00 61	998 (2200)
	C **	9/32"	13-50 05 27	9/32"	13-44 00 58	1/2"	13-44 00 61	998 (2200)
	D	3/8"	13-45 00 59	3/8"	13-44 00 59	5/8"	13-44 00 77	1724 (3800)
	E	3/8"	13-45 00 59	3/8"	13-44 00 59	11/16"	13-44 00 71	1996 (4400)
2	A	3/8"	13-45 00 59	3/8"	13-44 00 59	5/8"	13-44 00 60	1996 (4400)
3	A	5/8"	13-45 00 60	N/R	N/R	5/8"	13-44 00 60	4990 (11000)

* = DOES NOT MATCH SPROCKET ON CHAIN HOIST 13-43 00 06, SEE DOCUMENT No. ES 4.31.1

** = CALIBRATED CHAIN MAY BE USED WITH CHAIN HOST 13-43 00 06, SEE DOCUMENT No. ES 4.31.1

NOTES :

- 1 - CHAINS ARE :
 - SUITABLE FOR OVERHEAD LIFTING PER N.A.C.M.
 - MADE OF STAINLESS STEEL, 316 GRADE 50.
 - QUANTITY = BY METER
- 2 - CHAIN CONNECTORS ARE :
 - SUITABLE FOR OVERHEAD LIFTING PER N.A.C.M.
 - MADE OF STAINLESS STEEL, 316 GRADE 50.
 - QUANTITY = BY PIECE
- 3 - HANDLE CONNECTORS ARE :
 - SUITABLE FOR OVERHEAD LIFTING PER N.A.C.M.
 - MADE OF STAINLESS STEEL, 316 GRADE 50.
 - QUANTITY = BY PIECE
- 4 - SEE DOCUMENT NO. ES 4.40.3.1 & ES 4.40.4.1 FOR PROPER SELECTION.

N/R: NOT REQUIRED

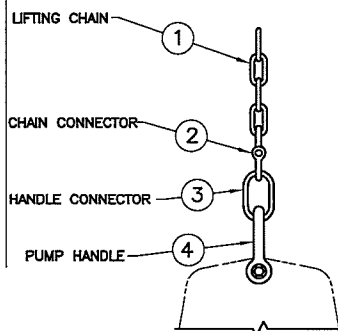
DIMENSIONS ARE IN mm

TYPE 1

PART NUMBER SIZE

13-45 00 57	3/16"
*13-45 00 58	9/32"
**13-50 05 27	9/32"
13-45 00 59	3/8"
13-44 00 57	3/16"
13-44 00 58	9/32"
13-44 00 59	3/8"
13-44 00 76	3/8"
13-44 00 61	1/2"
13-44 00 77	5/8"
13-44 00 71	1 1/16"

TOOLS REQUIRED : SEE NOTE 6



HANDLE MUST BE REMOVED

NOTES:

- 1) PASS ONE HALF OF THE CHAIN CONNECTOR 2 THROUGH THE CHAIN. 1
- 2) PASS THE OTHER HALF OF THE CHAIN CONNECTOR 2 THROUGH THE HANDLE CONNECTOR. 3
- 3) ASSEMBLE THE CHAIN CONNECTOR HALVES 2 TOGETHER.
- 4) REMOVE THE HANDLE 4 FROM THE PUMP AND PASS THE HANDLE 4 THROUGH THE HANDLE CONNECTOR. 3
- 5) REMOUNT THE PUMP HANDLE TO THE PUMP. 4
- 6) HAMMER REQUIRED TO ASSEMBLE CHAIN CONNECTOR HALVES 2 TOGETHER AND TO REMOVE THE HANDLE ON PUMP C-3085 USING PUNCH (SIZE 2mm TO 4mm DIA.). ALL OTHER PUMPS THAT USE 'TYPE 1' CHAIN INSTALLATION HAVE EITHER HX. HD. BOLTS OR SOCKET HD. CAP SCREWS HOLDING THE PUMP HANDLE IN PLACE.

* = DOES NOT MATCH SPROCKET ON CHAIN
HOIST 13-43 00 06, SEE DOCUMENT
No. ES 4.31.1

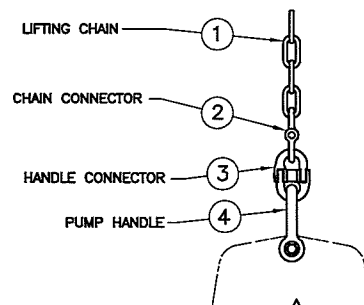
** = CALIBRATED CHAIN MAY BE USED WITH
CHAIN HOIST 13-43 00 06, SEE
DOCUMENT No. ES 4.31.1

TYPE 2

PART NUMBER SIZE

13-45 00 59	3/8"
13-44 00 59	3/8"
13-44 00 60	5/8"

TOOLS REQUIRED : HAMMER



DO NOT REMOVE HANDLE

NOTES:

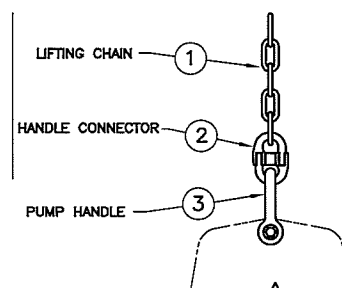
- 1) PASS ONE HALF OF THE CHAIN CONNECTOR 2 THROUGH THE CHAIN. 1
- 2) PASS THE OTHER HALF OF THE CHAIN CONNECTOR 2 THROUGH ONE HALF OF THE HANDLE CONNECTOR. 3
- 3) ASSEMBLE THE CHAIN CONNECTOR HALVES 2 TOGETHER.
- 4) PASS THE PUMP HANDLE 4 THROUGH THE OTHER HALF OF THE HANDLE CONNECTOR. 3
- 5) ASSEMBLE THE HANDLE CONNECTOR HALVES 3 TOGETHER.
- 6) HAMMER REQUIRED TO ASSEMBLE CHAIN CONNECTOR HALVES 2 AND HANDLE CONNECTOR HALVES 3 TOGETHER.

TYPE 3

PART NUMBER SIZE

13-45 00 60	5/8"
13-44 00 60	5/8"

TOOLS REQUIRED : HAMMER



HANDLE DOES NOT HAVE TO BE REMOVED

NOTES:

- 1) PASS ONE HALF OF THE HANDLE CONNECTOR 2 THROUGH THE CHAIN. 1
- 2) PASS THE PUMP HANDLE 3 THROUGH THE OTHER HALF OF THE HANDLE CONNECTOR. 2
- 3) ASSEMBLE THE HANDLE CONNECTOR HALVES 2 TOGETHER.
- 4) HAMMER REQUIRED TO ASSEMBLE HANDLE CONNECTOR HALVES 2 TOGETHER.



Station Equipment

DOCUMENT NO.

ES 4.40.3.1

Stainless Steel Lifting System Selection Chart Small & Medium Pumps in P/S/T Installation.

SUPERSEDES

ISSUED

10/2010

11/2012

PUMP MODEL NO. (DRIVE UNIT)	MAXIMUM PUMP WEIGHT	CHAIN LIFTING ARRANGEMENT			Flygt LIFT SYSTEM		
	kg (lbs)	TYPE	CATEGORY	DOC. NO.	CHAIN SLING P/N	GRIP EYE P/N	DOC. NO.
3041 (ALL)	23 (50)	1	A	ES 4.32.2.1	442 18 06 13-50 05 56	620 09 00	ES 4.33.1
3045 (ALL)	28 (62)	1	A	ES 4.32.2.1	442 18 06 13-50 05 56	620 09 00	ES 4.33.1
3057 (ALL)	34 (75)	1	A	ES 4.32.2.1	442 18 06 13-50 05 56	620 09 00	ES 4.33.1
3060 (ALL)	43 (95)	1	A	ES 4.32.2.1	442 18 06 13-50 05 56	620 09 00	ES 4.33.1
3068 (ALL)	50 (110)	1	A	ES 4.32.2.1	442 18 06 13-50 05 56	620 09 00	ES 4.33.1
3075 (ALL)	51 (112)	1	A	ES 4.32.2.1	442 18 06 13-50 05 56	620 09 00	ES 4.33.1
3076 (ALL)	68 (150)	1	A	ES 4.32.2.1	442 18 06 13-50 05 56	620 09 00	ES 4.33.1
3080 (ALL)	100 (220)	1	A	ES 4.32.2.1	442 18 06 13-50 05 56	620 09 00	ES 4.33.1
3085 (ALL)	88 (194)	1	A	ES 4.32.2.1	442 18 06 13-50 05 56	620 09 00	ES 4.33.1
3102 (ALL)	166 (366)	1	A	ES 4.32.2.1	442 18 06 13-50 05 56	620 09 00	ES 4.33.1
3126 (ALL)	215 (474)	1	A	ES 4.32.2.1	442 18 06 13-50 05 56	620 09 00	ES 4.33.1
3127 (ALL)	183 (403)	1	A	ES 4.32.2.1	442 18 06 13-50 05 56	620 09 00	ES 4.33.1
3152 (ALL)	486 (1071)	1	B & C	ES 4.32.2.1	442 18 06 13-50 05 56	620 09 00	ES 4.33.1
3153 (ALL)	437 (963)	1	A	ES 4.32.2.1	442 18 06 13-50 05 56	620 09 00	ES 4.33.1
3170 (ALL N, CP) (CS-HT/MT, CT-HT/MT)	570 (1257)	1	B & C	ES 4.32.2.1	442 18 05 13-50 05 55	620 09 01	ES 4.33.1
3170 (CS-LT, CT-LT)	660 (1455)	1	B & C	ES 4.32.2.1	442 18 05 13-50 05 55	620 09 01	ES 4.33.1
3171 (ALL)	525 91157)	1	B & C	ES 4.32.2.1	442 18 05 13-50 05 55	620 09 01	ES 4.33.1
3201 (CP-SH/HT/MT, NP, NS)	588 (1296)	1	B & C	ES 4.32.2.1	442 18 05 13-50 05 55	620 09 01	ES 4.33.1
3201 (CP-LT,CS-LT,CT-SH,NT)	779 (1717)	1	B & C	ES 4.32.2.1	442 18 05 13-50 05 55	620 09 01	ES 4.33.1
3202 (NP-HT/MT, NS-HT)	580 (1279)	1	B & C	ES 4.32.2.1	442 18 05 13-50 05 55	620 09 01	ES 4.33.1
3202 (NP-LT, NS-MT/LT, NT)	1030 (2271)	1	D	ES 4.32.2.1	442 18 05 13-50 05 55	620 09 01	ES 4.33.1
3300 (REGULAR) (CP,CS,NP,NS ALL)	1176 (2593)	2	A	ES 4.32.2.1	442 18 05 13-50 05 55	620 09 01	ES 4.33.1
3300 (REGULAR) (CT-HT)	1043 (2299)	2	A	ES 4.32.2.1	442 18 05 13-50 05 55	620 09 01	ES 4.33.1
3300 (REGULAR) (CT-MT/LT, NT-MT/LT)	1340 (2954)	2	A	ES 4.32.2.1	442 18 05 13-50 05 55	620 09 01	ES 4.33.1
3301 (NP-HT/MT,NP-LT 300) (NT-HT/MT,NS)	1110 (2447)	1	D	ES 4.32.2.1	442 18 05 13-50 05 55	620 09 01	ES 4.33.1
3301 (NP-LT 350,NT-LT)	1360 (2998)	1	D	ES 4.32.2.1	442 18 05 13-50 05 55	620 09 01	ES 4.33.1

- SEE DOCUMENT NO. ES 4.32.2.1 AND ES 4.33.1 FOR PART NUMBER SELECTION.
- FOR PUMPS NOT LISTED PLEASE CONSULT THE TECHNICAL DEPARTMENT.



Station Equipment

DOCUMENT NO.

ES 4.40.4.1

Stainless Steel Lifting System Selection Chart Large Pumps c/w Drive Unit in P Installation

SUPERSEDES

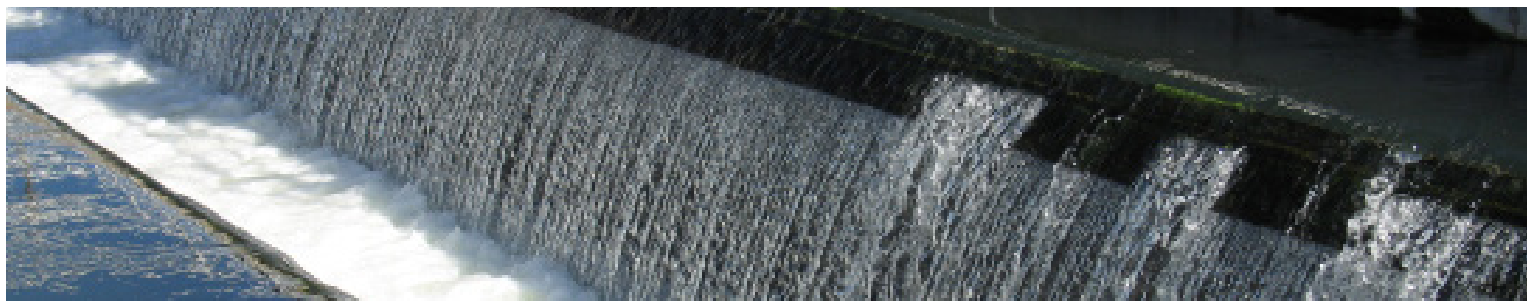
10/2010

ISSUED

11/2012

PUMP MODEL NO. (DRIVE UNIT)	MAXIMUM PUMP WEIGHT	CHAIN LIFTING ARRANGEMENT		
	kg (lbs)	TYPE	CATEGORY	DOC. NO.
3231 6X5	1410 (3109)	1	D	ES 4.32.2.1
3231 7X5 NOTE 1	1910 (4211)	N/A		
3231 7X5 NOTE 2	2060 (4542)	N/A		
3240 8xx	3900 (8598)	N/A		
3300 6X5	1490 (3285)	2	A	ES 4.32.2.1
3301 8X0	1085 (2392)	1	D	ES 4.32.2.1
3306 6X5	1440 (3175)	2	A	ES 4.32.2.1
3306 7X5 NOTE 1	1940 (4277)	N/A		
3306 7X5 NOTE 2	1920 (4233)	N/A		
3312 7X5 NOTE 1	2100 (4630)	N/A		
3312 7X5 NOTE 2	2250 (4960)	N/A		
3312 8XX	3850 (8488)	N/A		
3315	1530 (3373)	1	E	ES 4.32.2.1
3351 9XX	6800 (14991)	N/A		
3356 6X5	1550 (3417)	2	A	ES 4.32.2.1
3356 7X5 NOTE 1	2050 (4519)	N/A		
3356 7X5 NOTE 2	2200 (4850)	N/A		
3400 7X5 NOTE 1	2850 (6283)	N/A		
3400 7X5 NOTE 2	3000 (6614)	N/A		
3400 8XX	4650 (10,251)	N/A		
3501 7XX NOTE 1	3250 (7165)	N/A		
3501 7XX NOTE 2	3400 (7496)	N/A		
3501 8XX	5030 (11,089)	N/A		
3531 7X5 NOTE 1	3200 (7055)	N/A		
3531 7X5 NOTE 2	3350 (7385)	N/A		
3531 8XX	5000 (11,023)	N/A		
3531 9XX	6850 (15,102)	N/A		
3602 7X5 NOTE 1	3850 (8488)	N/A		
3602 7X5 NOTE 2	4000 (8818)	N/A		
3602 8XX	5650 (12,456)	N/A		
3602 9XX	6400 (14,109)	N/A		
3800 9XX	10,400 (22,928)	N/A		

NOTE:
 1- DRIVE WITH SMALL CONNECTION HOUSING FOR CABLE UP TO 70mm SQ.
 2- DRIVE WITH LARGE CONNECTION HOUSING FOR CABLE 95-120mm SQ.
 - FOR PUMPS NOT LISTED PLEASE CONSULT THE TECHNICAL DEPARTMENT.
 - N/A : NOT AVAILABLE, NO S/S ACCESSORIES (LARGE ENOUGH TO ACCEPT LIFTING
 HANDLE SIZE & PUMP WEIGHT) APPROVED FOR OVERHEAD LIFTING, EXIST ON MARKET.



Expert 3400

HYDROSTATIC LEVEL TRANSMITTER

2.77

General

MJK Expert™ 3400 Hydrostatic level transmitter has a ceramic capacitive measuring system and built-in breather tube for direct air pressure compensation. It is designed for use in waterworks, sewage treatment plants and industrial applications where accurate and long term stable level measuring is required. It is very robust and designed for applications in rough environments with aggressive chemicals and aggressive fluids.

A major design feature is the programming facility using a USB interface and a laptop, the servicing transmitter can be programmed on-the-fly to the required measuring range. The design without sharp edges prevents particles, textiles and paper from sticking to the housing or the diaphragm.

Features

- Ceramic diaphragm, PPS body,
- Surge resistance
- 2-wire 4-20 mA loop powered, 10-30VDC
- Accuracy better than $\pm 0.1\%$ and stability 0.1 % per year
- Field re-programmable for custom ranges and zero offset calibration
- Steel re-inforced PUR cable with high tensile strength (2000 lb)
- Measuring ranges from 0-1 ft. to 0-1000 ft.

Applications

- Waste water
- Process water
- Chemical tanks with a range of acids, bases and ferric chloride.
- Approved for mounting in explosion hazardous areas
- **UL Class 1, Division 1, Group A-D.**



DATASHEET

US 2.77 Expert 3400™ datasheet 2203



mjk
a xylem brand

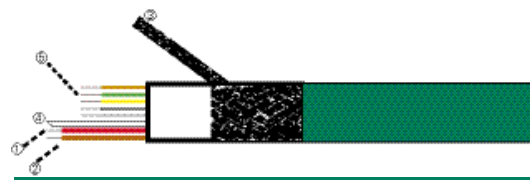
Expert™ 3400 Hydrostatic Level Transmitter

Specifications

Expert™ 3400 Level Transmitter							
Nominal Measuring Range	0-1 ft.	0-3 ft.	0-10 ft.	0-30 ft.	0-100 ft.	0-300 ft.	0-1000 ft.
Measuring principle	Ceramic capacitive, relative pressure					Sealed gate	
Min. programmable range	0-.9 ft.	0-.9 ft.	0-3 ft.	0-9 ft.	0-30 ft.	0-90 ft.	0-300 ft.
Max. programmable range	0-3 ft.	0-3 ft.	0-10 ft.	0-30 ft.	0-100 ft.	0-300 ft.	0-1000 ft.
Max. overpressure	75 psi	75 psi	100 psi	150 psi	275 psi	600 psi	600 psi
Temperature range	14 °F to 158 °F						
Temp. deviation, zero point	Better than $\pm 0.01\%$ / °F						
Temperature deviation, FS	Better than $\pm 0.005\%$ / °F						
Linearity / Stability	Better than $\pm 0.5\%$ FS / $\pm 0.5\%$ FS						
Measurement accuracy	Better than $\pm 0.1\%$ FS @ 50-85 °F Better than $\pm 0.2\%$ FS @ full temperature range						
Long time stability	Better than $\pm 0.1\%$ FS per year						
Materials	Housing: PPS (Ryton), Diaphragm: 99,9 % Al ₂ O ₃ Packing: Viton®						
Supply voltage	10-30 V DC (12-30 V DC for cable lengths above 315 ft.)						
Output signal	2-wire 4-20 mA (passive transmitter)						
Cable	2 x AWG 20 (pressure) + 5 x AWG 25 (data), shielded, PUR insulation						
Cable length	39 ft.	39 ft.	39 ft.	39 ft.	115 ft.	330 ft.	1030 ft.
Enclosure	NEMA 6P, withstands a static pressure equal to max. overpressure.						
Safety Approvals	UL® Class I, II and III, Division 1 Hazardous (Classified) Locations, UL 913 ATEX: II 2G EEx ia IIC T4...T6						
Test Approvals	CE directive: 89/336/EEC, 2004/108/EC, 1999/EC EN 61326-1, 61326-2-3, EN 55011, 61000-4-2, 61000-4-3, 61000-4-4, 61000-4-5, 61000-4-6, 61000-4-8 EN 61000-6-4 2007-02-19, EN 61000-6-2 2005-09-08,						

① The cable length may be extended by using the transmitter connection box (202922) and twisted shielded pair wire.

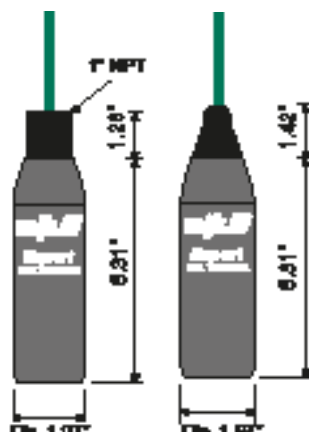
Electrical Connections



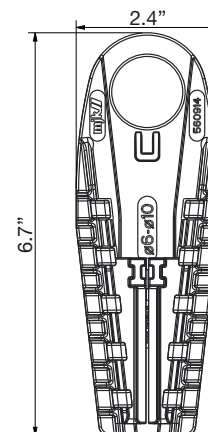
Designations		
1	Red	Positive (+) wire
2	Brown	Negative (-) wire
3	Black	Shield (NOT signal ground!)
4	Tube	Air pressure compensation tube and Moisture filter
5	Hidden	Programming wires

⑤ Programming	
Green	CS (Chip Select)
Brown	GND
Yellow	Data In
White	Clk
Grey	Data Out

Dimensions



MJK Expert™ 3400 level transmitter



Mounting bracket for MJK Expert™ 3400 level transmitter

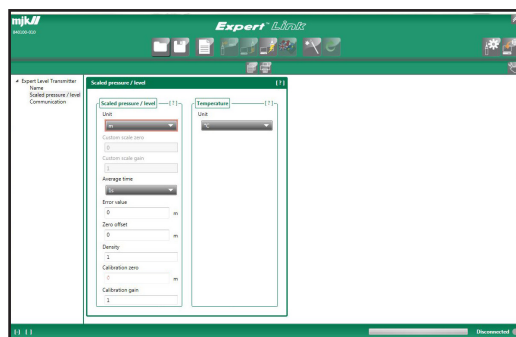
Expert™ 3400 Hydrostatic Level Transmitter

Accessories



Expert Link is free of charge.

Link to download center on www.mjk.com.



Expert Link Software

840100 Expert Link for Windows for support of the Expert series. Use with 203990 PC (USB) Interface. Information (pressure- and temperature range, version of software etc.):

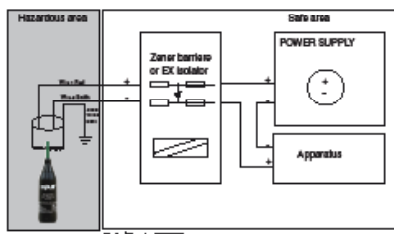
- Indication of actual pressure value and adjusting measuring ranges
- Selection of units
- Setting of a new zero and gain for the transmitter
- Reprogramming of the analog output (i.e. different unit, other pressure range)
- Setting of the instrument address (for Bus-operation and real-time value).



203992 PC Interface used for connecting Expert Hydrostatic Level Transmitters to a PC using Expert Link.



202993 Isolator, an integrated unit including active components with low power consumption. A power supply provides power to the pressure transmitter through a galvanic separator so that neither current nor voltage exceeds the specified values. A ground connection is not required.



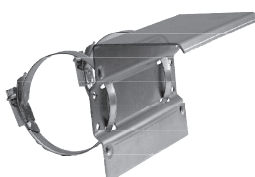
202995 Zener barrier, a passive unit requiring a safe grounding connection based on ATEX regulations. The Zener barrier contains a number of Zener diodes for voltage protection. A safety fuse will blow if the current exceeds the limits. A Zener barrier requires a safe ground connection.



202922 Transmitter connection box NEMA 6 enclosure for a safe connection of 4-20 mA cables.®



200126 Field indicator 531, display insert, 4-20mA loop powered, NEMA 6 enclosure. For instant local readout.®



200117 Rainroof/Sun Shield for 531, weather protection for display and connection box.

® For a complete display you need both the 20016 531 display insert and the 202922 transmitter connection box.

Expert™ 3400 Hydrostatic Level Transmitter

Order Numbers

Expert™ 3400, Hydrostatic Level Transmitter			
		With 1" NPT Thread	
MJK209921	Pressure Transmitter 3400, range 0-1 ft., 39 ft. cable	MJK209991	Pressure Transmitter 3400, range 0-1 ft., 39 ft. cable, NPT thread
MJK209922	Pressure Transmitter 3400, range 0-3 ft., 39 ft. cable	MJK209992	Pressure Transmitter 3400, range 0-3 ft., 39 ft. cable, NPT thread
MJK209923	Pressure Transmitter 3400, range 0-10 ft., 39 ft. cable	MJK209993	Pressure Transmitter 3400, range 0-10 ft., 39 ft. cable, NPT thread
MJK209923-50	Pressure Transmitter 3400, range 0-10 ft., 50 ft. cable	MJK209993-75	Pressure Transmitter 3400, range 0-10 ft., 75 ft. cable, NPT thread
MJK209923-100	Pressure Transmitter 3400, range 0-10 ft., 100 ft. cable	MJK209993-100	Pressure Transmitter 3400, range 0-10 ft., 100 ft. cable, NPT thread
MJK209923-125	Pressure Transmitter 3400, range 0-10 ft., 125 ft. cable	MJK209993-250	Pressure Transmitter 3400, range 0-10 ft., 250 ft. cable, NPT thread
MJK209925	Pressure Transmitter 3400, range 0-30 ft., 39 ft. cable	MJK209995	Pressure Transmitter 3400, range 0-30 ft., 39 ft. cable, NPT thread
MJK209925-50	Pressure Transmitter 3400, range 0-30 ft., 50 ft. cable	MJK209995-75	Pressure Transmitter 3400, range 0-30 ft., 75 ft. cable, NPT thread
MJK209925-75	Pressure Transmitter 3400, range 0-30 ft., 75 ft. cable	MJK209995-100	Pressure Transmitter 3400, range 0-30 ft., 100 ft. cable, NPT thread
MJK209925-100	Pressure Transmitter 3400, range 0-30 ft., 100 ft. cable	MJK209996	Pressure Transmitter 3400, range 0-100 ft., 105 ft. cable, NPT thread
MJK209925-125	Pressure Transmitter 3400, range 0-30 ft., 125 ft. cable		
MJK209925-150	Pressure Transmitter 3400, range 0-30 ft., 150 ft. cable		
MJK209925-300	Pressure Transmitter 3400, range 0-30 ft., 300 ft. cable		
MJK209925-450	Pressure Transmitter 3400, range 0-30 ft., 450 ft. cable		
MJK209926	Pressure Transmitter 3400, range 0-100 ft., 115 ft. cable		
MJK209926-150	Pressure Transmitter 3400, range 0-100 ft., 150 ft. cable		
MJK209927	Pressure Transmitter 3400, range 0-300 ft., 330 ft. cable		
MJK209928	Pressure Transmitter 3400, range 0-1000 ft., 1030 ft. cable		

Order Numbers Accessories

Expert™ 3400, Hydrostatic Level Transmitter Accessories	
MJK200117	Rainroof/Sun Shield for 531, display and connection box
MJK200126	Field indicator 531, display insert, 4-20mA loop powered
MJK202922	Transmitter connection box
MJK202993	Loop supplied Ex-isolator for DIN rail mounting
MJK202995	Zener barrier for hydrostatic transmitter 9002/13-280-110-00
MJK203990	PC Interface (USB) for Expert™ Transmitter 1400/3400
MJK209821	1" NPT thread for Expert 2100, 3400
MJK560914	Plastic Mounting Bracket for Expert™ Transmitter, cable Ø6-10mm
MJK691075-FT	Communication Cable 2x2x0.5mm2 (AWG24) shielded twisted pair

For support in North Americas,
please contact:

YSI
1700/1725 Brannum Lane
Yellow Springs, Ohio 45387
Tel. +1 937 767 7241 x 768

environmental@ysi.com
www.YSI.com



Flygt ENM-10 liquid level regulators

TRIED-AND-TRUE SIMPLICITY



So simple, yet so reliable

Proper level regulation is critical for operational reliability. Lack of proper regulation can cause improper operation, pump damage or, in the worst case, overflow that results in environmental and economic consequences. With over a decade of experience and countless satisfied users, the natural choice for level regulation is the Flygt ENM-10.

Simple, reliable and effective

The Flygt ENM-10 level regulators are the ideal choice for most level control applications, such as wastewater pumping stations and ground water or drainage pumping. When the liquid level reaches the regulator, the bulb tilts, activating the internal microswitch, which starts or stops a pump or triggers an alarm device.

Practically maintenance free

The outer casing of Flygt ENM-10 level regulators are resistant to most aggressive liquids. The cable material prevents buildup of deposits and other impurities. Rather than floating on the surface, ENM-10 hangs immersed in the liquid. This prevents the cables from tangling when several regulators are used.

Xylem testing indicates Flygt ENM-10 level regulators last several times longer than most standard level regulators. This ensures reliable operation and a low degree of maintenance, thereby keeping service costs down to a minimum over the life cycle.

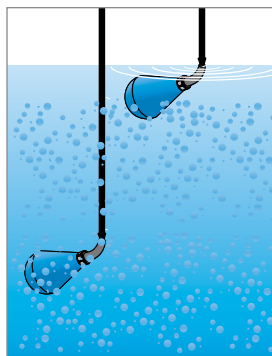
Environmentally friendly

The Flygt ENM-10 level regulator contains no harmful or toxic materials such as lead or mercury and does not require the use of any adhesives. All plastic components are welded and screwed together.

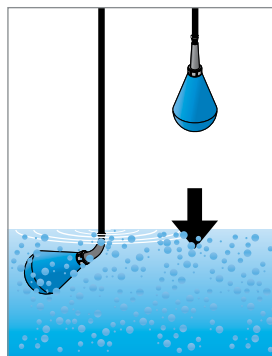


In a two-pump station, four level regulators are generally used. One is positioned at the stop level, one at each pump's start level, and the fourth regulator is used to trigger a high level alarm.

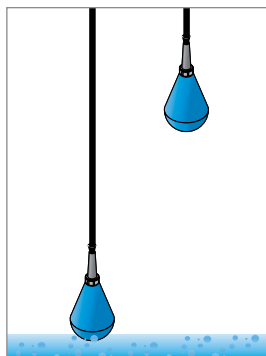
Operating principle



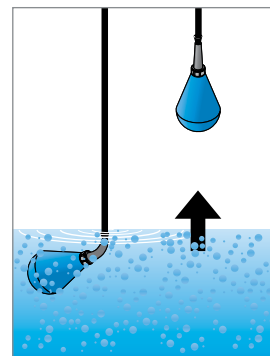
1. Media to be pumped reaches the pre-set level and tilts the upper regulator, which signals the start of the pumping process.



2. The pumping process drains the media.



3. The pumping process stops when the lower regulator returns to the vertical position and signals a stop.



4. The media level begins to rise, starting the process again.

With a variety of installations in mind

The Flygt liquid level regulator comes in two main versions, standard and Ex-approved. Depending upon the media and type of installation, the Flygt ENM-10 is available for a broad range of liquid densities and with many cable lengths from which to choose.

Features



Watertight cable

Encases and protects power supply to the microswitch.

Operational mechanism

Initiates pump start and pump stop through the microswitch when the Flygt ENM-10 tilts or resumes its original position. Delivers reliable performance under all operating conditions including the slow movements often found in stormwater or rainwater installations.

Bending relief (protective sleeve)

Enables watertight, oil-resistant cable entry.

Microswitch

Provides proven quality, performance and reliability for all operating conditions, including switching low currents.

Accessories



Flygt ENM-10 cable holder

Dedicated accessory that reduces the cable wear. Sold separately or as package with the Flygt ENM-10.

Flygt Ex safety barrier

This two-channel safety barrier protects against overvoltage for liquid level regulators. Required when Flygt ENM-10s are to be used in Ex-installations.

Technical data

	Standard version	Ex-approved version
Cable lengths	6, 13, 20, 30 or 50 m* (20, 42, 65, 100 or 167 ft.)	6, 13, or 20 m (20, 42 or 65 ft.)
Liquid density	Between 0.95 and 1.10 g/cm ³ *	Between 0.95 and 1.10 g/cm ³
Materials	Body: Polypropylene Bending relief: EPDM rubber Cable: Special compound PVC & NBR/PVC	Body: Conductive polypropylene/carbon black Bending relief: NBR/PVC rubber Cable: NBR/PVC
Liquid temperature	Min. 0°C (32°F) Max. 60°C (140°F)	Min. 0°C (32°F) Max. 60°C (140°F)
Ingress protection	IP 68	IP 68
Electrical range (microswitch)	Interrupting capacity: AC: 250 V/10 A resistive load AC: 250 V/3 A inductive load at $\cos\phi = 0.5$ DC: 24 V/ 10mA min., 6 A max. Note that local regulations may limit the voltage.	Interrupting capacity: AC: 250 V/10 A resistive load AC: 250 V/3 A inductive load at $\cos\phi = 0.5$ DC: 24 V/ 10mA min., 6 A max. Note that local regulations may limit the voltage.
Approvals	CE, CSA, SEMKO, NEMKO, DEMKO	CE, SEMKO, NEMKO, DEMKO, ATEX/IECEX

*Other cable lengths and liquid densities available upon request.

The Flygt ENM-10 is a part of the Level Sensor product range from Xylem, covering most level measuring needs.

Xylem ['zīləm]

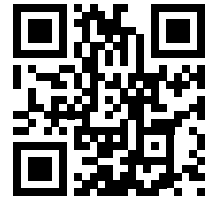
- 1) The tissue in plants that brings water upward from the roots
- 2) A leading global water technology company

We're 12,000 people unified in a common purpose: creating innovative solutions to meet our world's water needs. Developing new technologies that will improve the way water is used, conserved, and re-used in the future is central to our work. We move, treat, analyze, and return water to the environment, and we help people use water efficiently, in their homes, buildings, factories and farms. In more than 150 countries, we have strong, long-standing relationships with customers who know us for our powerful combination of leading product brands and applications expertise, backed by a legacy of innovation.

For more information on how Xylem can help you, go to xyleminc.com

Installation, Operation, and
Maintenance Manual

90050501_1.0



Flygt 3085

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1 Introduction and safety

1.1 Introduction

EN

Purpose of the manual

The purpose of this manual is to provide the necessary information for working with the unit. Read this manual carefully before starting work.

Read and keep the manual

Save this manual for future reference, and keep it readily available at the location of the unit.

Intended use



WARNING:

Operating, installing, or maintaining the unit in any way that is not covered in this manual could cause death, serious personal injury, or damage to the equipment and the surroundings. This includes any modification to the equipment or use of parts not provided by Xylem. If there is a question regarding the intended use of the equipment, please contact a Xylem representative before proceeding.

Other manuals

See also the safety requirements and information in the original manufacturer's manuals for any other equipment furnished separately for use in this system.

1.2 Safety terminology and symbols

About safety messages

It is extremely important that you read, understand, and follow the safety messages and regulations carefully before handling the product. They are published to help prevent these hazards:

- Personal accidents and health problems
- Damage to the product and its surroundings
- Product malfunction

Hazard levels

Hazard level	Indication
 DANGER:	A hazardous situation which, if not avoided, will result in death or serious injury
 WARNING:	A hazardous situation which, if not avoided, could result in death or serious injury
 CAUTION:	A hazardous situation which, if not avoided, could result in minor or moderate injury
NOTICE:	Notices are used when there is a risk of equipment damage or decreased performance, but not personal injury.

Special symbols

Some hazard categories have specific symbols, as shown in the following table.

Electrical hazard	Magnetic fields hazard
 Electrical Hazard:	 CAUTION:

1.3 User safety

All regulations, codes, and health and safety directives must be observed.

The site

- Observe lockout and tagout procedures before starting work on the product, such as transportation, installation, maintenance, or service.
- Pay attention to the risks presented by gas and vapors in the work area.
- Always be aware of the area surrounding the equipment, and any hazards posed by the site or nearby equipment.

Qualified personnel

This product must be installed, operated, and maintained by qualified personnel only.

Protective equipment and safety devices

- Use personal protective equipment as needed. Examples of personal protective equipment include, but are not limited to, hard hats, safety goggles, protective gloves and shoes, and breathing equipment.
- Make sure that all safety features on the product are functioning and in use at all times when the unit is being operated.

1.4 Ex-approved products

Follow these special handling instructions if you have an Ex-approved unit.

Personnel requirements

These are the personnel requirements for Ex-approved products in potentially explosive atmospheres:

- All work on the product must be carried out by certified electricians and Xylem authorized mechanics. Special rules apply to installations in explosive atmospheres.
- All users must know about the risks of electric current and the chemical and physical characteristics of the gas, the vapor, or both present in hazardous areas.
- Any maintenance for Ex-approved products must conform to international and national standards (for example, IEC/EN 60079-17).

Xylem disclaims all responsibility for work done by untrained and unauthorized personnel.

Product and product handling requirements

These are the product and product handling requirements for Ex-approved products in potentially explosive atmospheres:

- Only use the product in accordance with the approved motor data.
- The equipment must never run dry during operation. The volute must be filled with liquid during operation. Dry running during service and inspection is only permitted outside the classified area.
- Before you start work on the product, make sure that the product and the control panel are isolated from the power supply and the control circuit, so they cannot be energized.
- Do not open the product while it is energized or in an explosive gas atmosphere.

- Intrinsically safe circuits are normally required for the automatic level-control system by the level regulator if mounted in zone 0.
- The yield stress of fasteners must be in accordance with the approval drawing and the product specification.
- Do not modify the equipment without approval from an Ex-approved Xylem representative.
- Only use original Xylem spare parts that are provided by an Ex-approved Xylem representative.
- The thermal contacts that are fitted to the stator windings must be connected correctly to a separate motor control circuit and in use. The thermal contacts shall be connected to a monitoring device, which disconnects the power supply immediately upon activation. This action prevents the rise of temperatures above the temperature value for the approval classification.
- The width of flameproof joints is more than the values specified in the tables of the EN/IEC 60079-1 standard. For information contact the manufacturer.
- The gap of flameproof joints is less than the values specified in Table 2 of the EN/IEC 60079-1 standard. For information contact the manufacturer.
- It is NOT allowed to repair the flameproof joints.
- Ambient temperature: -20°C to 60°C

Guidelines for compliance

Compliance is fulfilled only when you operate the unit within its intended use. Do not change the conditions of the service without the approval of an Ex-approved Xylem representative. When you install or maintain explosion proof products, always comply with the directive and applicable standards (for example, IEC/EN 60079–14).

Monitoring equipment

For additional safety, use condition-monitoring devices. Examples of condition-monitoring devices include, but are not limited to, the following:

- Level indicators
- Temperature detectors in addition to the stator thermal detectors

Any thermal detectors or thermal protection devices delivered with the pump must be installed and in use at all times.

The site owner is responsible for selection, installation, and proper maintenance of functional monitoring equipment for motor protection.

1.5 Special hazards

1.5.1 Biological hazards

The product is designed for use in liquids that can be hazardous to your health. Observe these rules when you work with the product:

- Make sure that all personnel who may come into contact with biological hazards are vaccinated against diseases to which they may be exposed.
- Observe strict personal cleanliness.



WARNING: Biological Hazard

Infection risk. Rinse the unit thoroughly with clean water before working on it.

1.5.2 Wash the skin and eyes

Follow these procedures for chemicals or hazardous fluids that have come into contact with your eyes or your skin:

Condition	Action
Chemicals or hazardous fluids in eyes	1. Hold your eyelids apart forcibly with your fingers. 2. Rinse the eyes with eyewash or running water for at least 15 minutes. 3. Seek medical attention.
Chemicals or hazardous fluids on skin	1. Remove contaminated clothing. 2. Wash the skin with soap and water for at least 1 minute. 3. Seek medical attention, if necessary.

1.6 Protecting the environment

Emissions and waste disposal

Observe the local regulations and codes regarding:

- Reporting of emissions to the appropriate authorities
- Sorting, recycling and disposal of solid or liquid waste
- Clean-up of spills

Exceptional sites



CAUTION: Radiation Hazard

Do NOT send the product to Xylem if it has been exposed to nuclear radiation, unless Xylem has been informed and appropriate actions have been agreed upon.

1.7 Substances of very high concern

This product contains the following substances of very high concern according to REACH article 33:

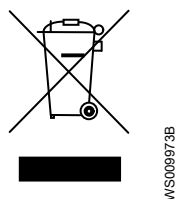
- Lead

1.8 End-of-life product disposal

Handle and dispose of all waste in compliance with local laws and regulations.

EU and UK only: Correct disposal of this product — waste electrical and electronic equipment

- EU: Directive 2012/19/EU on waste electrical and electronic equipment (WEEE)
- UK: SI 2013 No. 3113



This marking on the product, accessories or literature indicates that the product should not be disposed of with other waste at the end of its working life.

1.9 Spare parts



CAUTION:

Only use the manufacturer's original spare parts to replace any worn or faulty components. The use of unsuitable spare parts may cause malfunctions, damage, and injuries as well as void the warranty.

1.10 Warranty

For information about warranty, see the sales contract.

2 Transportation and storage

2.1 Examine the delivery

EN

2.1.1 Examine the package

1. Examine the package for damaged or missing items upon delivery.
2. Record any damaged or missing items on the receipt and freight bill.
3. If anything is out of order, then file a claim with the shipping company.
If the product has been picked up at a distributor, make a claim directly to the distributor.

2.1.2 Examine the unit

1. Remove packing materials from the product.
Dispose of all packing materials in accordance with local regulations.
2. To determine whether any parts have been damaged or are missing, examine the product.
3. If applicable, unfasten the product by removing any screws, bolts, or straps.
Use care around nails and straps.
4. If there is any issue, then contact a sales representative.

2.2 Transportation guidelines

2.2.1 Precautions



DANGER: Crush Hazard

Moving parts can entangle or crush. Always disconnect and lock out power before servicing to prevent unexpected startup. Failure to do so could result in death or serious injury.



Position and fastening

The unit can be transported either horizontally or vertically. Make sure that the unit is correctly fastened during transportation, and cannot roll or fall over.

2.2.2 Lifting

Always inspect the lifting equipment and tackle before starting any work.



WARNING: Crush Hazard

Always lift the unit by its designated lifting points.
Use suitable lifting equipment and ensure that the product is properly harnessed.
Wear personal protective equipment.
Stay clear of cables and suspended loads.

NOTICE:

Never lift the unit by its cables or hose.

Lifting points

The following images show the lifting points for the different installation types. For more information about the installation types, see [Installation](#) on page 18.

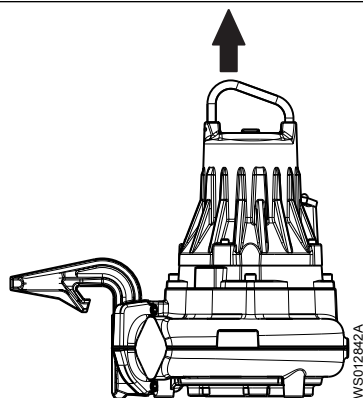


Figure 1: P-, T-, and X-installation

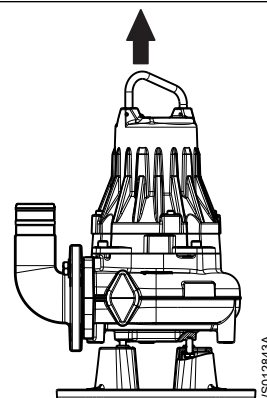


Figure 2: S- and F-installation

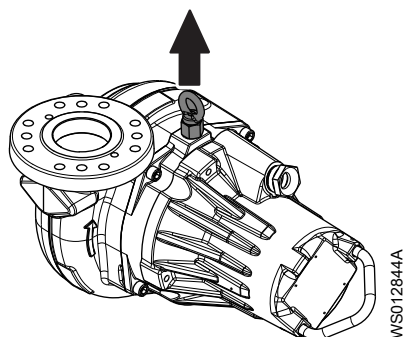


Figure 3: Z-installation, with hydraulic unit

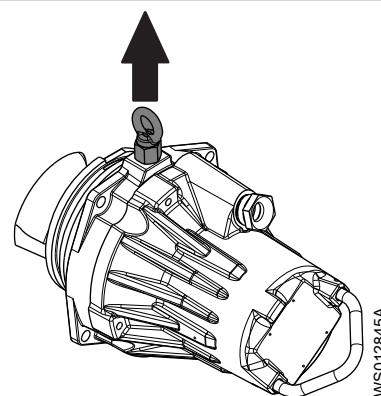


Figure 4: Z-installation, without hydraulic unit

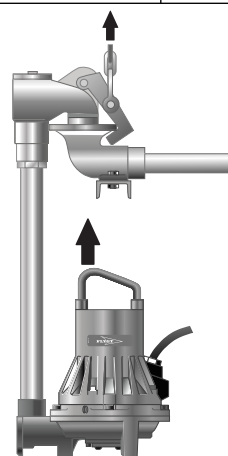


Figure 5: H-installation

Lifting equipment

Lifting equipment is always required to handle the unit. The lifting equipment must fulfill the following requirements:

- The minimum height between the lifting hook and the floor must be sufficient to lift the unit. Contact a Xylem representative for more information.
- The lifting equipment must be able to hoist the unit straight up and down, preferably without the need for resetting the lifting hook.
- The lifting equipment must be correctly anchored and in good condition.
- The lifting equipment must support the weight of the entire assembly. Only authorized personnel may use the lifting equipment.

- The lifting equipment must be dimensioned to lift the unit with any remaining pumped media in it.
- The lifting equipment must not be oversized.

**CAUTION: Crush Hazard**

Improperly-dimensioned lifting equipment can lead to injury. A site-specific risk analysis must be done.

EN

2.3 Temperature ranges for transportation, handling and storage

Handling at freezing temperature

At temperatures below freezing, the product and all installation equipment, including the lifting gear, must be handled with extreme care.

Make sure that the product is warmed up to a temperature above the freezing point before starting up. Avoid rotating the impeller/propeller by hand at temperatures below the freezing point. The recommended method to warm the unit up is to submerge it in the liquid which will be pumped or mixed.

NOTICE:

Never use a naked flame to thaw the unit.

Unit in as-delivered condition

If the unit is still in the condition in which it left the factory - all packing materials are undisturbed - then the acceptable temperature range during transportation, handling and storage is: -50°C (-58°F) to $+60^{\circ}\text{C}$ ($+140^{\circ}\text{F}$).

If the unit has been exposed to freezing temperatures, then allow it to reach the ambient temperature of the sump before operating.

Lifting the unit out of liquid

The unit is normally protected from freezing while operating or immersed in liquid, but the impeller/propeller and the shaft seal may freeze if the unit is lifted out of the liquid into a surrounding temperature below freezing.

Follow these guidelines to avoid freezing damage:

1. Empty all pumped liquid, if applicable.
2. Check all liquids used for lubrication or cooling, both oil and water-glycol mixtures, for the presence of unacceptable amounts of water. Change if needed.

Water-glycol mixtures: Units equipped with an internal closed-loop cooling system are filled with a mixture of water and 30% glycol. This mixture remains a flowing liquid at temperatures down to -13°C (9°F). Below -13°C (9°F), the viscosity increases such that the glycol mixture will lose its flow properties. However, the glycol-water mixture will not solidify completely and thus cannot harm the product.

2.4 Storage guidelines

Storage location

The product must be stored in a covered and dry location free from heat, dirt, and vibrations.

NOTICE:

Protect the product against humidity, heat sources, and mechanical damage.

NOTICE:

Do not place heavy weights on the packed product.

Long-term storage

If the unit is stored for more than six months, then the following apply:

- Before operating the unit after storage, it must be inspected. Special attention must be given to the seals and the cable entry.
- The impeller or propeller must be rotated every other month to prevent the seals from sticking together.

Packaging material stacking limit

If the packaging material has an indicated stacking limit, then it is valid for 23°C (73°F) and 50% relative humidity. Depending on the material, other temperature and humidity ranges can reduce the stacking limit.

3 Product description

Products included

EN

Pump	Non-explosion proof drive unit	Explosion proof drive unit	High Efficiency motor (LSPM)	D-hydraulic	F-hydraulic	M-hydraulic (Grinder)	Adaptive N™ hydraulic
3085.060	X						X ¹
3085.070		X					X ¹
3085.092		X		X	X		
3085.160	X						X ²
3085.172	X					X	
3085.183	X			X	X		
3085.190		X					X ²
3085.891		X				X	
3085.900	X		X				X ²
3085.910		X	X				X ²
3085.920	X		X				X ¹
3085.930		X	X				X ¹
¹ Hard-Iron™							
² Cast iron, gray							

Pump-specific information

For the specific weight, current, voltage, power ratings, and speed of the pump, see the data plate of the pump.

3.1 Pump design

The pump is submersible, and driven by an electric motor.

Intended use

The product is intended for moving wastewater, sludge, raw and clean water. Always follow the limits that are given in [Technical reference](#) on page 73. If there is a question regarding the intended use of the equipment, please contact a local sales and service representative before proceeding.



DANGER: Explosion/Fire Hazard

Special rules apply to installations in explosive or flammable atmospheres. Do not install the product or any auxiliary equipment in an explosive zone unless it is rated explosion-proof or intrinsically-safe. If the product is rated explosion-proof or intrinsically-safe, then see the specific explosion-proof information in the safety chapter before taking any further actions.

NOTICE:

Do NOT use the unit in highly corrosive liquids.

Spare parts

- Modifications to the unit or installation should only be carried out after consulting with Xylem.
- Original spare parts and accessories that are authorized by Xylem are essential for compliance. The use of other parts can invalidate any claims for warranty or compensation. For more information contact your Xylem representative.

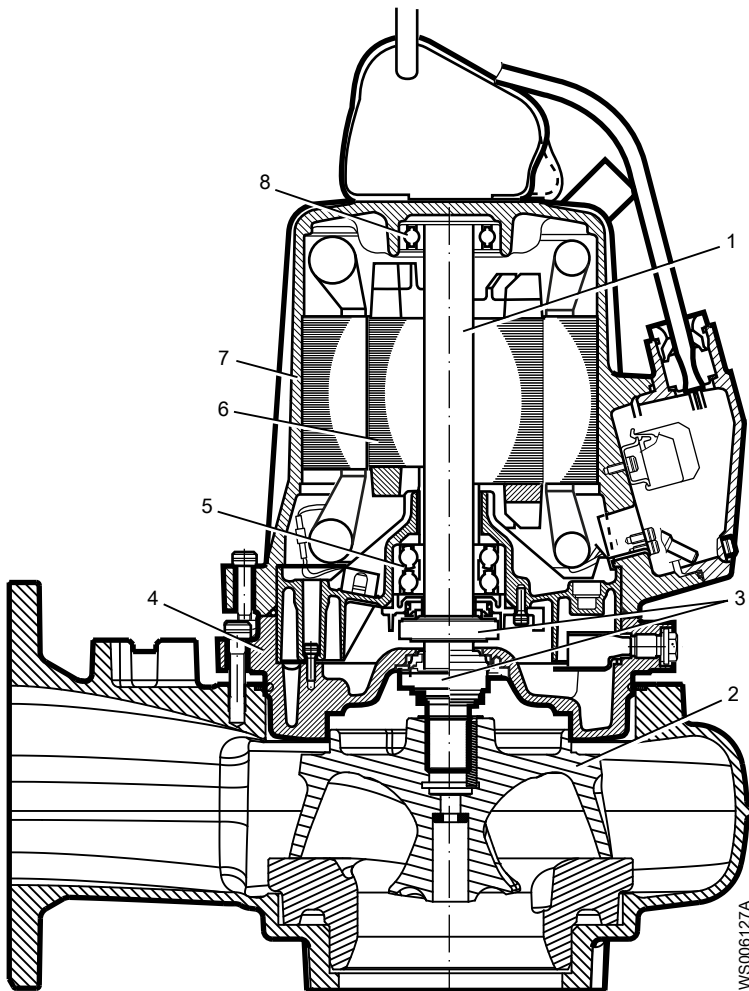
Pressure class

LT	Low head
MT	Medium head
HT	High head
SH	Super high head

Exterior product concept

Exterior is a product concept including N-technology, Premium efficiency motor, and the intelligent control SmartRun.

Parts



Position	Denomination	Description
1	Shaft	The shaft is made of stainless steel with an integrated rotor.
2	Impeller	There are multiple types of impellers. For information about the pumps impellers, see Parts List.

Position	Denomination	Description
3	Mechanical seals	One inner and one outer seal in a combination of materials: - Aluminium oxide Al_2O_3 - Silicon carbide $RSiC$ - Corrosion-resistant cemented carbide WCCR For information about the pumps mechanical seals, see Parts List.
4	Oil housing	The oil housing includes a coolant that lubricates and cools the seals; the housing acts as a buffer between the pumped fluid and the drive unit.
5	Main bearing	The bearing consisting of a two-row angular contact ball bearing.
6	Motor	For information about the motor, see Technical reference on page 73.
7	Stator housing	The pump is cooled by the ambient liquid/air.
8	Support bearing	The bearing consisting of a single-row ball bearing.

3.2 Sensors and monitoring equipment

The following applies to the monitoring equipment of the pump:

- The stator incorporates three thermal contacts connected in series that activate the alarm and stops the pump at overtemperature
- The thermal contacts open at 125°C (257°F).
- Ex-approved pumps must have thermal contacts connected to the control panel.
- The sensors must be connected to either the MiniCAS II monitoring equipment or an equivalent equipment.
- The monitoring equipment must be of a design that makes automatic restart impossible.
- Information in the junction box shows if the pump is equipped with optional sensors.

Optional sensors

FLS FLS is a miniature float switch for detection of liquid in the stator housing. Due to its design it is best suited for pumps in a vertical position. The FLS sensor is installed in the bottom of the stator housing.

CLS CLS is a sensor for detection of water in the oil housing. The sensor initiates an alarm when the oil contains approximately 35% water. The sensor is installed in the bearing housing/bearing holder with its sensing part in the oil housing. The CLS sensor is not applicable to Ex-approved pumps.

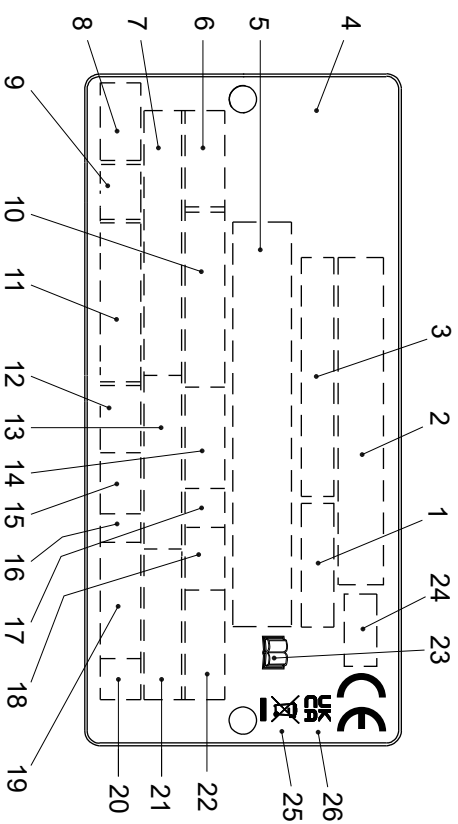
NOTICE:

The CLS sensor body is made of glass. Handle the sensor with care.

One CLS and one FLS sensor can be used in the same pump, if they are connected in parallel.

3.3 The data plate

The data plate is a metal label that is located on the main body of the products. The data plate lists key product specifications. Specially approved products also have an approval plate.



WS006257D

1. Curve code or Propeller code
2. Serial number
3. Product number
4. Country of origin
5. Additional information
6. Phase; type of current; frequency
7. Rated voltage
8. Thermal protection
9. Thermal class
10. Rated shaft power
11. International standard
12. Degree of protection
13. Rated current
14. Rated speed
15. Maximum submergence
16. Direction of rotation: L=left, R=right
17. Duty class
18. Duty factor
19. Product weight
20. Locked rotor code letter
21. Power factor
22. Maximum ambient temperature
23. Read installation manual
24. Notified body, only for EN-approved Ex products
25. WEEE-Directive symbol
26. UKCA marking

Figure 6: The data plate

3.4 Motor regulation

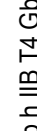
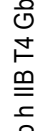
This product is submersible and therefore exempted from the motor efficiency requirement, in accordance with EU commission regulation 2019/1781 Article 2(2)(e).

3.5 Approvals

3.5.1 Product approvals for hazardous locations 3085

This table shows product approvals for the following products:

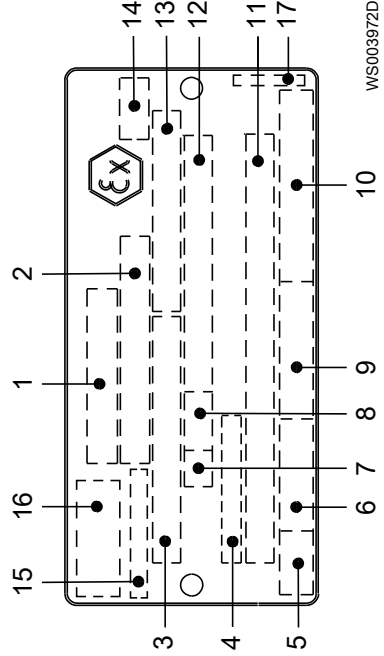
- 3085.070
- 3085.092
- 3085.190
- 3085.891
- 3085.910
- 3085.930

Type	Approval
European Norm (EN)	• ATEX Directive 2014/34/EU • EN IEC 60079-0:2018, EN 60079-1:2014, EN ISO 80079-36:2016, EN ISO 80079-37:2016 •  II 2 G Ex db h IIB T4 Gb
IEC	• IECEx scheme 02 • IEC 60079-0:2017, IEC 60079-1:2014-06, ISO 80079-36:2016, ISO 80079-37:2016 • Ex db h IIB T4 Gb
FM (FM Approvals)	• Explosion proof for use in Class I, Div. 1, Group C and D • Dust ignition proof for use in Class II, Div. 1, Group E, F and G • Suitable for use in Class III, Div. 1, Hazardous Locations
CSA Ex	• Explosion proof for use in Class I, Div. 1, Group C and D
UKEx	• UK SI 2016 No. 1107 • EN IEC 60079-0:2018, EN 60079-1:2014, EN ISO 80079-36:2016, EN ISO 80079-37:2016
	•  II 2 G Ex db h IIB T4 Gb

Protection mode h is insured by concept of protection c, as defined in clause 5 of standard ISO 80079-37.

EN approval plate

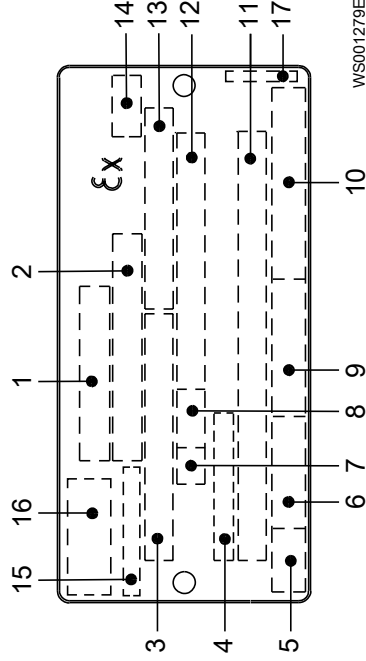
This illustration describes the EN approval plate and the information that is contained in its fields.



1. Approval number
2. Approval authority and Approval number
3. Approved drive unit
4. Cable entry temperature
5. Stall time
6. Starting current or Rated current
7. Duty class
8. Duty factor
9. Input power
10. Rated speed
11. Additional information
12. Maximum ambient temperature
13. Serial number
14. ATEX marking
15. Country of origin
16. Brand
17. Part number

IEC approval plate

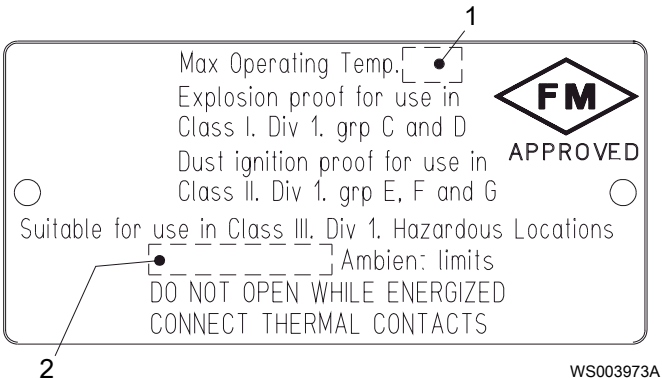
This illustration describes the IECEx approval plate and the information that is contained in its fields.



1. Approval number
2. Approval authority and Approval number
3. Approved drive unit
4. Cable entry temperature
5. Stall time
6. Starting current or Rated current
7. Duty class
8. Duty factor
9. Input power
10. Rated speed
11. Additional information
12. Maximum ambient temperature
13. Serial number
14. ATEX marking, not used for IECEx approved products
15. Country of origin
16. Brand
17. Part number

FM approval plate

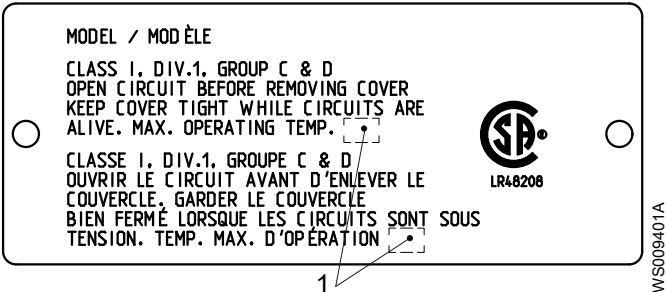
This illustration describes the FM approval plate and the information that is contained in its fields.



- 1. Temperature class
- 2. Maximum ambient temperature

CSA approval plate

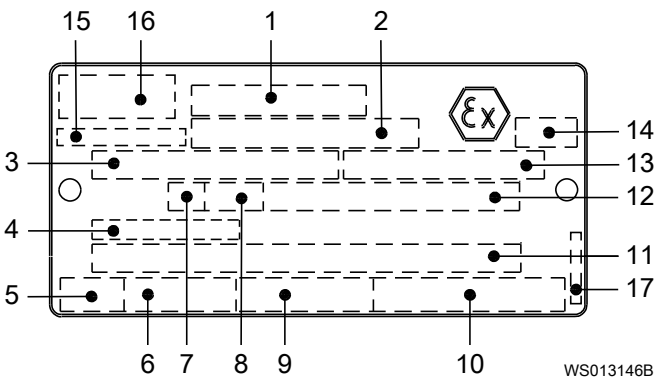
This illustration describes the CSA approval plate and the information that is contained in its fields.



- 1. Temperature class

United Kingdom: UKEx approval plate

This illustration describes the UKEx approval plate and the information that is contained in its fields.



- 1. Approval
- 2. Approval authority and Approval number
- 3. Approved drive unit
- 4. Cable entry temperature
- 5. Stall time
- 6. Starting current or Rated current
- 7. Duty class
- 8. Duty factor
- 9. Input power
- 10. Rated speed
- 11. Additional information
- 12. Maximum ambient temperature
- 13. Serial number
- 14. UKEx marking
- 15. Country of origin
- 16. Brand
- 17. Part number

3.6 Product denomination

Reading instruction

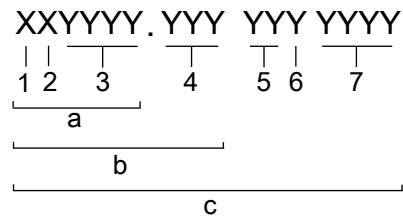
In this section, code characters are illustrated accordingly:

X = letter

Y = digit

The different types of codes are marked up with a, b and c. Code parameters are marked up with numbers.

Codes and parameters



WS006265B

EN

Type of Callout	Number	Indication
Type of code	a	Sales denomination
	b	Product code
	c	Serial number
Parameter	1	Hydraulic end
	2	Type of installation
	3	Sales code
	4	Version
	5	Production year
	6	Production cycle
	7	Running number

4 Installation

4.1 Precautions



Before starting work, make sure that the safety instructions have been read and understood.

DANGER: Electrical Hazard

Before starting work on the unit, make sure that the unit and the control panel are isolated from the power supply and cannot be energized. This applies to the control circuit as well.



DANGER: Inhalation Hazard

Before entering the work area, make sure that the atmosphere contains sufficient oxygen and no toxic gases.



WARNING: Electrical Hazard

The permanent magnet synchronous motor generates voltage on internal components when the shaft rotates, even if the power supply is disconnected. Make sure that the shaft cannot rotate if the unit is disassembled.

4.1.1 Hazardous atmospheres



DANGER: Explosion/Fire Hazard

Special rules apply to installations in explosive or flammable atmospheres. Do not install the product or any auxiliary equipment in an explosive zone unless it is rated explosion-proof or intrinsically-safe. If the product is rated explosion-proof or intrinsically-safe, then see the specific explosion-proof information in the safety chapter before taking any further actions.

General requirements

These requirements apply:

- Use the pump dimensional drawing in order to ensure proper installation.

Before installing the pump, do the following:

- Provide a suitable barrier around the work area, for example, a guard rail.
- Make sure that equipment is in place so that the unit cannot roll or fall over during the installation process.
- Check the explosion risk before you weld or use electric hand tools.
- Check that the cable and cable entry have not been damaged during transport.
- Always remove all debris and waste material from the sump, inlet piping, and discharge connection, before you install the pump.
- If the unit has a permanent magnet motor, then ensure that you have read and understood all safety instructions regarding permanent magnet motors.

NOTICE:

Do not run the pump dry.

NOTICE:

Never force piping to make a connection with a pump.

Authority regulation

Vent the tank of a sewage station in accordance with local plumbing codes.

Fasteners

- Only use fasteners of the correct size and material.
- Replace all corroded or damaged fasteners.
- Make sure that all the fasteners are correctly tightened and that there are no missing fasteners.

EN

4.2 Install with P-installation

In the P-installation, the pump is installed on a stationary discharge connection, and operates either completely or partially submerged in the pumped liquid. These requirements and instructions only apply when the installation is made according to the dimensional drawing.

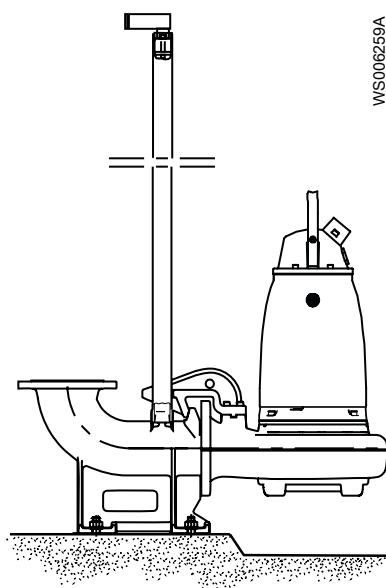


Figure 7: P-installation

These items are required:

- Guide bars
- Guide bar bracket for attaching the guide equipment to the access frame or to the upper part of the sump
- Cable holder for holding the cable
- Access frame (with covers) to which the upper guide bar bracket and cable holder can be attached
- Discharge connection for connecting the pump to the discharge line

The discharge connection has a flange which fits the pump housing flange and a bracket for attaching the guide equipment.

- Fasteners for the discharge connection
- Anchor bolts

1. Install the access frame:

- a) Place the access frame in position and align it horizontally.
- b) Grout the frame in place.

2. Grout the anchor bolts in place.

Be careful when you align and position the discharge connection in relation to the access frame.

3. Place the discharge connection in position, and tighten the nuts.

4. Install the guide bars:
 - a) Secure the guide bars in the bracket.
 - b) Check that the guide bars are placed vertically. Use a level or a plumb line.
5. Connect the discharge pipe to the discharge connection.
6. Lower the pump along the guide bars.

When it reaches the bottom position, the pump automatically connects to the discharge connection.
7. Secure the motor cable:
 - a) Fasten the permanent lifting device to the pump and to the access frame. For example, you can use a stainless-steel lifting chain with shackles.
 - b) Fasten the cable to the cable holder.

Make sure that the cable cannot be sucked into the pump inlet or that it is neither sharply bent, or pinched. Support straps are required for deep installations.
 - c) Connect the motor cable and the starter and monitoring equipment according to the separate instructions.

Make sure that the impeller rotation is correct. For more information, see [Check the impeller rotation](#) on page 39.

Clean all debris from the sump before starting the pump.

4.3 Install with S-installation

In the S-installation, the pump is transportable and intended to operate either completely or partially submerged in the pumped liquid. The pump is equipped with a connection for hose or pipe and stands on a base stand.

These requirements and instructions only apply when the installation is made according to the dimensional drawing. For information about the different installation types, see Parts List.

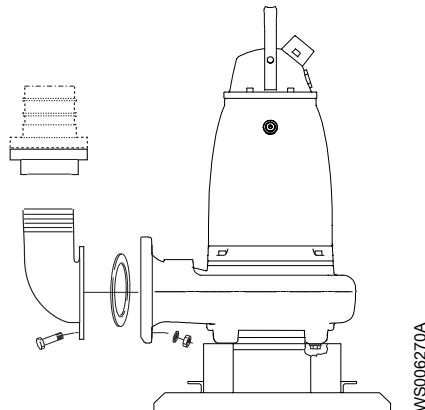


Figure 8: S-installation

1. Run the cable so that it has no sharp bends. Make sure that it is not pinched, and cannot be sucked into the pump inlet.
2. Connect the discharge line.
3. Lower the pump into the sump.
4. Place the pump on the base and make sure it cannot fall over or sink.

Alternatively, the pump can be suspended with chains just above the sump bottom. Make sure that the pump cannot rotate at start-up or during operation.
5. Connect the motor cable and the starter and monitoring equipment according to the separate instructions.

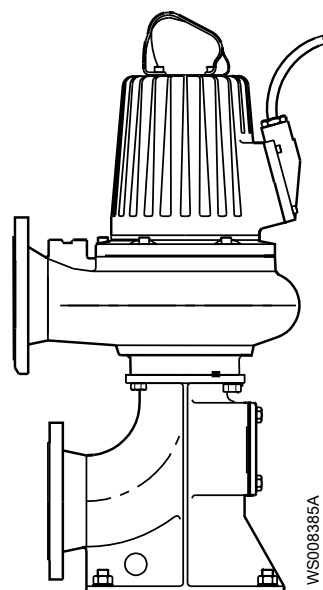
Make sure that the impeller rotation is correct. For more information, see [Check the impeller rotation](#) on page 39.

4.4 Install with T-installation

This installation is not applicable for these versions:

- .172
- .891

In the T-installation, the pump is installed in a vertical position in a dry well next to the wet well. These requirements and instructions only apply when the installation is made according to the dimensional drawing.



These items are required:

- Anchor bolts for anchoring the pump to a base.
- Shut-off valves that allow you to remove the pump from service

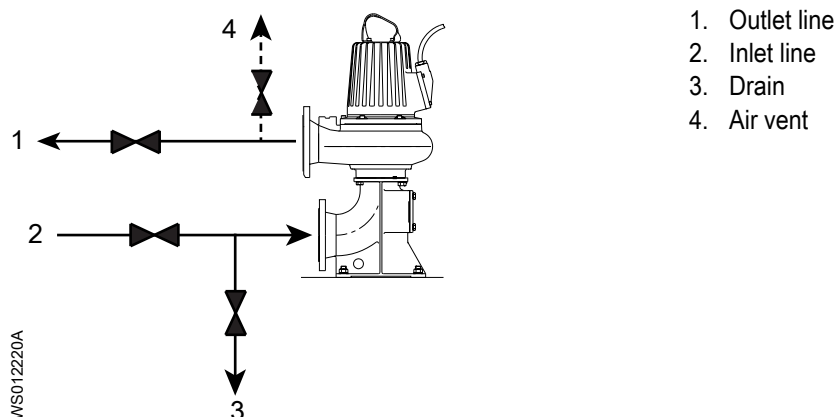


Figure 9: Shutoff and air vent valves (generic pump shown)

NOTICE:

The risk of freezing is particularly high in T- or Z-installations.

1. Fasten the pump:
 - a) Bolt the stationary suction connection to the concrete base.
 - b) Bolt the pump to the suction connection.
2. Make sure that the pump is vertical for the T-installation.
3. Connect the suction line and discharge line.

4. Connect the motor cable and the starter and monitoring equipment according to the separate instructions.

Make sure that the impeller rotation is correct. For more information, see [Check the impeller rotation](#) on page 39.

5. Make sure that the weight of the pump does not put strain on the piping.

4.5 Install with Z-installation

This installation is not applicable for these versions:

- .172
- .891

In the Z-installation, the pump is installed in a horizontal position on a support stand in a dry well next to the wet well. The following requirements and instructions are for Z-installations that comply to the dimensional drawing.

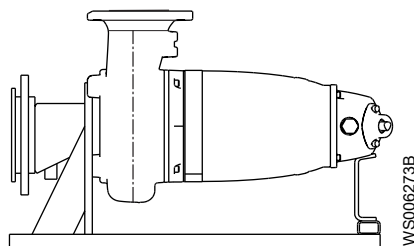


Figure 10: Z-installation

These items are required:

- Anchor bolts for anchoring the pump to a base.
- Shut-off valves that allow you to remove the pump from service

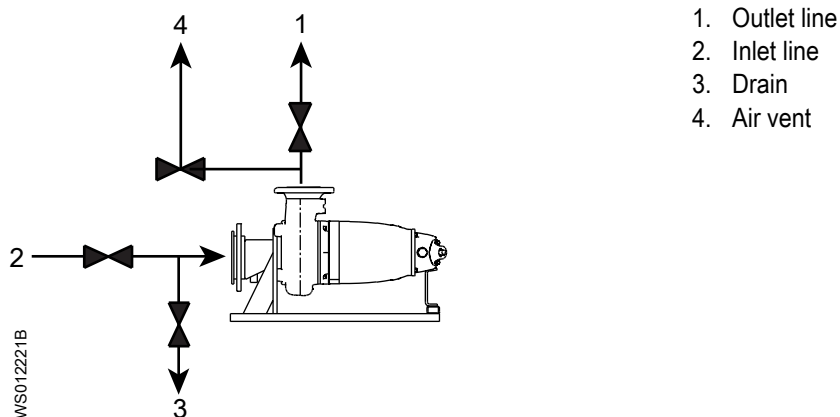


Figure 11: Shutoff and air vent valves (generic pump shown)

NOTICE:

The risk of freezing is particularly high in T- or Z-installations.

1. Fasten the pump:
 - a) Bolt the stationary suction connection to the concrete base.
 - b) Bolt the pump to the suction connection.
2. Make sure that the pump is vertical for the T-installation or horizontal for the Z-installation.
3. Connect the suction line and discharge line.
4. Connect the motor cable and the starter and monitoring equipment according to the separate instructions.

Make sure that the impeller rotation is correct. For more information, see [Check the impeller rotation](#) on page 39.

5. Make sure that the weight of the pump does not put strain on the piping.

4.6 Install with F-installation

EN

In the F-installation, the pump is free standing and installed primarily in a small sump on a firm surface. The pump is intended to operate completely or partially submerged in the pumped liquid. The pump is equipped with a connection for hose or pipe and with supporting legs and/or a strainer. These requirements and instructions are for F-installations that comply to the dimensional drawing.

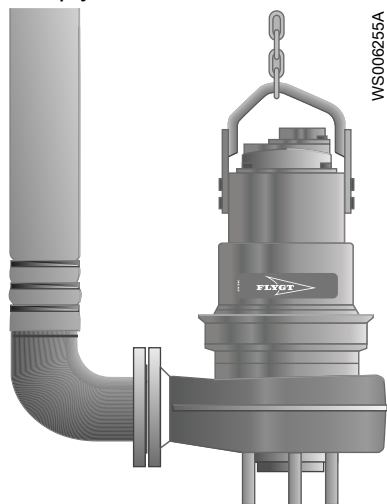


Figure 12: F-installation

These items are required:

- Nipple pipe/Flange/Coupling
 - Pipe or hose
1. Run the cable so that it has no sharp bends, is not pinched, and cannot be sucked into the pump inlet.
 2. Fit the nipple pipe/flange/coupling.
 3. Lower the pump into the sump.
 4. Place the pump on the sump bottom and make sure it cannot fall over.
 5. Connect the motor cable and the starter and monitoring equipment according to the separate instructions.

Make sure that the impeller rotation is correct. For more information, see [Check the impeller rotation](#) on page 39.

4.7 Install with H-installation

This installation is only applicable for the following versions:

- 172
- 891

In the H-installation, the pump is installed with a permanent, quick-connection, suspended arrangement that incorporates an integral non-return valve. The pump is intended to operate completely or partially submerged in the pumped liquid. These requirements and instructions are for H-installations that comply to the dimensional drawing.

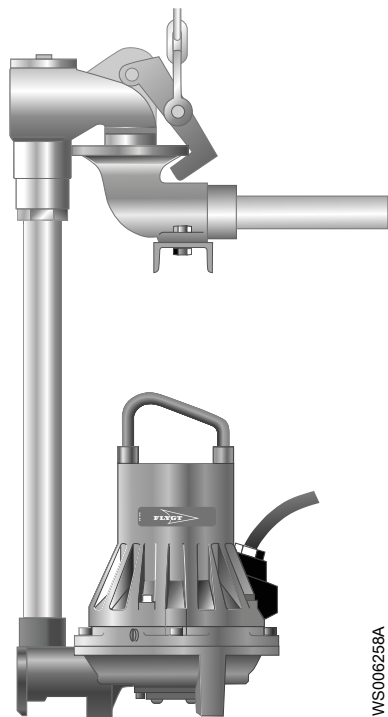


Figure 13: H-installation

These items are required:

- Discharge connection
 - Pipe
1. Run the cable so that it has no sharp bends, is not pinched, and cannot be sucked into the pump inlet.
 2. Fit the pipe and connect the discharge connection.
 3. Lower the pump into the sump.
 4. Make sure that the pump is hanging just above the sump bottom.
 5. Connect the motor cable and the starter and monitoring equipment according to the separate instructions.

Make sure that the impeller rotation is correct. For more information, see [Check the impeller rotation](#) on page 39.

Clean all debris from the sump before you start the pump.

4.8 Install with X-installation

In the X-installation, the pump has no pre-determined mechanical connection. The flange is drilled.

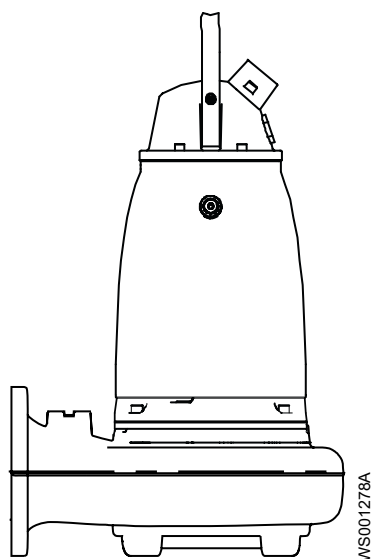


Figure 14: X-installation

NOTICE:

A pump prepared for X-installation is only approved to be used in P- or S-installation. Never use it in T- or Z-installation

For installation instructions, see appropriate mechanical accessories.

4.9 Make the electrical connections

4.9.1 General precautions

**DANGER: Electrical Hazard**

Before starting work on the unit, make sure that the unit and the control panel are isolated from the power supply and cannot be energized. This applies to the control circuit as well.

**WARNING: Electrical Hazard**

Risk of electrical shock or burn. A certified electrician must supervise all electrical work. Comply with all local codes and regulations.

**WARNING: Electrical Hazard**

There is a risk of electrical shock or explosion if the electrical connections are not correctly carried out, or if there is fault or damage on the product. Visually inspect equipment for damaged cables, cracked casings or other signs of damage. Make sure that electrical connections have been correctly made.

**WARNING: Electrical Hazard**

The permanent magnet synchronous motor generates voltage on internal components when the shaft rotates, even if the power supply is disconnected. Make sure that the shaft cannot rotate if the unit is disassembled.

**WARNING: Crush Hazard**

Risk of automatic restart.

**CAUTION: Electrical Hazard**

Prevent cables from becoming sharply bent or damaged.

NOTICE:

Leakage into the electrical parts can cause damaged equipment or a blown fuse. Keep the cable ends dry at all times.

Requirements

These general requirements apply for the electrical installation:

- If the pump will be connected to the public mains, then the supply authority must be notified before installing the pump. When the pump is connected to the public power supply, it can cause flickering of incandescent lamps when started.
- The mains voltage and frequency must agree with the specifications on the data plate. If the pump can be connected to different voltages, then follow the specified voltage on the yellow sticker close to the cable entry.
- If the operation can be intermittent, such as S3 periodic duty, then the pump must be supplied with monitoring equipment supporting such operation.
- If stated on the data plate, then the motor is convertible between different voltages.
- The thermal contacts or thermistors must be in use.
- If the temperature of the pumped liquid is above 40°C (104°F), then do not connect the T1 and T2 leads to thermal contacts. This does not apply to Ex-versions of the product.
- For FM-approved pumps, a leakage sensor must be connected and in use to meet approval requirements.

Motor and short-circuit protection**NOTICE:**

A qualified electrician must select the size of motor protection breakers and fuses. The size must be chosen for the specific motor data such as rated current and starting current.

It is important that the short-circuit protection is not over-dimensioned. Over-dimensioned fuses or motor protection breakers decrease the protection for the motor.

- The fuse rating and the cables must be in accordance with the local rules and regulations.
- The fuses and circuit breakers must have the correct rating.
- The pump overload protection must be connected and set to the rated current.

The starting current in direct-on-line start can be up to six times higher than the rated current.

For more information, see the data plate and if applicable, the cable chart for the rated current.

Cables

When cables are installed, these requirements must be followed:

- The cables must be in good condition, not have any sharp bends, and not be pinched.
- The cables must not be damaged and must not have indentations.
- The cables must not be embossed at the cable entry.
- The cable entry seal sleeve and washers must conform to the outer diameter of the cable.

- The minimum bend radius must not be smaller than the accepted value.
- If a cable is reused, a short piece at the end must be peeled off when the cable is refitted. This action is necessary so that the seal sleeve of the cable entry does not close around the cable at the same point again. If the outer jacket of the cable is damaged, then the cable must be replaced.
Contact a sales or authorized service representative.
- The voltage drop in long cables must be considered. The rated voltage of the drive unit is the voltage that is measured at the cable connection point in the pump.
- If a variable frequency drive (VFD) is used, the screened cable must be used according to the European CE and EMC requirements. For more information, contact a sales or authorized service representative.
- The cable length must be long enough for maintenance work.
- For SUBCAB™ cables, the twisted pair copper foil must be trimmed.
- All unused conductors must be insulated.

4.9.2 Grounding (earthing)

Grounding (earthing) must be done in compliance with all local codes and regulations.



DANGER: Electrical Hazard

All electrical equipment must be grounded (earthed). Test the ground (earth) lead to verify that it is connected correctly and that the path to ground is continuous.



WARNING: Electrical Hazard

Risk of electrical shock. The ground (earth) lead must be sufficiently longer than the phase leads to make sure that the ground lead is the last to become disconnected if the cable is jerked loose.

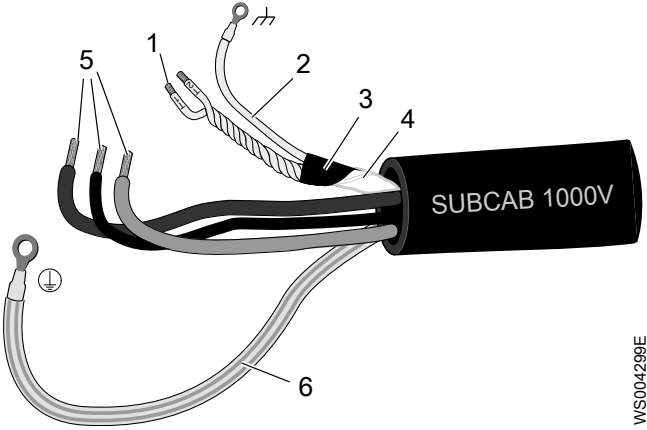
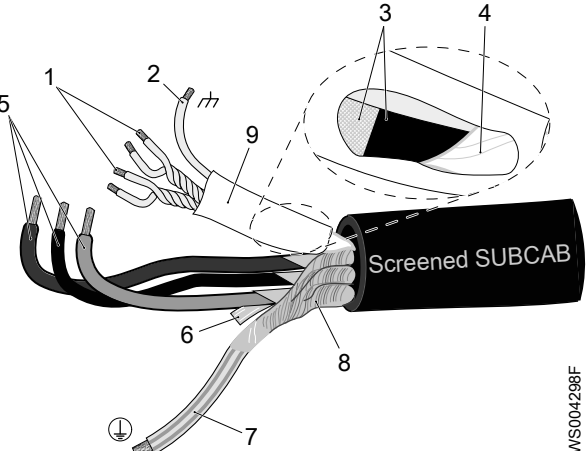


WARNING: Electrical Hazard

Risk of electrical shock or burn. You must connect an additional ground- (earth-) fault protection device to the grounded (earthed) connectors if persons are likely to come into contact with liquids that are also in contact with the pump or pumped liquid.

4.9.3 Prepare the SUBCAB cables

This section applies to SUBCAB cables with twisted-pair control conductors.

The prepared SUBCAB cable	The prepared screened SUBCAB cable, without cable lugs
 1. T1+T2 twisted pairs in control element 2. Drain wire in control element (tinned copper strands) with shrink tube 3. Aluminum and textile layers 4. Insulation jacket or plastic jacket, for the control element 5. Power conductors 6. Ground (earth) conductor with yellow-green shrink tube WS004299E	 1. T1+T2 and T3+T4 twisted pairs in control element 2. Drain wire in control element (tinned copper strands) with shrink tube 3. Aluminum and textile layers 4. Insulation jacket or plastic jacket, for the control element 5. Power conductors 6. Plastic laminated aluminum foil, screen 7. Ground (earth) conductor with yellow-green shrink tube 8. Uncovered screen/braided wire 9. shrink tube WS004298F

1. Peel off the outer jacket at the end of the cable.

2. Prepare the control element:

- Peel the insulation jacket or plastic jacket.
- Peel the aluminum and textile layers.

The aluminum foil is a conductive screen. Do not peel more than necessary, and remove the peeled foil.

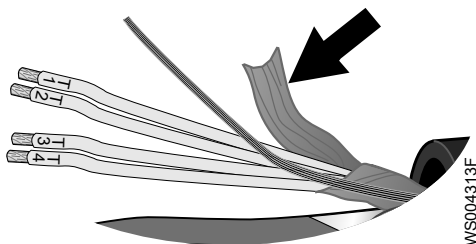


Figure 15: Aluminum foil on the control element.

- Put a white shrink tube over the drain wire.
- Twist T1+T2 and T3+T4.
- Put a shrink tube over the control element.

Make sure that the conductive aluminum foil and drain wire are covered.

3. Prepare the ground (earth) conductor of the SUBCAB cable:

- Peel the yellow-green insulation from the ground (earth) conductor.
- Check that the ground (earth) conductor is at least 10% longer than the phase conductors in the cabinet.
- If applicable, put a cable lug on the ground conductor.

4. Prepare the ground (earth) conductor of the screened SUBCAB cable:

- Untwist the screens around the power conductors.
- Twist all power conductor screens together to create a ground (earth) conductor.
- Put a yellow-green shrink tube over the ground (earth) conductor.

- Leave a short piece uncovered.
- d) Check that the connected ground (earth) conductor has sufficient slack. The conductor must stay connected even if the power conductors are pulled loose.
5. Prepare the power conductors:
- Remove the aluminum foil around each power conductor.
 - Peel the insulation from each power conductor.
6. Prepare the ends of the ground (earth) conductor, the power conductors, and the drain wire:

Connection type	Action
Screw	Fit cable lugs to the ends.
Terminal block	Fit end sleeves or leave the ends as they are.

4.9.4 Connect the motor cable to the pump

NOTICE:

Leakage into the electrical parts can cause damaged equipment or a blown fuse. Keep the end of the motor cable dry at all times.

- Remove the entry gland screw from the stator housing.
- Remove the stator housing.
This provides access to the terminal board/closed end splices.
- Remove the terminal board.
- Check the data plate to see which connections are required for the power supply.
- Arrange the connections on the terminal board/closed end splices in accordance with the required power supply.
- Connect the mains leads (L1, L2, L3, and ground (earth)) according to applicable cable chart.
The ground (earth) lead must be 160 mm (6.3 in.) longer than the phase leads in the junction box of the unit.
- Make sure that the pump is correctly connected to ground (earth).
- Make sure that any thermal contacts incorporated in the pump are properly connected to the terminal block/closed end splices.
- Install the stator housing.
- Install the entry gland screw.

4.9.5 Connect the motor cable to the starter and monitoring equipment



DANGER: Explosion/Fire Hazard

Special rules apply to installations in explosive or flammable atmospheres. Do not install the product or any auxiliary equipment in an explosive zone unless it is rated explosion-proof or intrinsically-safe. If the product is rated explosion-proof or intrinsically-safe, then see the specific explosion-proof information in the safety chapter before taking any further actions.

NOTICE:

Thermal contacts are incorporated in the pump.

NOTICE:

Thermal contacts must never be exposed to voltages higher than 250 V, breaking current maximum 4 A.

- If thermal contacts are included in the pump installation, then connect the T1 and T2 control conductors to the MiniCAS II monitoring equipment.

If the temperature of the pumped liquid is above 40°C (104°F), then do not connect the T1 and T2 leads to thermal contacts. This does not apply to Ex-versions of the product.

NOTICE:

The thermal contacts are incorporated in the stator. The thermal contacts must be connected to protect the unit.

2. Connect the mains leads (L1, L2, L3, and ground (earth)) to the starter equipment.
For information about the phase sequence and the color codes of the leads, see Cable charts.
3. Check the functionality of the monitoring equipment:
 - a) Check that the signals and the tripping function work correctly.
 - b) Check that the relays, lamps, fuses, and connections are intact.
 Replace any defective equipment.

4.9.6 Cable charts

Description

This topic contains general connection information. It also provides cable charts that show connection alternatives for use with different cables and power supply.

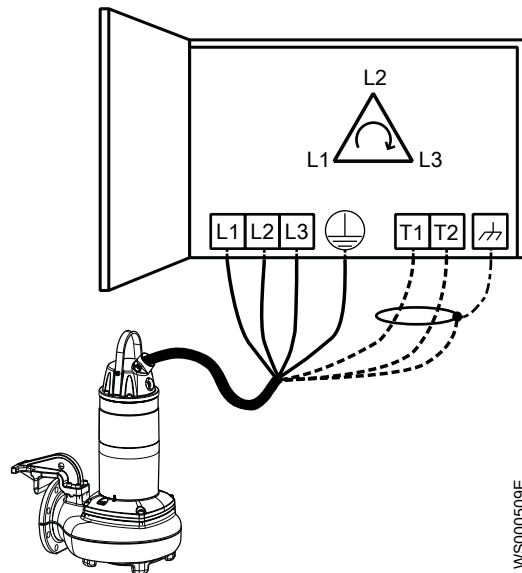
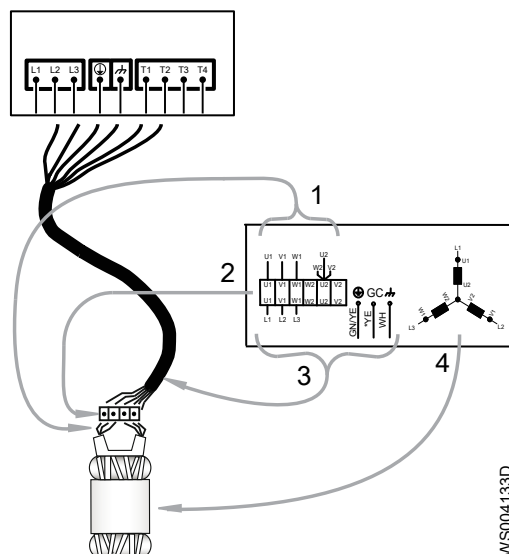


Figure 16: Phase sequence

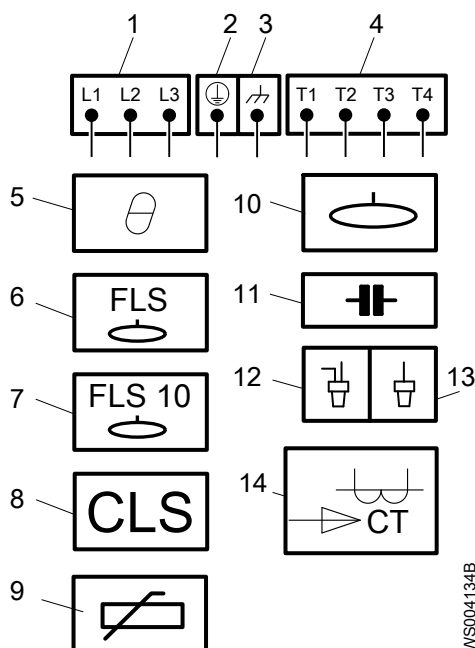
4.9.6.1 Connection locations

The figures in this section illustrate how to interpret the connection strip symbols.



1. Stator leads
2. Terminal board
3. Power cable leads
4. Stator (internal connection illustrated)

EN



1. Starter equipment and mains leads (L1, L2, L3)
2. Ground (earth)
3. Functional ground
4. Control leads (T1, T2, T3, T4)
5. Thermal contact
6. FLS
7. FLS 10
8. CLS
9. Thermistor
10. Level sensor
11. Capacitor
12. Crimp connection
13. Crimp isolation
14. Current transformer

Color code standard

Code	Description
BN	Brown
BK	Black
WH	White
OG	Orange
GN	Green
GNYE	Green-Yellow
RD	Red
GY	Grey
BU	Blue
YE	Yellow

4.9.6.2 Colors and markings of leads

Motor connection		Mains		SUBCAB 7GX Screenflex 7GX	SUBCAB 4GX Screenflex 4GX	SUBCAB AWG	SUBCAB Screened
Colours and marking of main leads		1~	3~				
COLOUR STANDARD BN=Brown BK=Black WH=White OG=Orange GN=Green GNYE=Green-Yellow RD=Red GY=Grey BU=Blue YE=Yellow *SUBCAB AWG * * Ground Conductor is stranded around cores GC=Ground Check	STATOR LEADS U1,U5 RD U2,U6 GN V1,V5 BN V2,V6 BU W1,W5 YE W2,W6 BK T1,T2 WH/YE	1	L1	BK 1	BN	RD	BN
		2	L2	BK 2	BK	BK	BK
		3	L3	BK 3	GY	WH	GY
			L1	BK 4	—	—	—
			L2	BK 5	—	—	—
			L3	BK 6	—	—	—
				GNYE	GNYE	GNYE or GN	**Screen/PE from cores
				Screen (WH)	Screen (WH)	—	Screen (WH)
		GC		—	—	YE	—
		863 17 50 (REV1)					

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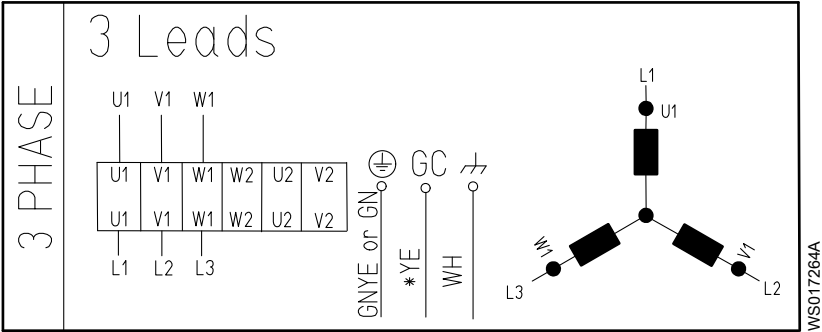
Figure 17: 3-phase and 1-phase without built-in starter

For markings on sensor leads, see [Sensors connection](#) on page 37.

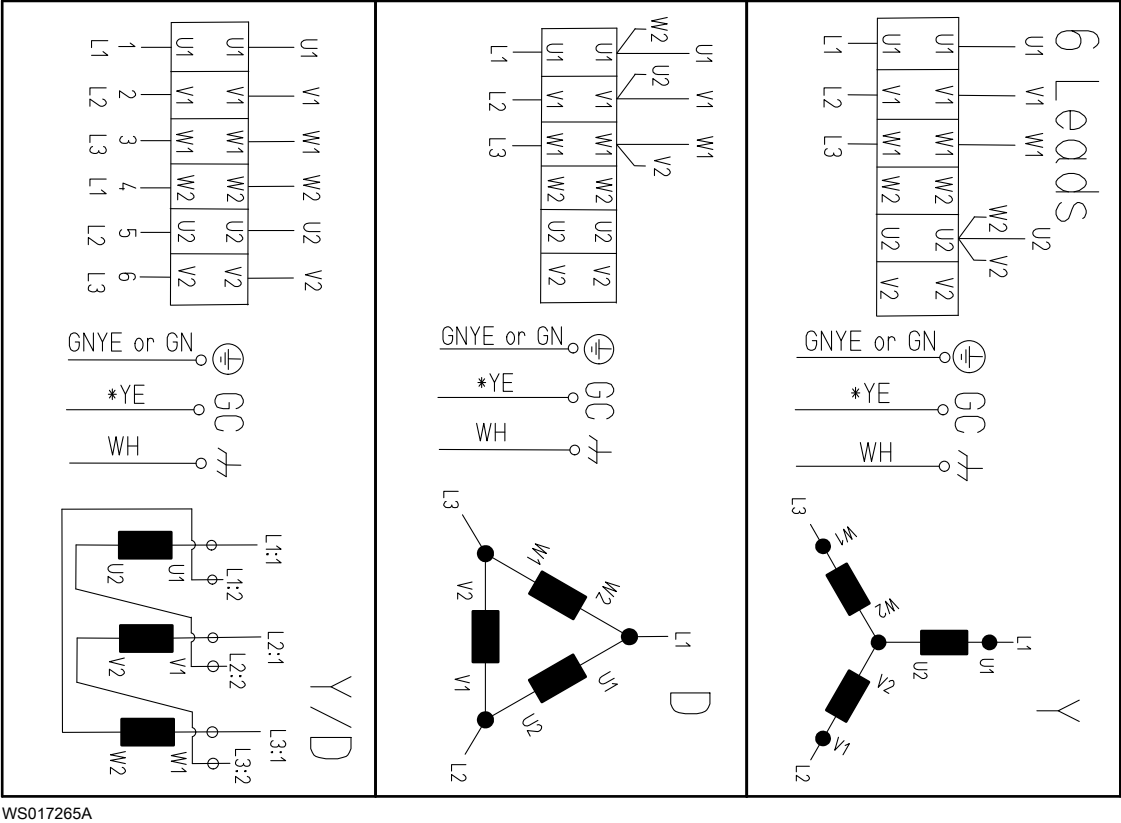
4.9.6.3 Connections included

- [3-phase connection](#) on page 32
- [1-phase connection](#) on page 36
- [Sensors connection](#) on page 37
- [Screened cable connection](#) on page 37

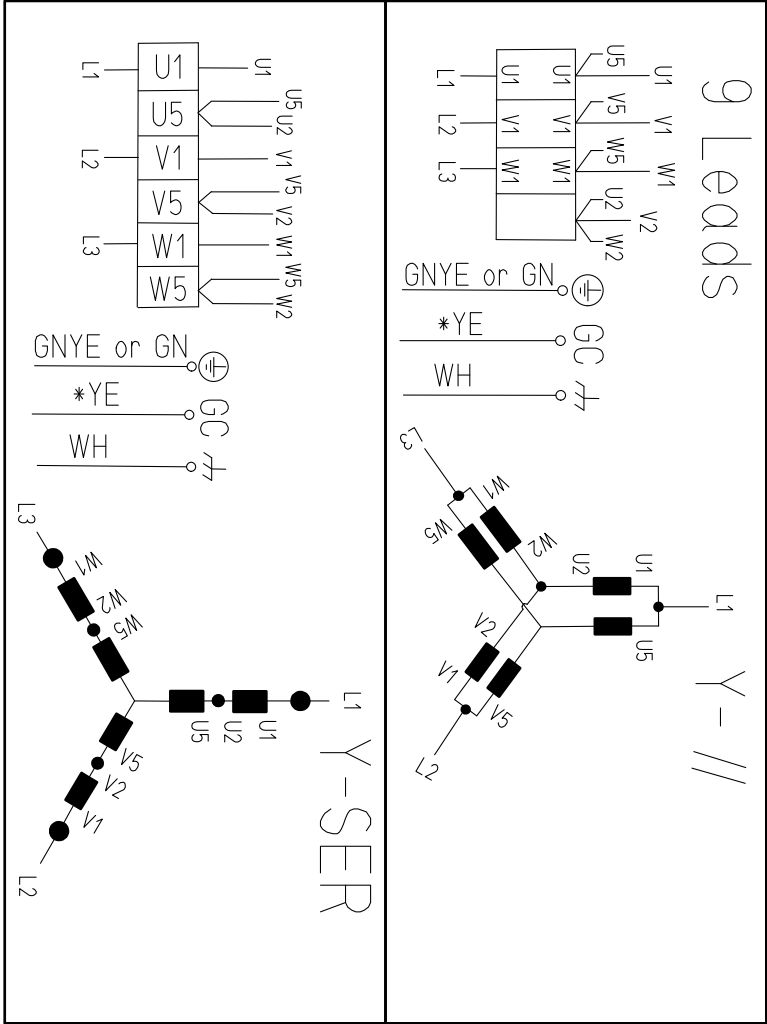
4.9.6.4 3-phase connection



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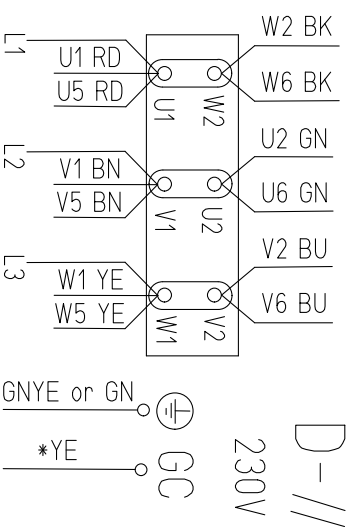
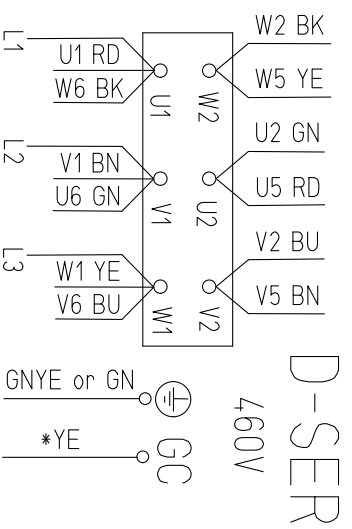
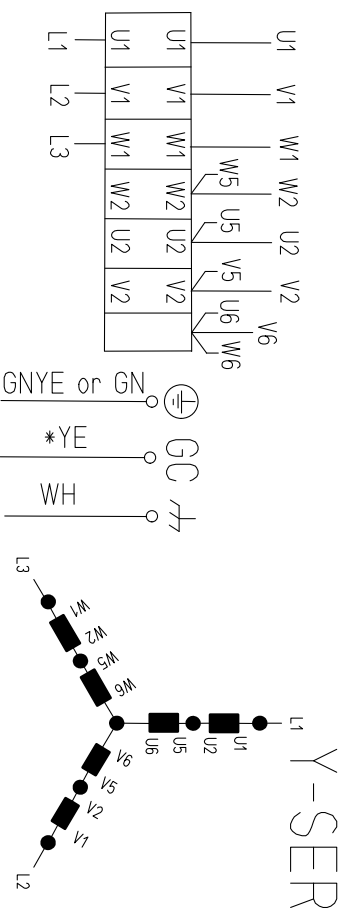
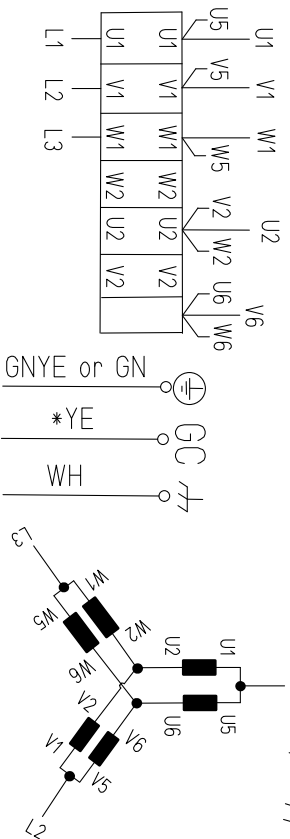


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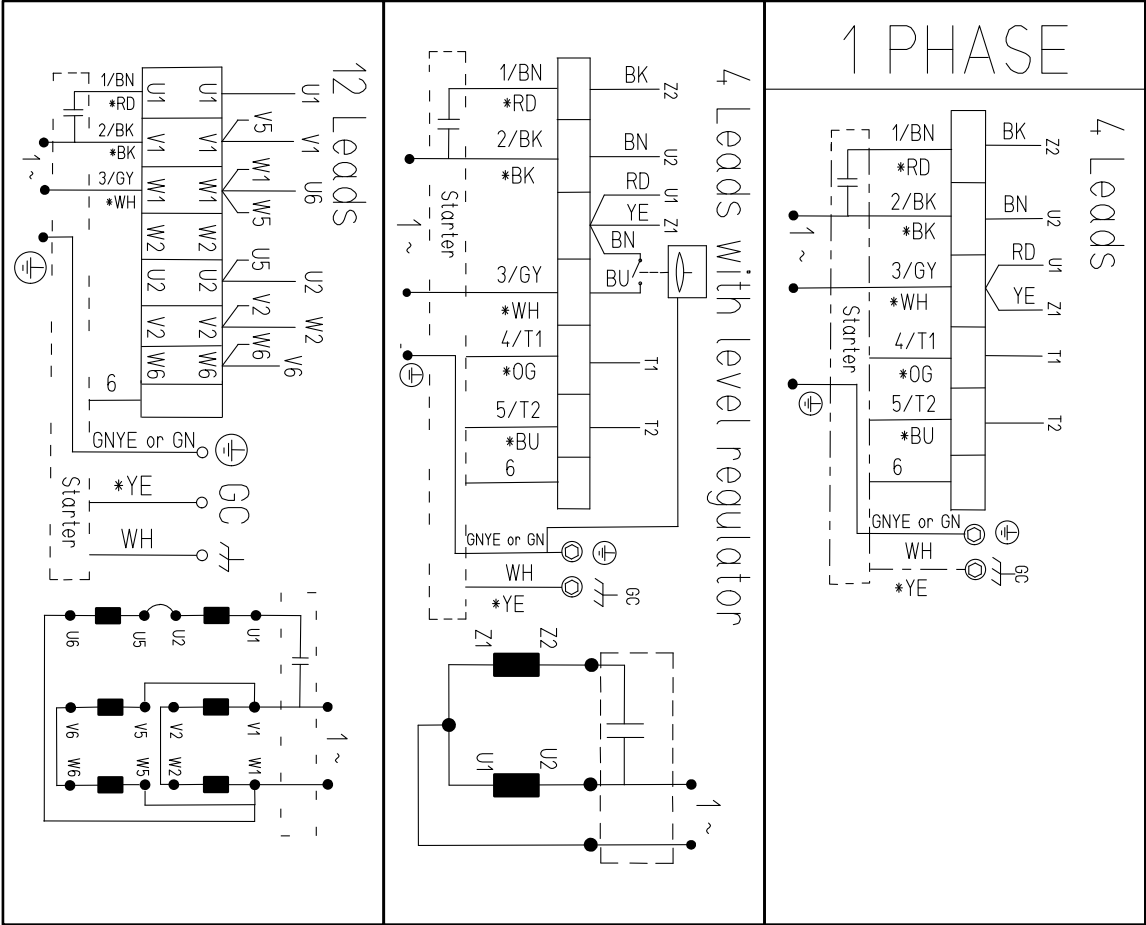
12 Leads



WS017267A

4.9.6.5 1-phase connection

EN



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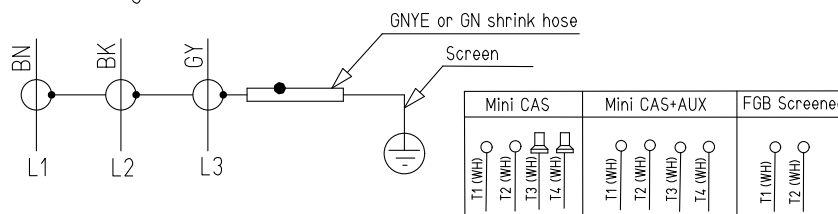
Figure 18: Without built-in starter

For more information, see [Colors and markings of leads](#) on page 32.

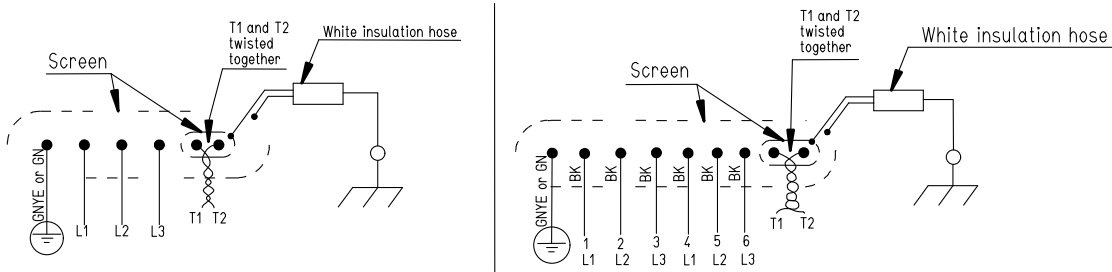
4.9.6.6 Screened cable connection

Screened connection SUBCAB & FGB Screened

Cable without sep. ground conductor
Screen as ground conductor



Screen - SUBCAB and Screenflex



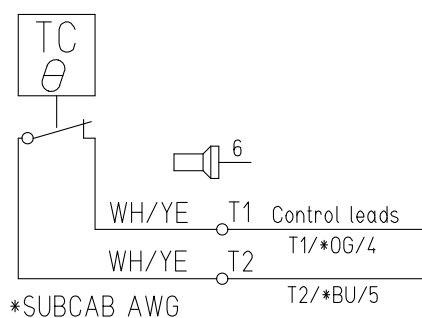
Sensors connection

These sensor connections are applicable for 3-phase and 1-phase without built-in starter:

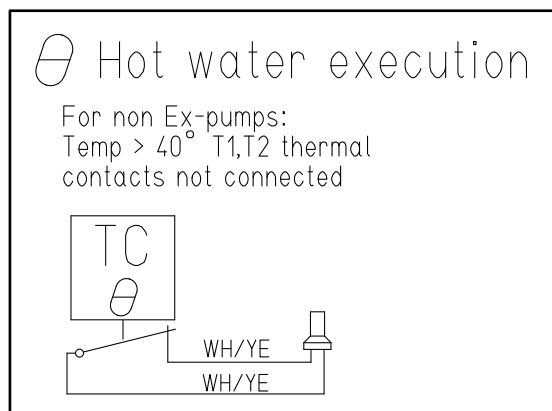
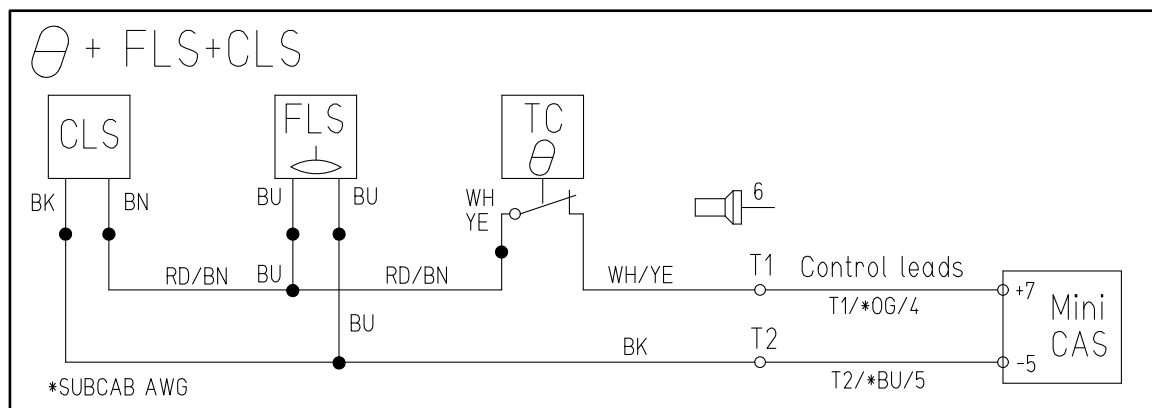
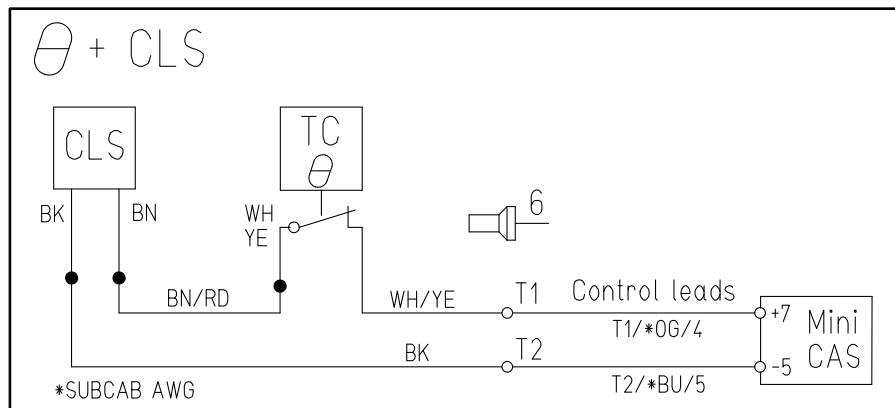
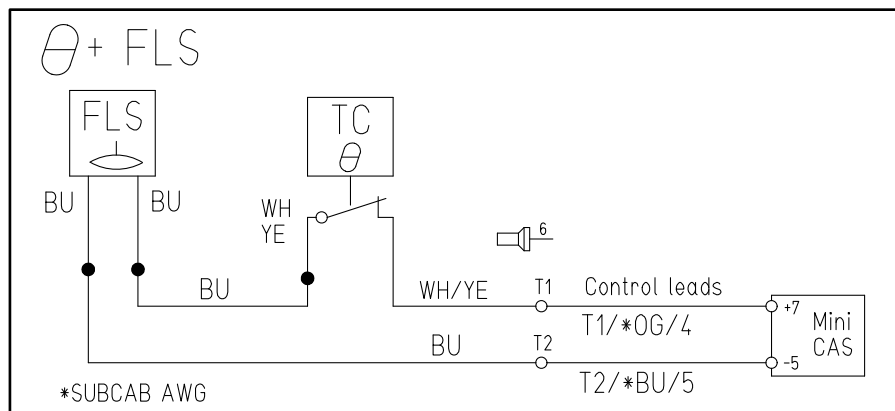
SENSORS	Control	SUBCAB 7GX & 4GX Screenflex	SUBCAB AWG	SUBCAB screened
	T1	WH T1	OG	WH T1
	T2	WH T2	BU	WH T2
	T3	-	-	WH T3
	T4	-	-	WH T4

* If Y/D-start, sensors not connected

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 θ (Thermal Contacts)

WS017270A



Sensor connection characteristics

The values have a 10% tolerance.

Sensors	Value (mA)	Definition
FLS and thermal contact	0	Overtemperature
	7.8	OK
	36	Leakage
CLS and thermal contact	0	Overtemperature
	5.5	OK
	29	Leakage (5 seconds delay)
CLS, FLS, and thermal contact	0	Overtemperature
	13.3	OK
	36–42	Leakage (0/5 seconds delay)

EN

4.10 T-installation: Bleed air before starting pump

1. Open the valve on the air vent line and bleed out the air. See the following figure.

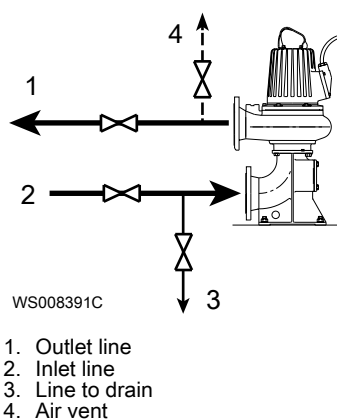


Figure 19: T-installation

2. Close the valve on the air vent line before the pump is started.

4.11 Check the impeller rotation

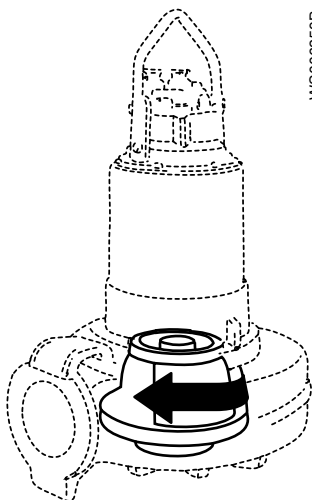


CAUTION: Crush Hazard

The starting jerk can be powerful. Make sure nobody is close to the unit when it is started.

1. Start the motor.
2. Stop the motor after a few seconds.
3. Check that the impeller rotates according to this illustration.

WS006253B



The correct direction of impeller rotation is clockwise when you look at the pump from above.

4. If the impeller rotates in the wrong direction, then do one of these steps:
 - If the motor has a 1-phase connection, then contact a sales or authorized service representative.
 - If the motor has a 3-phase connection, then transpose two phase leads and do this procedure again.

5 Operation

5.1 Precautions

EN

Before taking the unit into operation, check the following:

- All recommended safety devices are installed.
- The cable and cable entry have not been damaged.
- All debris and waste material has been removed.

NOTICE:

Never operate the pump with the discharge line blocked, or the discharge valve closed.



WARNING: Crush Hazard

Risk of automatic restart.

Distance to wet areas



WARNING: Electrical Hazard

Risk of electrical shock or burn. You must connect an additional ground- (earth-) fault protection device to the grounded (earthed) connectors if persons are likely to come into contact with liquids that are also in contact with the pump or pumped liquid.



CAUTION: Electrical Hazard

Risk of electrical shock or burn. The equipment manufacturer has not evaluated this unit for use in swimming pools. If used in connection with swimming pools then special safety regulations apply.

Noise level

NOTICE:

The sound power level of the product is lower than 70 dB(A). However, in some installations the resulting sound pressure level may exceed 70 dB(A) at certain operating points on the performance curve. Make sure that you understand the noise level requirements in the environment where the product is installed. Failure to do so may result in hearing loss or violation of local laws.

5.2 Estimate zinc anode replacement intervals

The mass and surface area of the zinc anodes are designed to protect the pump surface for 1 year in sea water with an average temperature of 20°C (68°F). Shorter inspection intervals and anode replacement can be required, depending upon the water temperature and the chemical composition as well as the presence of other metals in the vicinity of the pump.

The rate of zinc consumption, and the appropriate inspection intervals, can be estimated by measuring how much zinc is consumed during the first two months following installation.

Anodes are replaced when the anode mass is reduced to a selected fraction of its initial mass. The recommended interval for the selection fraction is 0.25–0.50 (25–50%).

1. Remove, weigh, and reinstall one or more of the exterior zinc anodes before starting up the pump.
2. After two months, remove and weigh the same zinc anode or anodes again.
3. Divide the lapsed time in days (between steps 1 and 2) by the anode weight loss in grams to get the calculated anode consumption rate (days/gram).

If multiple anodes were weighed, then use the anode which has lost the most weight for this calculation.

4. Calculate future replacement intervals so that they occur when the selected fraction of zinc is remaining.

5.3 Minimum water level

During continuous operation, it is recommended that the pump is fully submerged. The water level must be monitored to maintain a set minimum level. The minimum water level is at the interface of the pump housing and the drive unit.

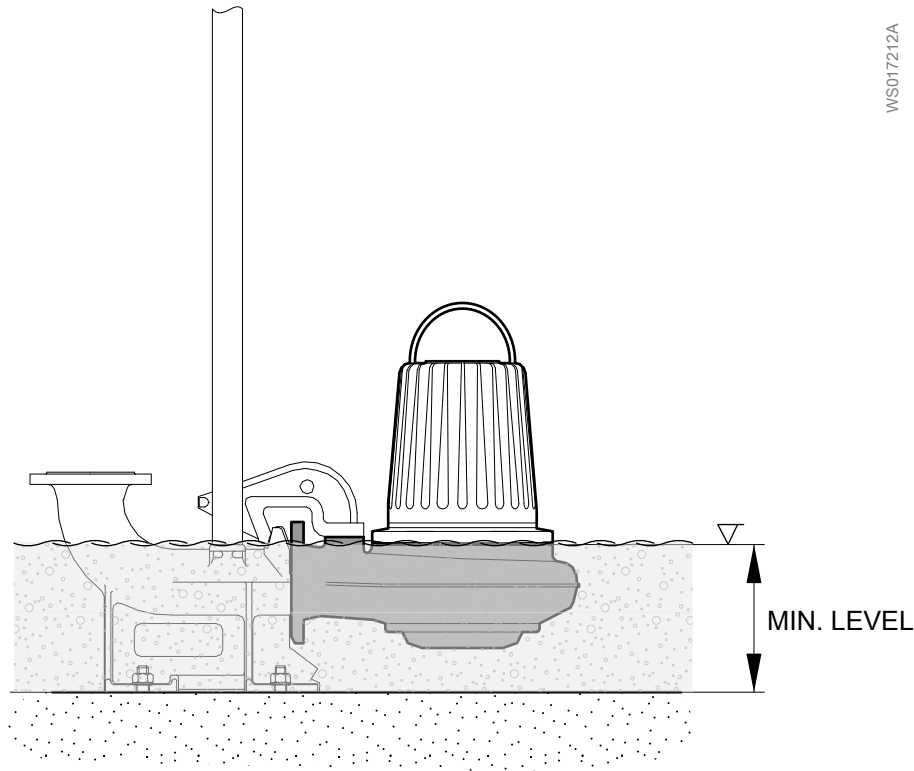


Figure 20: Minimum water level

- The pump must not operate constantly at the minimum level.
- The water level can be lower than the minimum level for short periods of time, for example during cleaning of the well.
- For operating temperatures above 40°C (104°F), contact the sales and service representative.
- For Ex-approved products, always follow any special handling instructions, see [Ex-approved products](#) on page 4.

5.4 Start the pump



CAUTION: Crush Hazard

The starting jerk can be powerful. Make sure nobody is close to the unit when it is started.



CAUTION: Thermal Hazard

The surfaces or parts of the unit may become hot during operation. Allow surfaces to cool before starting work, or wear heat-protective clothing.

NOTICE:

Make sure that the rotation of the impeller is correct.

1. Check the oil level in the oil housing.
2. Remove the fuses or open the circuit breaker, and check that the impeller can rotate freely.



WARNING: Crush Hazard

Never put your hand into the pump housing.

3. Conduct insulation test phase to ground. To pass, the value must exceed 5 megaohms.
4. Check that the monitoring equipment works.
5. Start the pump.

EN

6 Maintenance

6.1 Precautions



Before starting work, make sure that the safety instructions have been read and understood.

DANGER: Crush Hazard

Moving parts can entangle or crush. Always disconnect and lock out power before servicing to prevent unexpected startup. Failure to do so could result in death or serious injury.



DANGER: Inhalation Hazard

Before entering the work area, make sure that the atmosphere contains sufficient oxygen and no toxic gases.



WARNING: Biological Hazard

Infection risk. Rinse the unit thoroughly with clean water before working on it.



CAUTION: Crush Hazard

Make sure that the unit cannot roll or fall over and injure people or damage property.

Make sure that you follow these requirements:

- Check the explosion risk before you weld or use electrical hand tools.
- Allow all system and pump components to cool before you handle them.
- Make sure that the product and its components have been thoroughly cleaned.
- Before starting work, make sure that the work area is well-ventilated.
- Do not open any vent or drain valves or remove any plugs while the system is pressurized. Make sure that the pump is isolated from the system and that pressure is relieved before you disassemble the pump, remove plugs, or disconnect piping.

Ground continuity verification

A ground (earth) continuity test must always be performed after service.

6.2 Maintenance guidelines

During the maintenance and before reassembly, always remember to perform these tasks:

- Clean all parts thoroughly, particularly O-ring grooves.
- Change all O-rings, gaskets, and seal washers.
- Lubricate all springs, screws, O-rings with grease.

During the reassembly, always make sure that existing index markings are in line.

The reassembled drive unit must always be insulation-tested and the reassembled pump must always be test-run before normal operation.

6.3 Torque values

All screws and nuts must be lubricated to achieve correct tightening torque. Screws that are screwed into stainless steel must have the threads coated with applicable lubricants to prevent seizing.

If there is a question regarding the tightening torques, then contact a sales or authorized service representative.

Screws and nuts

Table 1: Stainless steel, A2 and A4, torque Nm (lbf-ft)

Property class	M4	M5	M6	M8	M10	M12	M16	M20	M24	M30
50	1.0 (0.74)	2.0 (1.5)	3.0 (2.2)	8.0 (5.9)	15 (11)	27 (20)	65 (48)	127 (93.7)	220 (162)	434 (320)
70, 80	2.7 (2)	5.4 (4)	9.0 (6.6)	22 (16)	44 (32)	76 (56)	187 (138)	364 (268)	629 (464)	1240 (915)
100	4.1 (3)	8.1 (6)	14 (10)	34 (25)	66 (49)	115 (84.8)	248 (183)	481 (355)	—	—

Table 2: Steel, torque Nm (lbf-ft)

Property class	M4	M5	M6	M8	M10	M12	M16	M20	M24	M30
8.8	2.9 (2.1)	5.7 (4.2)	9.8 (7.2)	24 (18)	47 (35)	81 (60)	194 (143)	385 (285)	665 (490)	1310 (966.2)
10.9	4.0 (2.9)	8.1 (6)	14 (10)	33 (24)	65 (48)	114 (84)	277 (204)	541 (399)	935 (689)	1840 (1357)
12.9	4.9 (3.6)	9.7 (7.2)	17 (13)	40 (30)	79 (58)	136 (100)	333 (245)	649 (480)	1120 (825.1)	2210 (1630)

Table 3: Brass, torque Nm (lbf-ft)

M5	M8	M10
2.7 (2.0)	11 (8.1)	22 (16.2)

Hexagon screws with countersunk heads

For hexagon socket head screws with countersunk head, maximum torque for all property classes must be 80% of the values for property class 8.8.

Round nuts with set screws

Table 4: Set screw, torque Nm (lbf-ft)

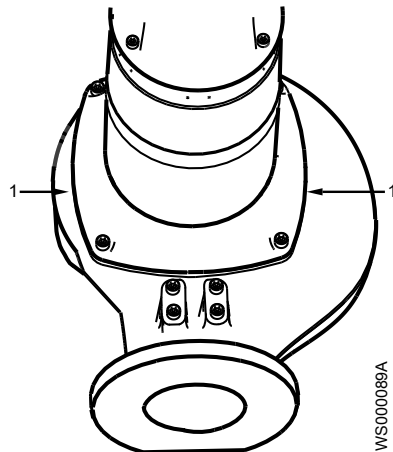
The torque values are only valid for the set screw, and not for the round nut.

M8	M10
18 (13)	35 (26)

6.4 Change the oil

This image shows the plugs that are used to change the oil.

EN



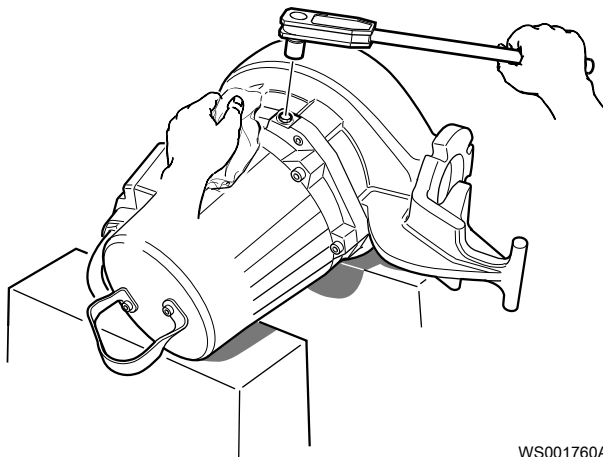
1. Oil plug

Empty the oil**CAUTION: Compressed Gas Hazard**

Air inside the chamber may cause parts or liquid to be propelled with force. Be careful when opening. Allow the chamber to de-pressurize before removal of the plug.

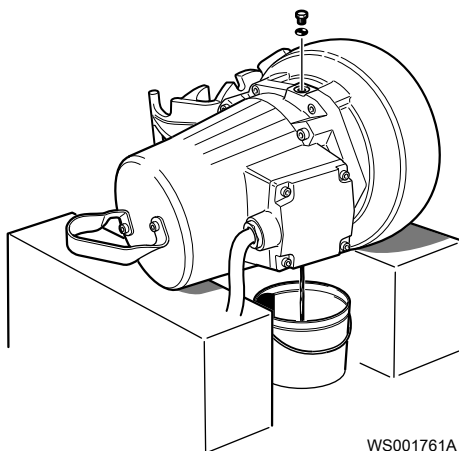
1. Place the pump in a horizontal position and unscrew the oil plug.

If the pump has a hole with the markings "oil out" it is important that this hole is used for drainage.



2. Place a container under the pump and turn the pump.
3. Unscrew the other oil plug.

If this hole has the markings "oil in", raise the pump upright for a short period of time during drainage in order to drain all the oil.



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Fill with oil

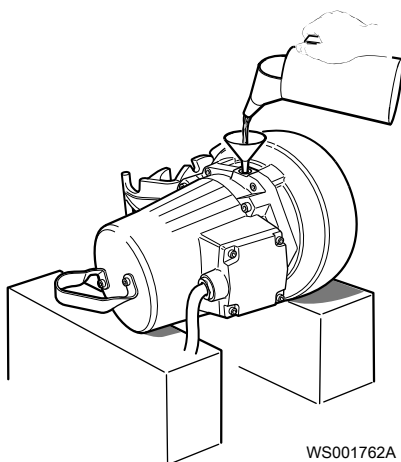
The oil should be a medical white oil of paraffin type that fulfills FDA 172.878 (a) and viscosity close to VG32. Examples of suitable oil types are the following:

- Statoil MedicWay 32™
- BP Enerpar M 004™
- Shell Ondina 927™
- Shell Ondina X430™

1. Replace the O-rings of the oil plugs.
2. Refit an oil plug in the hole that faces downwards or is marked "oil out", and tighten.
Tightening torque: 10-40 Nm (7.5-29.5 ft-lb)
3. Fill with oil through the hole on the opposite side or the hole marked "oil in".

If the hole is marked "oil in", slightly tilt the pump and lower it again in order to fill the pump with the correct quantity.

Quantity: approximately 1.0 L (1.1 qt).



WS001762A

4. Refit the oil plug and tighten.
Tightening torque: 10-40 Nm (7.5-29.5 ft-lb).

6.5 Service the pump

Type of maintenance	Purpose	Inspection interval
Initial inspection	A Xylem-authorized personnel checks the pump condition. From the results, the personnel recommends the intervals for the periodical inspection and overhaul for the installation.	Within the first year of operation.

Type of maintenance	Purpose	Inspection interval
Periodical inspection	The inspection prevents operational interruptions and machine breakdowns. The measures to increase performance and pump efficiency are decided for each application. They can include such things as impeller trimming, wear part control and replacement, control of zinc-anodes and control of the stator.	Up to every year. Applies to normal applications and operating conditions at media (liquid) temperatures <40°C (104°F).
Overhaul	The overhaul lengthens the operating lifetime of the product. It includes the replacement of key components and the measures that are taken during an inspection.	Up to 24,000 operational hours or six years, whichever comes first. Applies to normal applications and operating conditions at media (liquid) temperatures <40°C (104°F).

Xylem recommends to enter into a maintenance agreement. For more information, contact the local sales and service representative.

NOTICE:

Shorter intervals may be required when the operating conditions are extreme, for example with very abrasive or corrosive applications or when the liquid temperatures exceed 40°C (104°F).

6.5.1 Inspection

Service item	Action
Cable	1. If the outer jacket is damaged, then replace the cable. 2. Check that the cables do not have any sharp bends and are not pinched.
Connection to power	Check that the connections are properly secured.
Electrical cabinets	Check that they are clean and dry.
Impeller	1. Check the impeller clearance. 2. Adjust the impeller, if necessary.
Stator housing	1. Drain all liquid, if any. 2. Check the resistance of the leakage sensor. Normal value approximately 1500 ohms, alarm approximately 430 ohms.
Insulation	Use a megger maximum 1000 V. 1. Check that the resistance between the ground (earth) and phase lead is more than 5 megohms. 2. Conduct a phase-to-phase resistance check.
Junction box	Check that it is clean and dry.
Lifting device	Check that the local safety regulations are followed.
Lifting handle	1. Check the screws. 2. Check the condition of the lifting handle and the chain. 3. If necessary, replace.
O-rings	1. Replace the oil plug O-rings. 2. Replace the O-rings at the entrance or junction cover. 3. Grease the new O-rings.
Overload protection and other protections	Check the correct settings.
Personnel safety devices	Check the guard rails, covers, and other protections.

Service item	Action
Rotation direction	Check the impeller rotation.
Oil housing	Fill with new oil, if necessary.
Terminal block/closed end splice	Check that the connections are properly secured.
Thermal contacts	Normally closed circuit; interval 0–1 ohm.
Voltage and amperage	Check the running values.

6.5.2 Overhaul

The basic repair kit includes O-rings, seals, and bearings.

For an overhaul, do the following in addition to the tasks listed under Inspection.

Service item	Action
Support and main bearing	Replace the bearings with new bearings.
Mechanical seal	Replace with new seal units.

6.5.3 Service in case of alarm

For information about indication values for sensors, see [Sensors connection](#) on page 37.

Alarm source	Action
CLS	Check for water in the oil housing. If the oil contains too much water, then do the following: 1. Drain the oil and water. 2. Replace with new oil.
FLS	1. Check for liquid in the stator housing. 2. Drain all liquid, if any. 3. Check the mechanical seal unit, the O-rings, and the cable entry, if liquid was found.
Thermal contact	Check the start and stop levels.
The overload protection	Check that the impeller can rotate freely.

6.6 Replace the impeller

Required tools:

- 6 mm hexagon bit adapter with an extension of at least 125 mm (4.92 in)
- Impeller puller
- If applicable, contact your local sales and service representative for correct type and size.
- Rod (wood or copper) for locking the impeller in position, if applicable
- Two crowbars, if applicable



CAUTION: Cutting Hazard

Worn parts can have sharp edges. Wear protective clothing.

NOTICE:

When laying the pump on its side, do not allow the weight of the pump to rest on any portion of the impeller. The impeller must not be allowed to make contact with the concrete floor or other hard and rough surfaces.

If you fail with the impeller installation, then you must redo the installation procedure from the beginning.

6.6.1 Replace the D-impeller

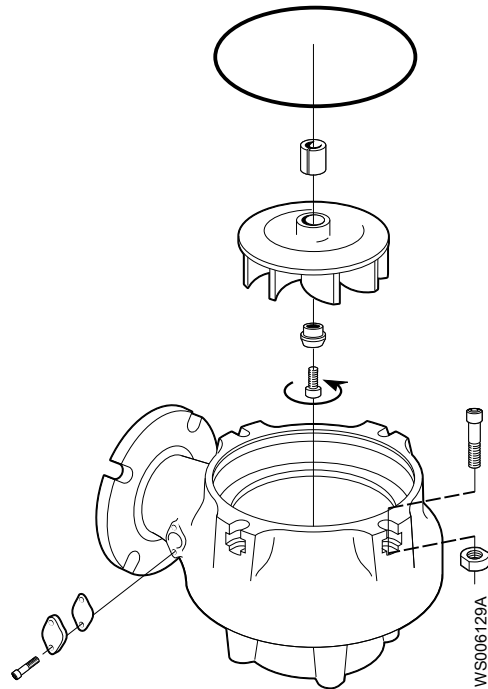


Figure 21: D-impeller

6.6.1.1 Remove the D-impeller



CAUTION: Cutting Hazard

Worn parts can have sharp edges. Wear protective clothing.

1. Remove the pump housing.



2. Remove the impeller screw.
If applicable, use the rod.



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3. Remove the impeller.
Use the impeller puller or the crowbars.



WS001978A

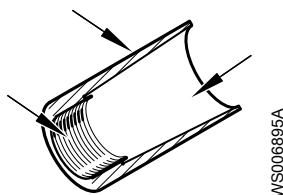
4. Remove the conical sleeve.

6.6.1.2 Install the D-impeller

1. Prepare the shaft:
 - a) Polish off any flaws by using a fine emery cloth.
The end of the shaft must be clean and free from burrs.
 - b) Coat the inner conical, the outer cylindrical surfaces, and the thread of the conical sleeve with a thin layer of grease.
The correct lubrication is grease for bearings, for example Exxon Mobil Unirex N3, Mobil Mobilith SHC 220 or equivalent.

NOTICE:

Surplus grease can cause the impeller to become loose. Remove surplus grease from conical and/or cylindrical surfaces of shafts and/or sleeves.



WS006895A

2. Mount the impeller:
 - a) Fit the conical sleeve onto the shaft.



WS002022A

- b) Lubricate the impeller screw threads and contact surface.
Ensure that all parts are clean.
 - c) Press the impeller onto the shaft with the impeller screw.
3. Tighten the impeller screw.
Tightening torque: 34 Nm (25 ft-lb).
Tighten a further 1/8 turn, 45° after tightening to the correct torque.
If applicable, use the rod.

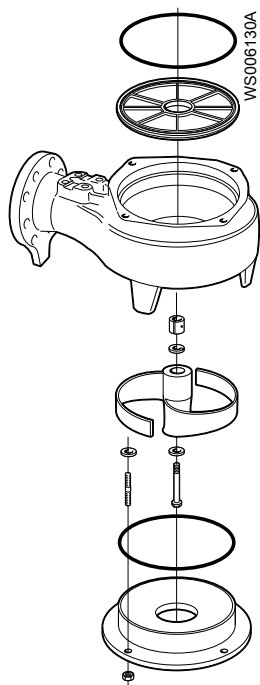


WS001941A

- Check that the impeller can rotate freely.
4. Mount the pump housing:
- a) Fit the pump housing.
 - b) Fit and tighten the lubricated screws.
Tightening torque: 57 Nm (42 lbf·ft)



6.6.2 Replace the F-impeller



6.6.2.1 Remove the F-impeller

**CAUTION: Cutting Hazard**

Worn parts can have sharp edges. Wear protective clothing.

1. Remove the suction cover.
If necessary, then pry off the suction cover.
2. Remove the impeller screw, the ring, and the washers.
If applicable, use the rod.



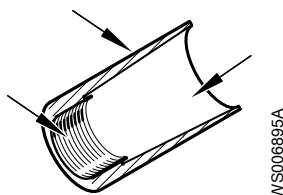
3. Remove the impeller.
Use the impeller puller or the crowbars.
Use the impeller screw to push the impeller.
4. Remove the conical sleeve.

6.6.2.2 Install the F-impeller

1. Prepare the shaft:
 - a) Polish off any flaws by using a fine emery cloth.
The end of the shaft must be clean and free from burrs.
 - b) Coat the inner conical, the outer cylindrical surfaces, and the thread of the conical sleeve with a thin layer of grease.
The correct lubrication is grease for bearings, for example Exxon Mobil Unirex N3, Mobil Mobilith SHC 220 or equivalent.

NOTICE:

Surplus grease can cause the impeller to become loose. Remove surplus grease from conical and/or cylindrical surfaces of shafts and/or sleeves.



2. Mount the impeller:
 - a) Fit the conical sleeve onto the shaft.
 - b) Fit one adjustment washer with a thickness of 0.25 mm (0.01 in) and two adjustment washers with a thickness of 1 mm (0.04 in) onto the impeller.



WS002021A

- c) Lubricate the impeller screw threads and contact surface.
Ensure that all parts are clean.
 - d) Press the impeller onto the shaft with the impeller screw.
3. Fasten the impeller:
- a) Prevent the impeller from rotating by inserting the rod through the pump housing outlet.
 - b) Tighten the impeller screw.
If applicable, use the rod.
Tightening torque: 30 Nm (22 ft-lb).
Tighten a further 1/8 turn, 45° after tightening to the correct torque.



WS001940A

Check that the impeller can rotate freely.

4. Adjust the impeller:
- a) Measure the distance between the edge of the impeller and the pump housing cover.
The correct distance should be 0.5-1.5 mm (0.02-0.06 in). Add or remove the appropriate number of adjusting washers in order to achieve correct distance.



- b) Tighten the impeller screw.
If applicable, use the rod.
Tightening torque: 30 Nm (22 ft-lb).
Tighten a further 1/8 turn, 45° after tightening to the correct torque.
Check that the impeller can rotate freely.
5. Fit a new lubricated O-ring to the into the groove in the pump housing.



6. Mount the suction cover:
- a) Fit the studs on the pump housing.
Use Loctite 603 locking liquid in order to secure the studs.
- b) Fit two adjustment washers with a thickness of 1 mm (0.04 in) and six adjustment washers with a thickness of 0.25 mm (0.01 in) onto each stud.



- c) Fit the suction cover to the studs.
- d) Fit the lock nuts onto the studs.
- e) Tighten the nuts.

Tightening torque: 17 Nm (13 ft-lb)



7. Adjust the suction cover:

- a) Measure the distance between the impeller and the suction cover.

The correct distance should be 0.5-1 mm (0.02-0.04 in). Add or remove the appropriate number of adjusting washers in order to achieve the correct distance.



- b) Tighten the nuts.

Tightening torque: 17 Nm (13 ft-lb).



8. Raise the pump to a vertical position.

Check that the impeller can rotate freely.

6.6.3 Replace the M-impeller

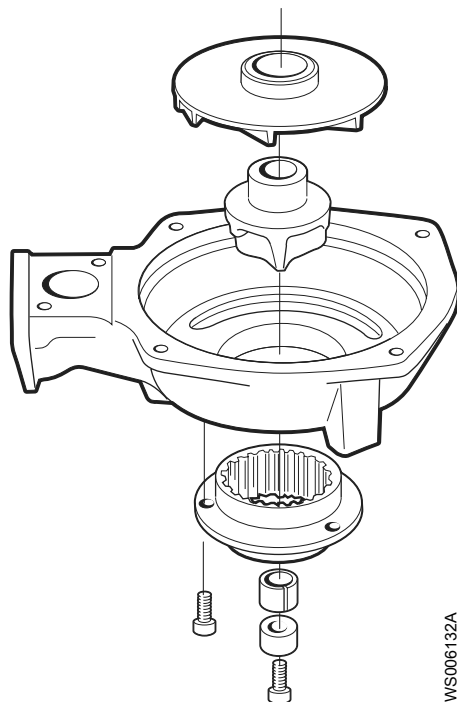


Figure 22: M-impeller

6.6.3.1 Remove the M-impeller



CAUTION: Cutting Hazard

Worn parts can have sharp edges. Wear protective clothing.

1. Remove the pump housing.
2. Remove the impeller screw and outer sleeve.
If applicable, use the rod.

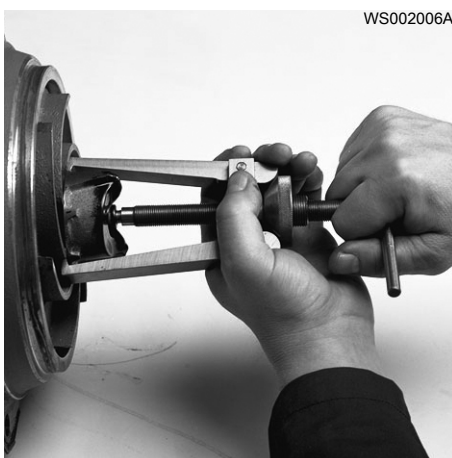


3. Refit the impeller screw.



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4. Fit the impeller puller and pull off the impeller and cutting wheel. Place a protector between the screw head and the impeller puller.



WS002006A

5. Remove the conical sleeve.



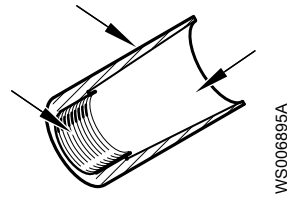
WS001987A

6.6.3.2 Install the M-impeller

1. Prepare the shaft:
 - a) Polish off any flaws by using a fine emery cloth.
The end of the shaft must be clean and free from burrs.
 - b) Coat the inner conical, the outer cylindrical surfaces, and the thread of the conical sleeve with a thin layer of grease.
The correct lubrication is grease for bearings, for example Exxon Mobil Unirex N3, Mobil Mobilith SHC 220 or equivalent.

NOTICE:

Surplus grease can cause the impeller to become loose. Remove surplus grease from conical and/or cylindrical surfaces of shafts and/or sleeves.

**2. Mount the impeller:**

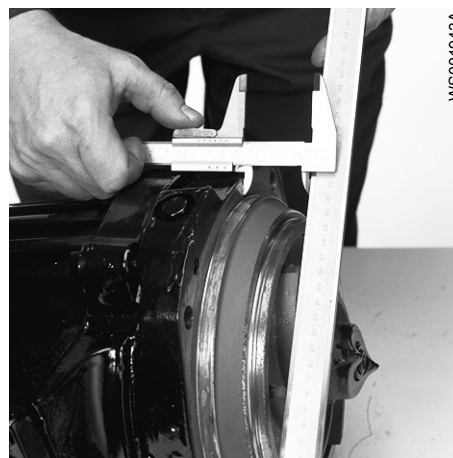
- a) Fit the conical sleeve onto the shaft.



- b) Fit the cutting wheel into the impeller.
c) Fit the outer sleeve onto the lubricated impeller screw.
d) Fit the impeller with the cutting wheel onto the shaft with the impeller screw.
Do not tighten the impeller screw.

3. Adjust the impeller:

- a) Use a straightedge and a vernier to measure the distance between the vanes of the impeller and the shoulder for the pump housing on the oil housing bottom.
The correct distance for pressure class LT and HT is 43.5 ± 0.3 mm (1.71 ± 0.01 in.).
b) Slide the impeller on to the shaft until you reach the correct distance.

**4. Tighten the impeller screw.**

If applicable, use the rod.

Tightening torque: 34 Nm (25 ft-lbs)



WS001939A

Check that the impeller can rotate freely.

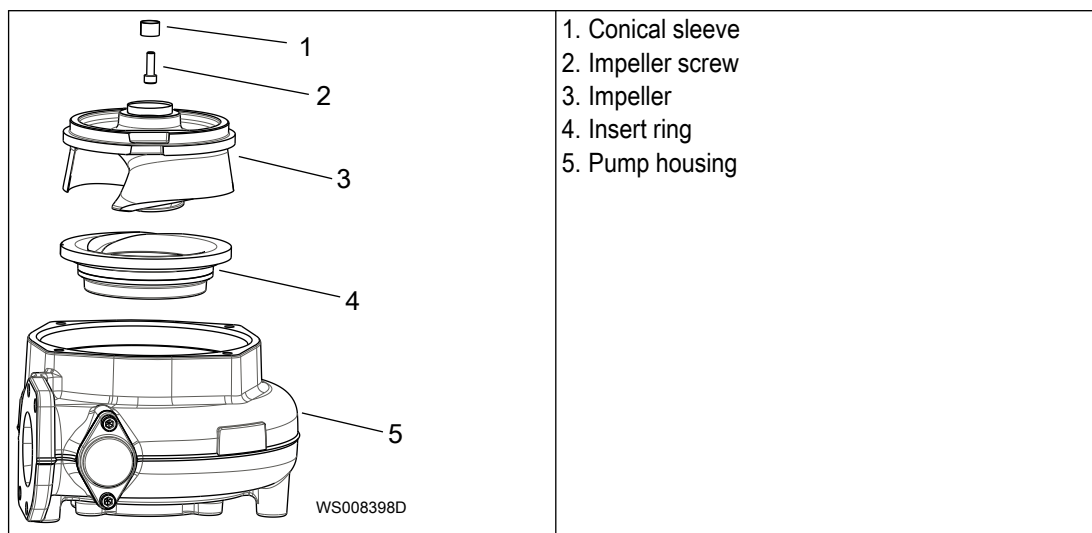
5. Mount the pump housing:
 - a) Fit the pump housing.
 - b) Fit and tighten the lubricated screws.

Tightening torque: 57 Nm (42 lbf·ft)

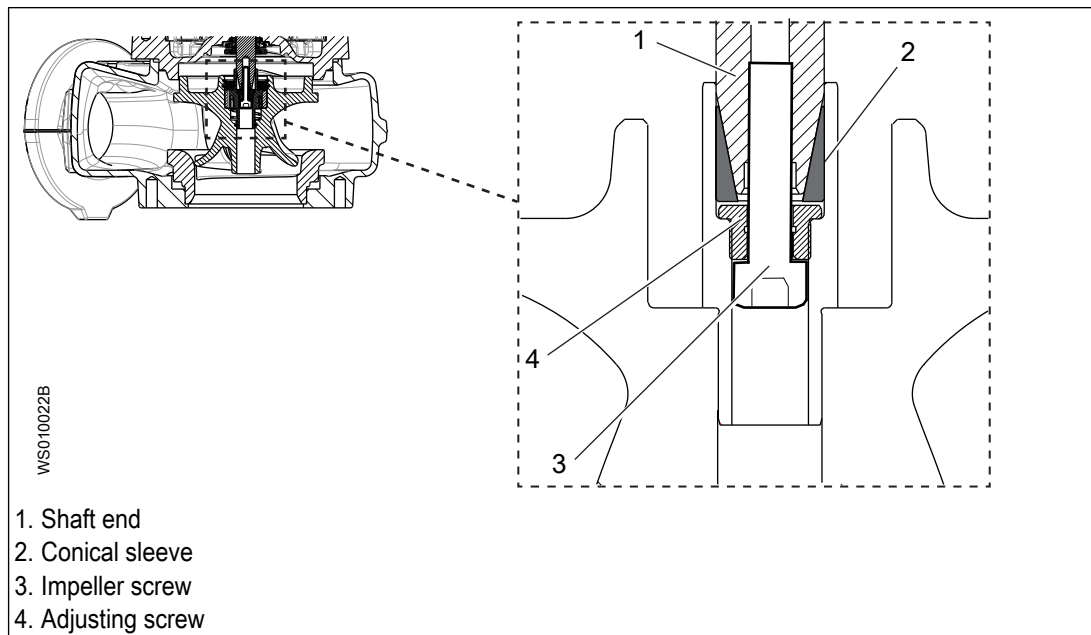


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6.6.4 Replacing the Adaptive N™ impeller



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**CAUTION: Cutting Hazard**

Worn parts can have sharp edges. Wear protective clothing.

NOTICE:

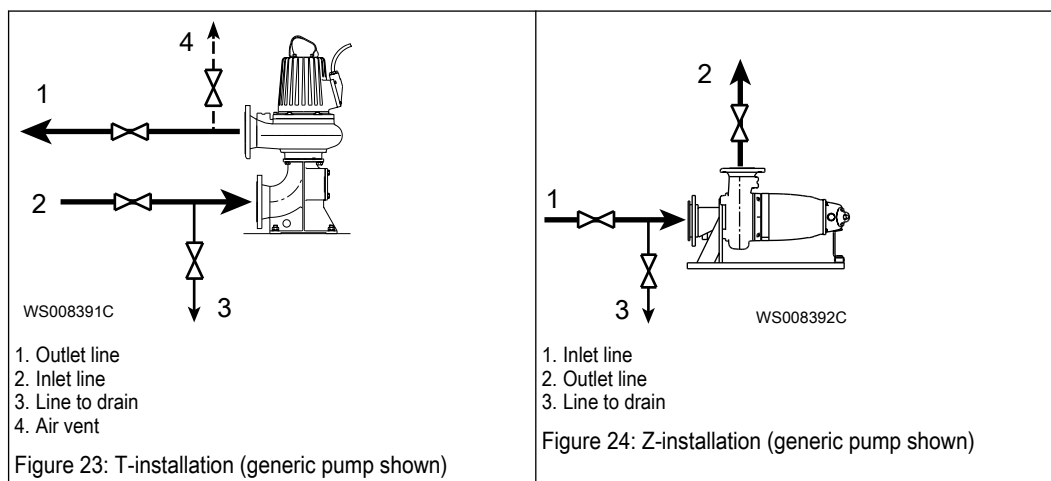
When laying the pump on its side, do not allow the weight of the pump to rest on any portion of the impeller. The impeller must not be allowed to make contact with the concrete floor or other hard and rough surfaces.

6.6.4.1 Prepare the pump for removal: T-, Z-installations

The pump must be removed from the installation to change the impeller.

1. Close the valves on the inlet and outlet lines.

See the following figures.

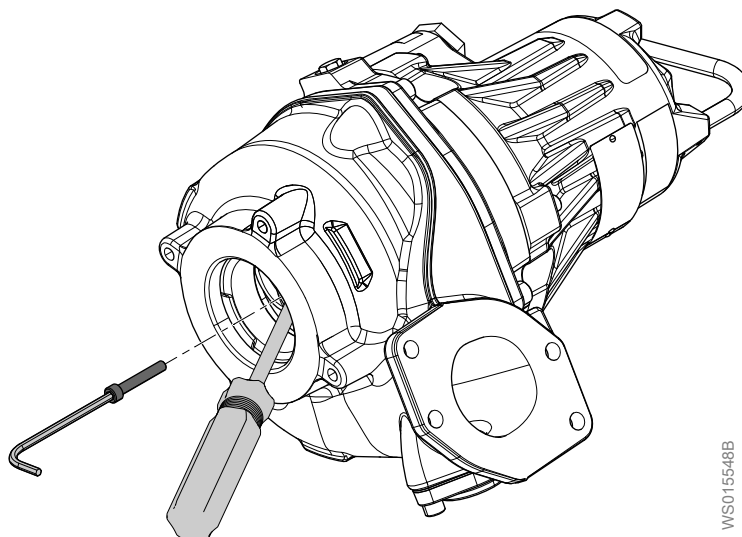


2. Drain the pump by opening the valve on the drain line.
3. Remove the pump from the installation.

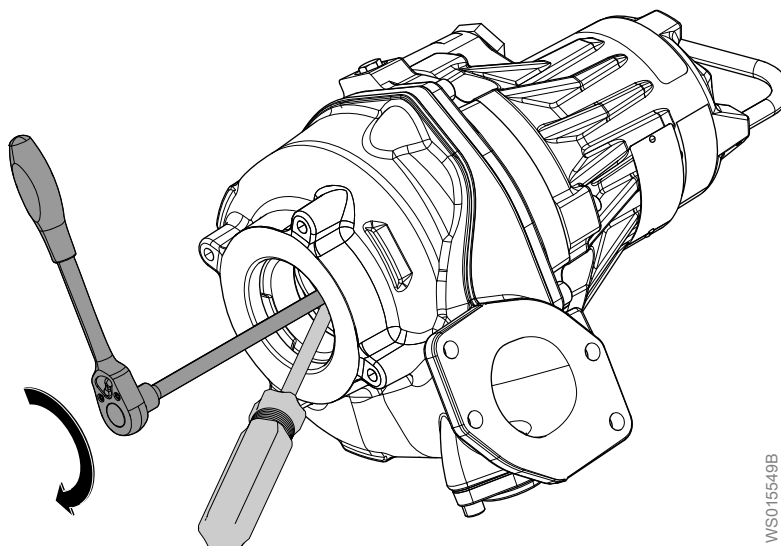
6.6.4.2 Remove the Adaptive N™ impeller: P, S, T, Z installations

To see which pumps are Adaptive N™, see [Product description](#) on page 11.

1. Lock the impeller with a screwdriver through the inlet.
2. Remove the impeller screw.



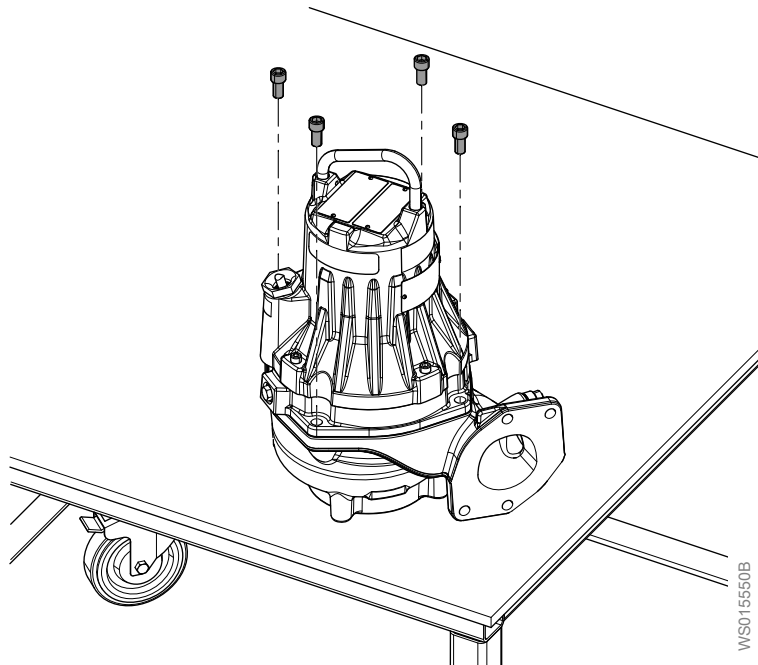
3. Insert a hexagon wrench and turn the adjusting screw clockwise until the impeller comes loose from the shaft.



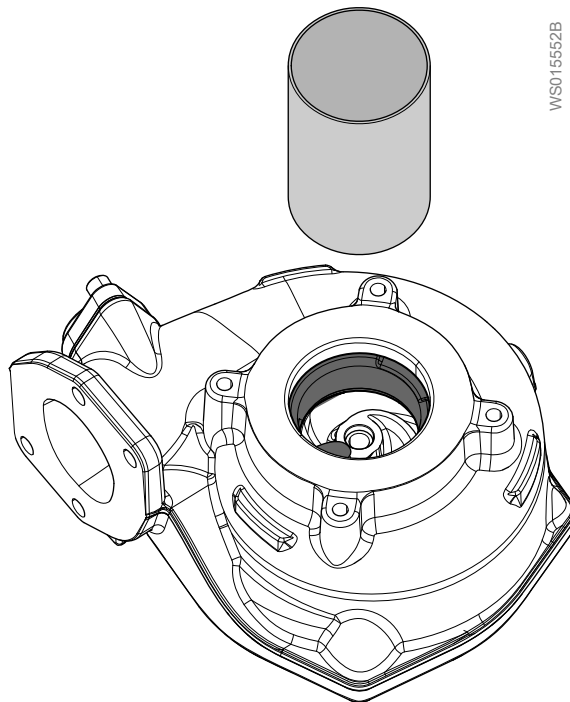
4. Put the pump in a vertical position.
5. Remove the four screws that attach the drive unit to the pump housing.

**CAUTION: Crush Hazard**

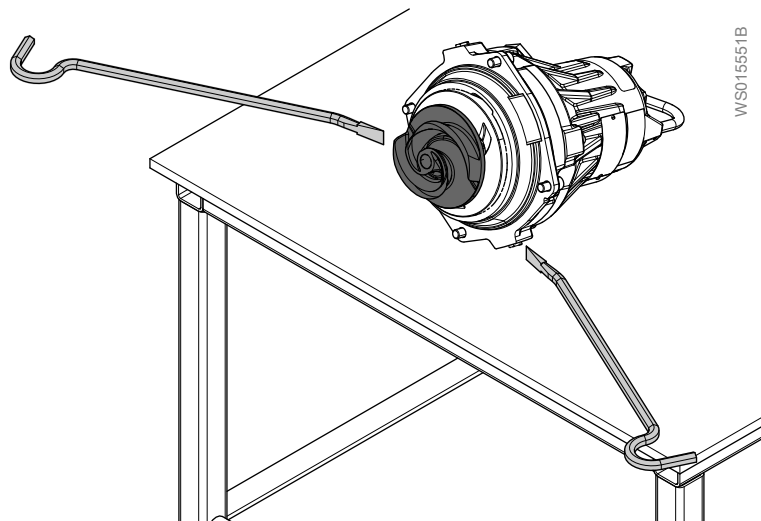
Make sure that the unit cannot roll or fall over and injure people or damage property.



6. Remove the drive unit from the pump housing.
If the impeller is loose from the shaft, then the impeller rests in the pump housing.
7. Remove the insert ring only if it needs to be replaced:



- a) Turn the pump housing upside down and put it into a hydraulic press.
 - b) Put a retention ring on the inlet.
 - c) Use a suitable sleeve to press out the insert ring.
 - d) Discard the insert ring.
8. If the impeller stays on the shaft, then remove the impeller by force:
 - a) Put the drive unit in a horizontal position.
 - b) Use crowbars to remove the impeller.



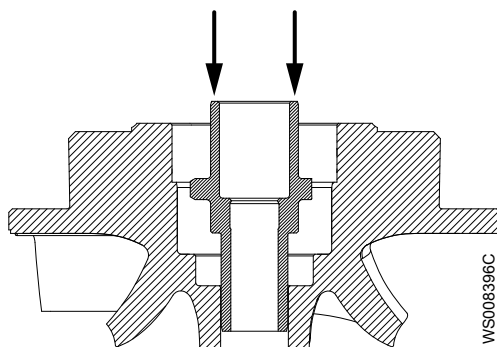
9. Remove the adjusting screw and the conical sleeve.

6.6.4.3 Install the Adaptive N™ impeller: P, S, T, Z installations

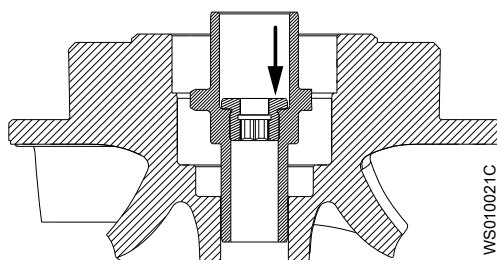
To see which pumps are Adaptive N™, see [Product description](#) on page 11.

1. Before installing the impeller, check that the sleeve moves freely up and down by pushing it.

When the sleeve is released, it should be fully pushed out again. If the sleeve does not move freely, or does not come fully out, then replace the impeller unit.



2. Adjust the adjusting screw in the new impeller.
The correct position is at the bottom without any tightening.

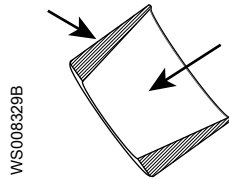


3. Prepare the shaft:

- a) Polish off any flaws by using a fine emery cloth.

The end of the shaft must be clean and free from burrs.

- b) Coat the inner conic and the outer cylindrical surfaces of the sleeve with a thin layer of grease.

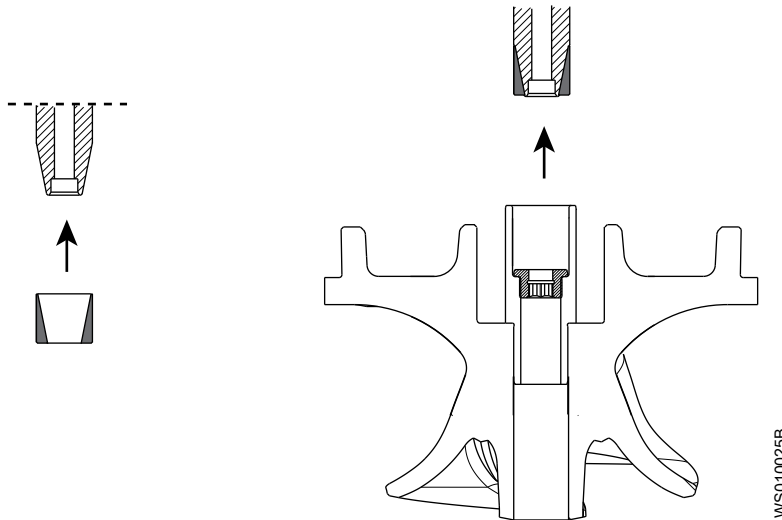


The correct lubrication is grease for bearings, for example Exxon Mobil Unirex N3, Mobil Mobilith SHC 220 or equivalent.

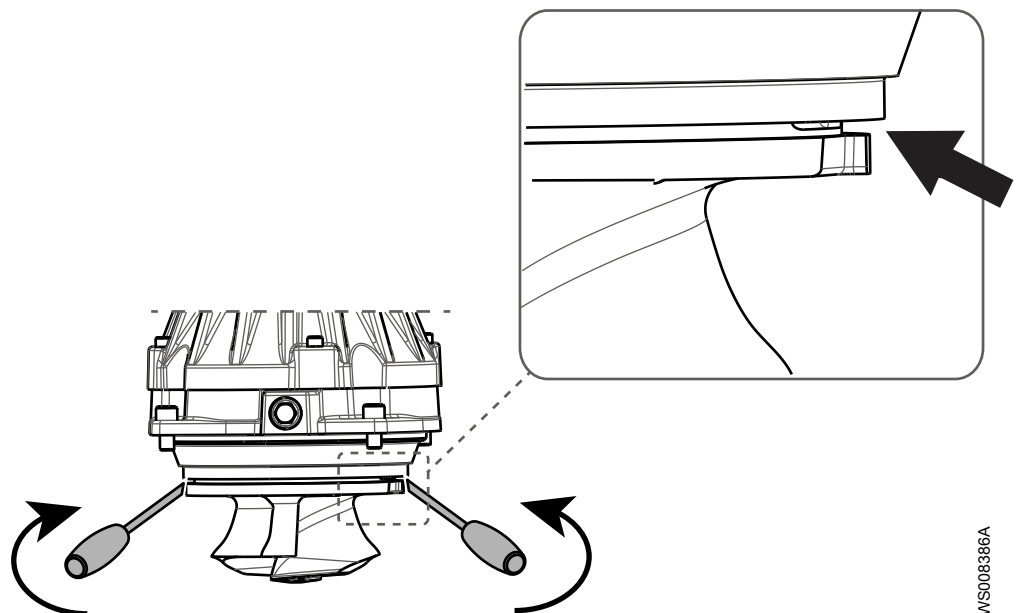
NOTICE:

Surplus grease can cause the impeller to become loose. Remove surplus grease from conical and/or cylindrical surfaces of shafts and/or sleeves.

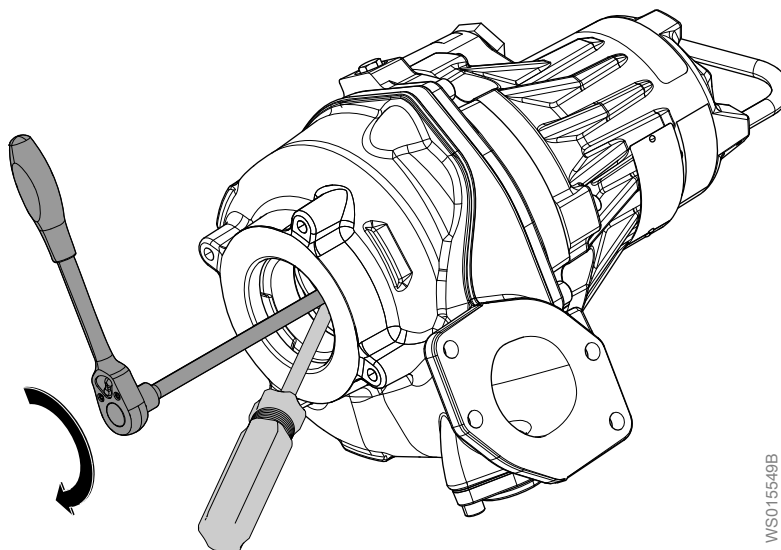
4. Install the impeller sleeve and the impeller onto the shaft.



5. Apply grease on the impeller screw and use the screw to attach the impeller in position. Hand-tighten the screw.
6. Check that the impeller is in its lowest position:
 - a) Insert two screwdrivers as shown in the following figure, and push the impeller downwards.



7. If the insert ring has been removed, then use a hydraulic press to install a new insert ring into the pump housing.
Apply oil on the insert ring.
8. Mount the pump housing:
 - a) Fit the pump housing.
 - b) Fit and tighten the lubricated screws.
 Tightening torque: 57 Nm (42 lbf·ft)
9. Put the pump in a horizontal position.
10. Remove the impeller screw.
11. Lock the impeller with a screwdriver through the inlet.
12. Turn the adjustment screw clockwise until the impeller barely touches the insert ring.



13. Check that the impeller can rotate freely.


WARNING: Crush Hazard

Beware of the pinch point hazard between the rotating impeller and the guide pin.

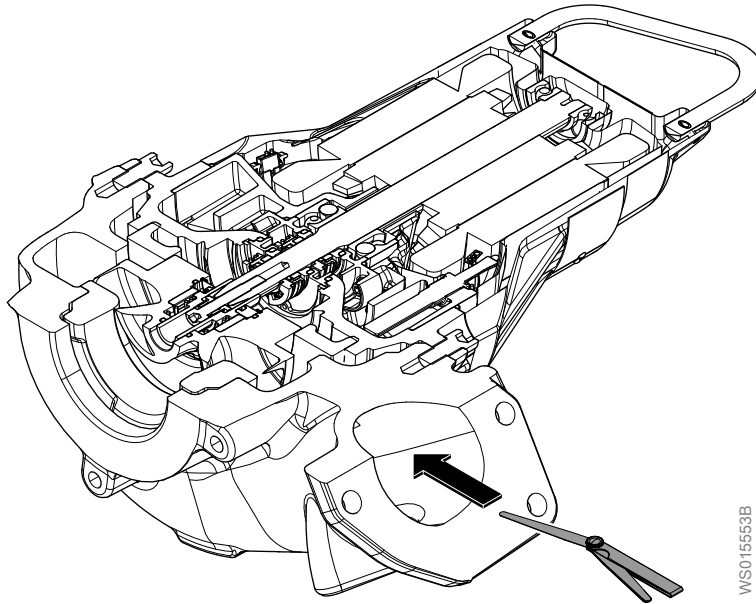
14. Tighten the impeller screw.
Tightening torque: 34 Nm (25 lbf·ft)
Tighten a further 1/8 turn, 45°, after tightening to the correct torque.
15. Check that the impeller can rotate freely.


WARNING: Crush Hazard

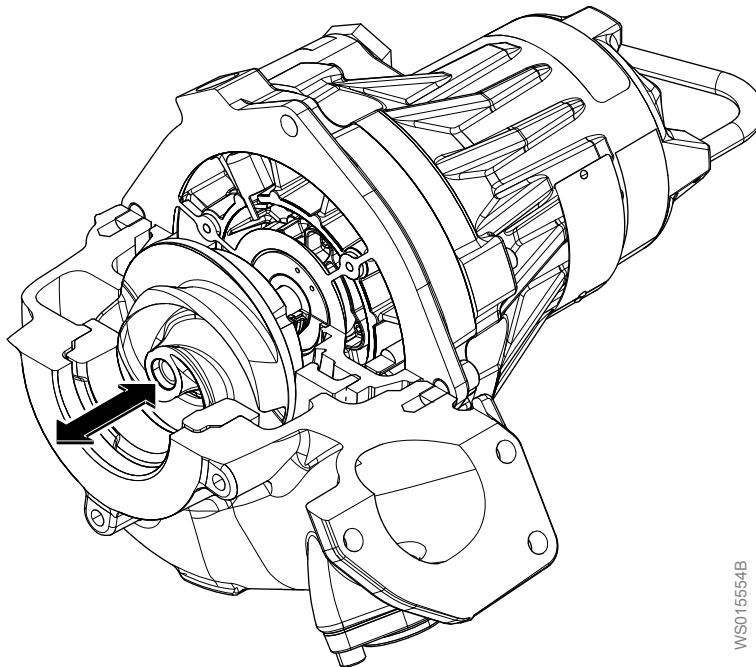
Beware of the pinch point hazard between the rotating impeller and the guide pin.

16. Check with an extended feeler gauge that the impeller clearance is 0.1–0.6 mm (0.004–0.024 in).

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17. To check the adaptive function, press the impeller from the inlet.
Make sure that the impeller moves both away from the inlet and back again.



18. Reinstall the pump.
T-installations: See the instructions in *T-installation: Bleed air before starting pump* on page 39.

7 Troubleshooting

7.1 Electrical troubleshooting

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DANGER: Electrical Hazard

Troubleshooting a live control panel exposes personnel to hazardous voltages. Electrical troubleshooting must be done by a qualified electrician.

Follow these guidelines when troubleshooting:

- Disconnect and lock out the power supply except when conducting checks that require voltage.
- Make sure that no one is near the unit when the power supply is reconnected.
- When troubleshooting electrical equipment, use the following:
 - Universal instrument multimeter
 - Test lamp (continuity tester)
 - Wiring diagram

7.2 The pump does not start



DANGER: Crush Hazard

Moving parts can entangle or crush. Always disconnect and lock out power before servicing to prevent unexpected startup. Failure to do so could result in death or serious injury.



WARNING: Electrical Hazard

The permanent magnet synchronous motor generates voltage on internal components when the shaft rotates, even if the power supply is disconnected. Make sure that the shaft cannot rotate if the unit is disassembled.

NOTICE:

Do NOT override the motor protection repeatedly if it has tripped. Doing so may result in equipment damage.

Cause	Remedy
An alarm signal has been triggered on the control panel.	Check that: • The impeller rotates freely. • The sensor indicators do not indicate an alarm. • The overload protection is not tripped.
The pump does not start automatically, but can be started manually.	Check that: • The start level regulator is functioning. Clean or replace if necessary. • All connections are intact. • The relay and contactor coils are intact. • The control switch (Man/Auto) makes contact in both positions. Check the control circuit and functions.

Cause	Remedy
The installation is not receiving voltage.	Check that: • The main power switch is on. • There is control voltage to the start equipment. • The fuses are intact. • There is voltage in all phases of the supply line. • All fuses have power and that they are securely fastened to the fuse holders. • The overload protection is not tripped. • The motor cable is not damaged.
The impeller is stuck.	Clean: • The impeller • The sump in order to prevent the impeller from clogging again.

If the problem persists, then contact a sales or authorized service representative.

Always state the serial number of the product, see [Product description](#) on page 11.

7.3 The pump does not stop when a level sensor is used



DANGER: Crush Hazard

Moving parts can entangle or crush. Always disconnect and lock out power before servicing to prevent unexpected startup. Failure to do so could result in death or serious injury.



WARNING: Electrical Hazard

The permanent magnet synchronous motor generates voltage on internal components when the shaft rotates, even if the power supply is disconnected. Make sure that the shaft cannot rotate if the unit is disassembled.

Cause	Remedy
The pump is unable to empty the sump to the stop level.	Check that: • There are no leaks from the piping and/or discharge connection. • The impeller is not clogged. • The non-return valve(s) are functioning properly. • The pump has adequate capacity. For information: Contact a sales or authorized service representative.
There is a malfunction in the level-sensing equipment.	• Clean the level regulators. • Check the functioning of the level regulators. • Check the contactor and the control circuit. • Replace all defective items.
The stop level is set too low.	Raise the stop level.

If the problem persists, then contact a sales or authorized service representative.

Always state the serial number of the product, see [Product description](#) on page 11.

7.4 The pump starts-stops-starts in rapid sequence

Cause	Remedy
The pump starts due to back-flow which fills the sump to the start level again.	Check that: • The distance between the start and stop levels is sufficient. • The non-return valve(s) work(s) properly. • The length of the discharge pipe between the pump and the first non-return valve is sufficiently short.
The self-holding function of the contactor malfunctions.	Check: • The contactor connections. • The voltage in the control circuit in relation to the rated voltages on the coil. • The functioning of the stop-level regulator. • Whether the voltage drop in the line at the starting surge causes the contactor's self-holding malfunction.

If the problem persists, then contact a sales or authorized service representative.

Always state the serial number of the product, see [Product description](#) on page 11.

7.5 The pump runs but the motor protection trips



DANGER: Crush Hazard

Moving parts can entangle or crush. Always disconnect and lock out power before servicing to prevent unexpected startup. Failure to do so could result in death or serious injury.



WARNING: Electrical Hazard

The permanent magnet synchronous motor generates voltage on internal components when the shaft rotates, even if the power supply is disconnected. Make sure that the shaft cannot rotate if the unit is disassembled.

NOTICE:

Do NOT override the motor protection repeatedly if it has tripped. Doing so may result in equipment damage.

Cause	Remedy
The motor protection is set too low.	Set the motor protection according to the data plate and if applicable the cable chart.
The impeller is difficult to rotate by hand.	• Clean the impeller. • Clean out the wet well. • Check that the impeller is correctly trimmed.
The drive unit cannot receive full voltage on all three phases.	• Check the fuses. Replace fuses that have tripped. • If the fuses are intact, then notify a certified electrician.
The phase currents change, or they are too high.	Contact a sales or authorized service representative.
The insulation between the phases and ground in the stator is defective.	1. Use an insulation tester. Use a 1000 VDC insulation and continuity tester to check that the insulation between the phases, and between any phase and ground, is > 5 megohms. 2. If the insulation is less, then do the following: Contact a sales or authorized service representative.

Cause	Remedy
The density of the pumped fluid is too high.	Make sure that the maximum density is 1100 kg/m ³ (9.2 lb/US gal) • Change the impeller, or • Change to a more applicable pump • Contact a sales or authorized service representative.
There is a malfunction in the overload protection.	Replace the overload protection.

If the problem persists, then contact a sales or authorized service representative.

Always state the serial number of the product, see [Product description](#) on page 11.

7.6 The pump delivers too little or no water



DANGER: Crush Hazard

Moving parts can entangle or crush. Always disconnect and lock out power before servicing to prevent unexpected startup. Failure to do so could result in death or serious injury.



WARNING: Electrical Hazard

The permanent magnet synchronous motor generates voltage on internal components when the shaft rotates, even if the power supply is disconnected. Make sure that the shaft cannot rotate if the unit is disassembled.

NOTICE:

Do NOT override the motor protection repeatedly if it has tripped. Doing so may result in equipment damage.

Cause	Remedy
The impeller rotates in the wrong direction.	• If it is a 3-phase pump, then transpose two phase leads. • If it is a 1-phase pump, then do the following: Contact a sales or authorized service representative.
One or more of the valves are set in the wrong positions.	• Reset the valves that are set in the wrong position. • Replace the valves, if necessary. • Check that all valves are correctly installed according to media flow. • Check that all valves open correctly.
The impeller is difficult to rotate by hand.	• Clean the impeller. • Clean out the sump. • Check that the impeller is properly trimmed.
The pipes are obstructed.	To ensure a free flow, clean out the pipes.
The pipes and joints leak.	Find the leaks and seal them.
There are signs of wear on the impeller, pump, and casing.	Replace the worn parts.
The liquid level is too low.	• Check that the level sensor is set correctly. • Depending on the installation type, add a means for priming the pump, such as a foot valve.

If the problem persists, then contact a sales or authorized service representative.

Always state the serial number of the product, see [Product description](#) on page 11.

8 Technical reference

8.1 Application limits

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Data	Description
Liquid temperature	40°C (104°F) maximum 3085.060, 3085.160, 3085.183: • Warm-liquid version: 70°C (158°F) maximum
pH of the pumped media (liquid)	5.5–14
Liquid density	1100 kg/m ³ (9.2 lb for each US gal) maximum
Depth of immersion	Maximum 20 m (65 ft)
Other	For the specific weight, current, voltage, power ratings, and speed of the pump, see the data plate of the pump.

8.2 Motor data

Feature	Description
Motor type	3085.060/.070/.092/.160/.172/.183/.190/.891: • Squirrel-cage induction 3085.900/.910/.920/.930: • Line-started, permanent-magnet synchronous motor
Frequency	50 Hz or 60 Hz
Supply	1-phase or 3-phase
Starting method	• Direct on-line • Star-delta • Soft starter Not recommended for LSPM motors • Variable frequency drive (VFD)
Maximum starts for each hour	30 evenly spaced starts for each hour
Code compliance	IEC 60034-1
Voltage variation without overheating	±10%, if it does not run continuously at full load
Voltage imbalance tolerance	2%
Stator insulation class	H (180°C [356°F])

Motor encapsulation

Motor encapsulation is in accordance with IP68.

Xylem |'zīləm|

- 1) The tissue in plants that brings water upward from the roots;
- 2) a leading global water technology company.

We're a global team unified in a common purpose: creating advanced technology solutions to the world's water challenges. Developing new technologies that will improve the way water is used, conserved, and re-used in the future is central to our work. Our products and services move, treat, analyze, monitor and return water to the environment, in public utility, industrial, residential and commercial building services settings. Xylem also provides a leading portfolio of smart metering, network technologies and advanced analytics solutions for water, electric and gas utilities. In more than 150 countries, we have strong, long-standing relationships with customers who know us for our powerful combination of leading product brands and applications expertise with a strong focus on developing comprehensive, sustainable solutions.

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Xylem Water Solutions Global
Services AB 556782-9253
361 80 Emmaboda
Sweden
Tel: +46-471-24 70 00
Fax: +46-471-24 74 01
<https://tpi.xylem.com>

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